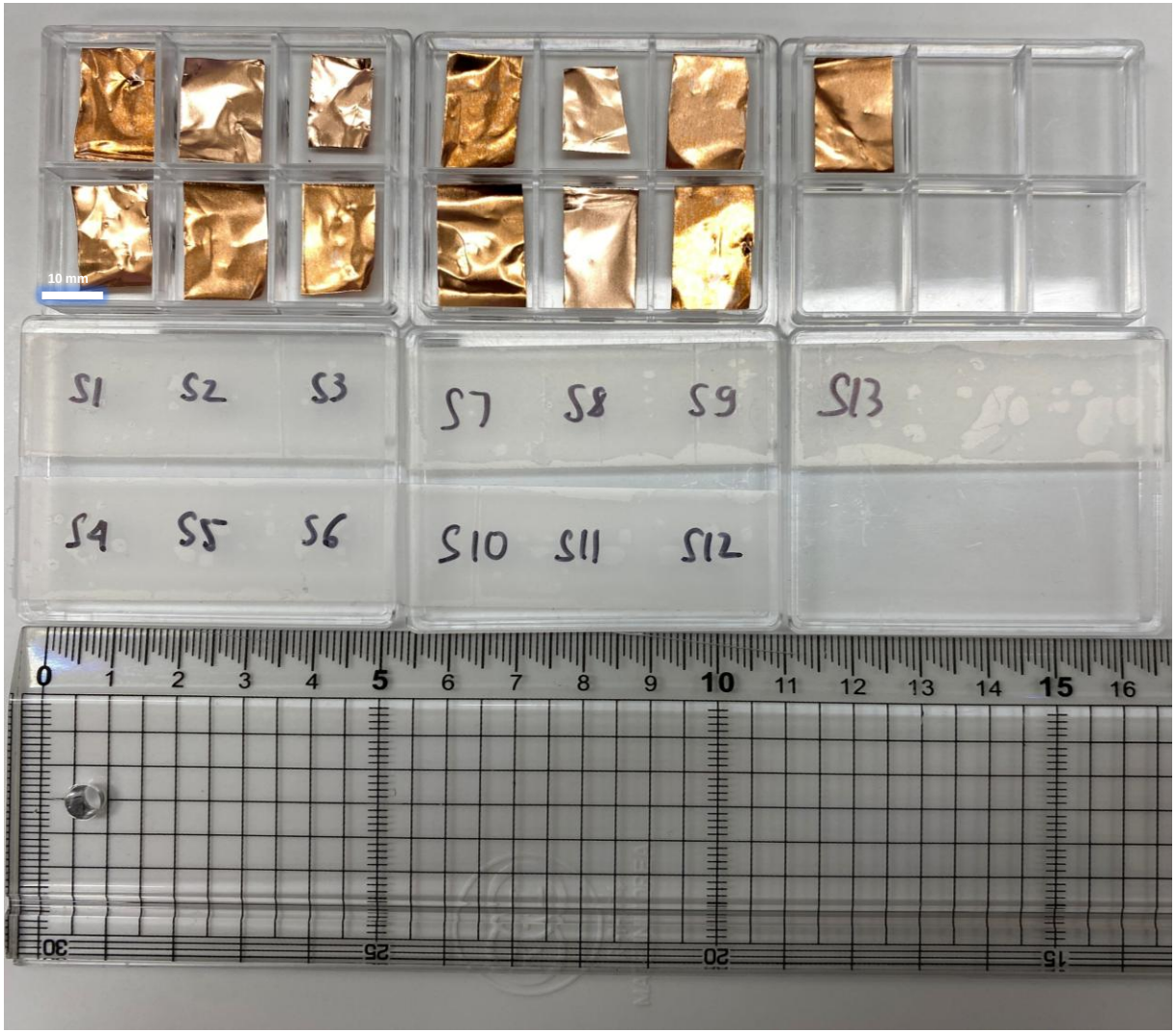
J-CVD Gr on Cu Foil Sample of Cu Probed by XRD FWHM Analysis

06 Oct 2026

Ho Huu Loi and 이승모

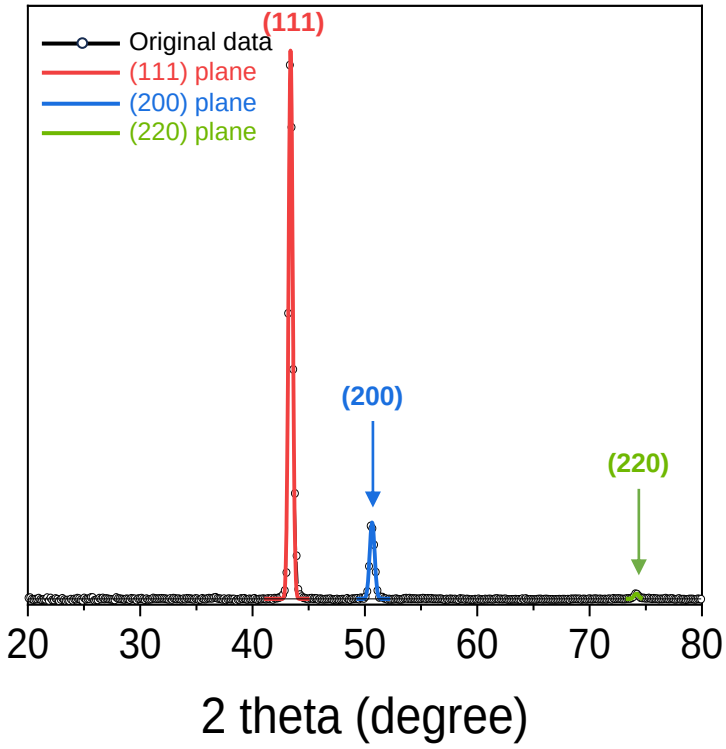
Sample Information

Total 13 samples, measured 2 positions

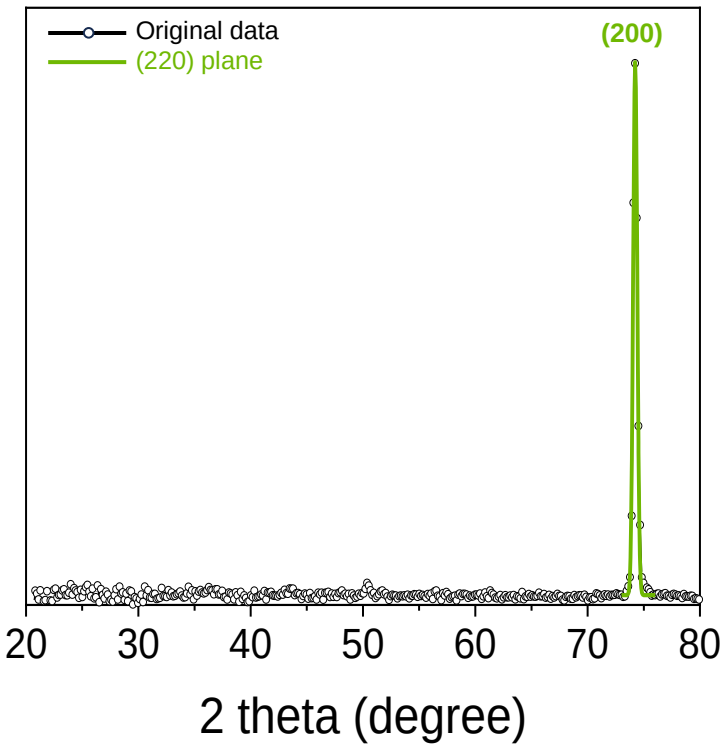


Original data XRD_S1

Position 1

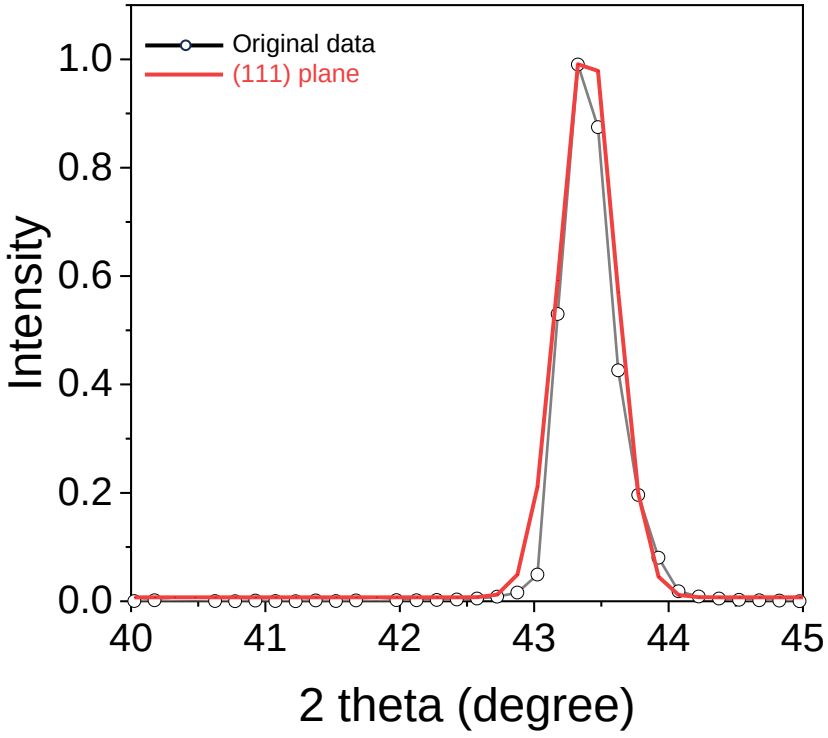


Position 2



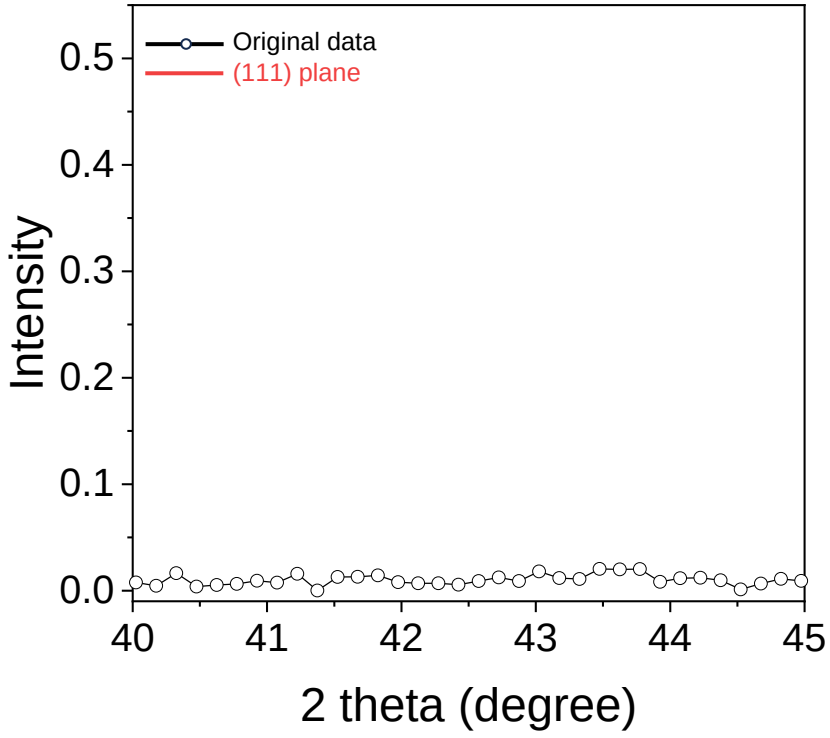
Peak Fitting Results (111) plane _S1

Position 1



FWHM:0.48386 ± 0.00336
Height: 1

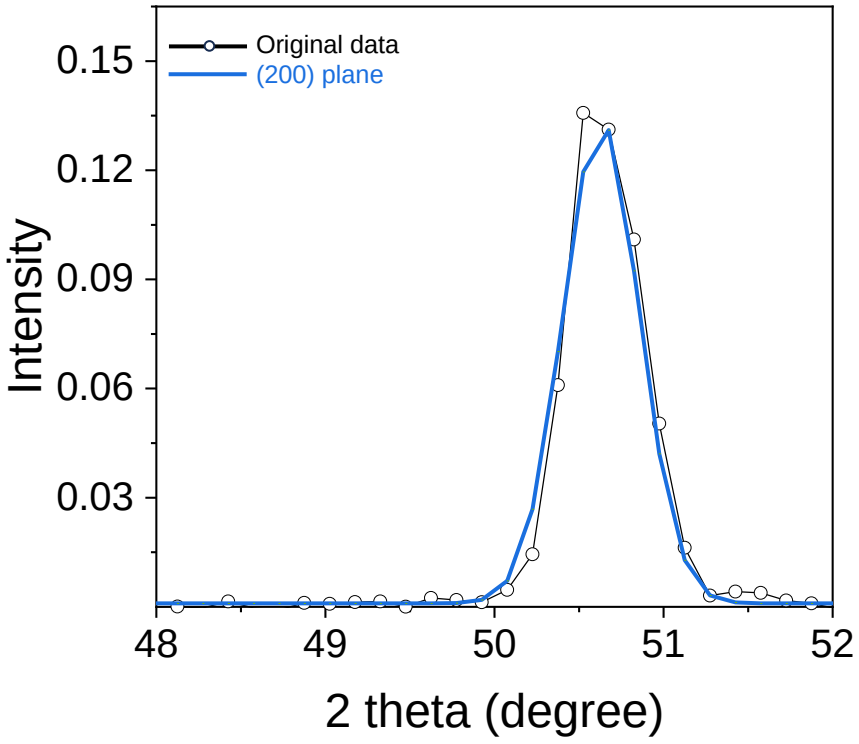
Position 2



No peak found

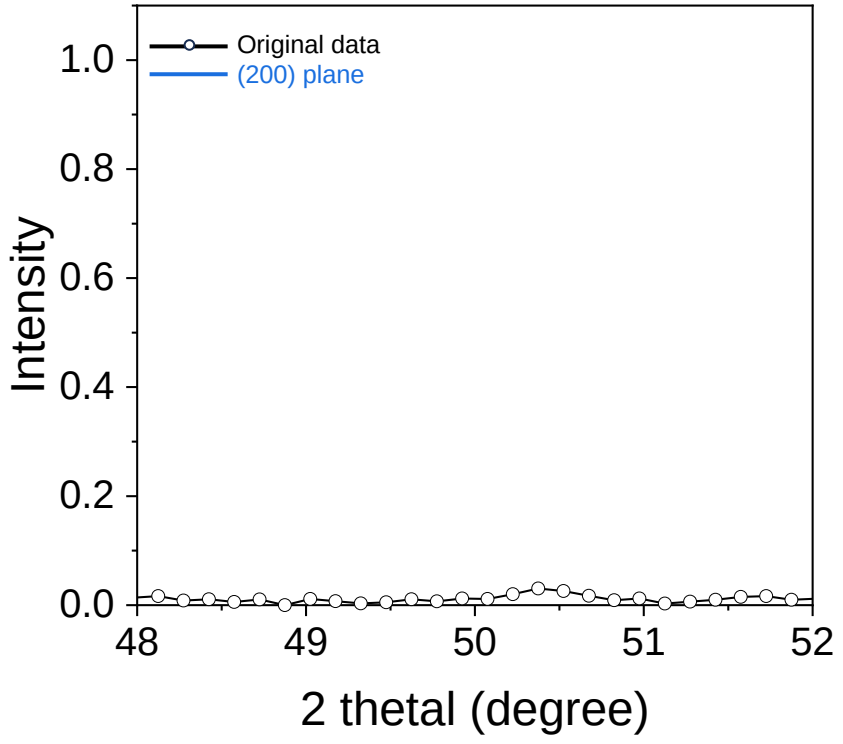
Peak Fitting Results (200) plane _S1

Position 1



FWHM: 0.52916 ± 0.01211
Height: 0.13259

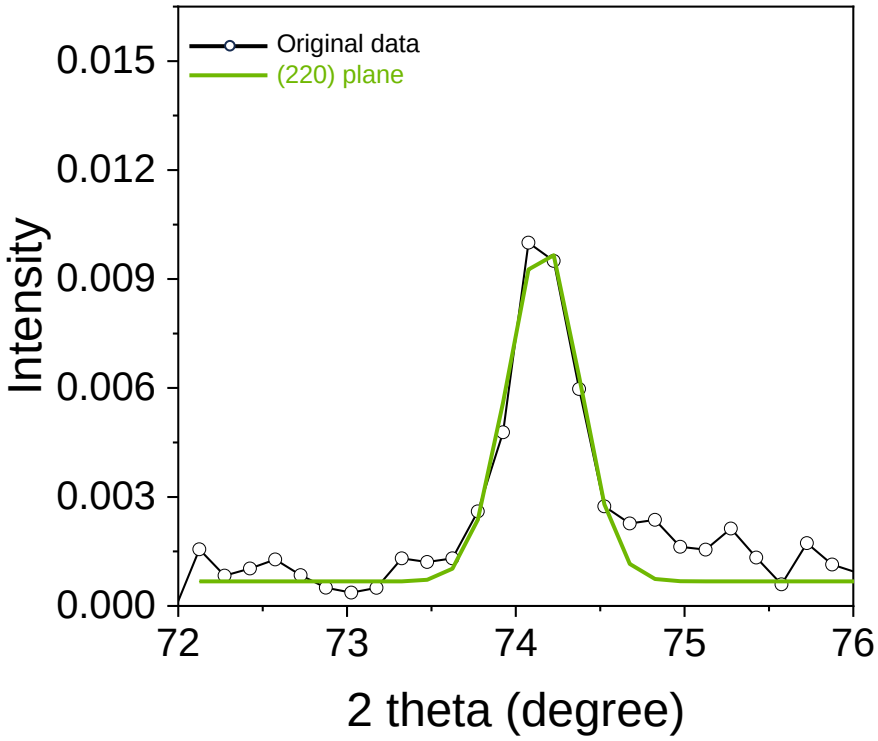
Position 2



No peak found

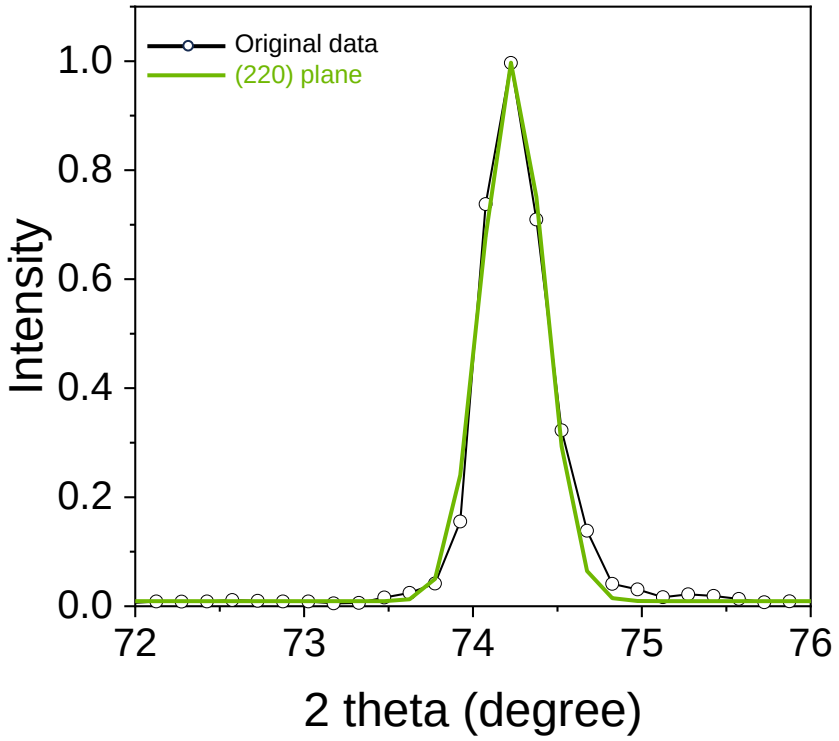
Peak Fitting Results (220) plane

Position 1



FWHM: 0.49436 ± 0.03321
Height: 0.00938

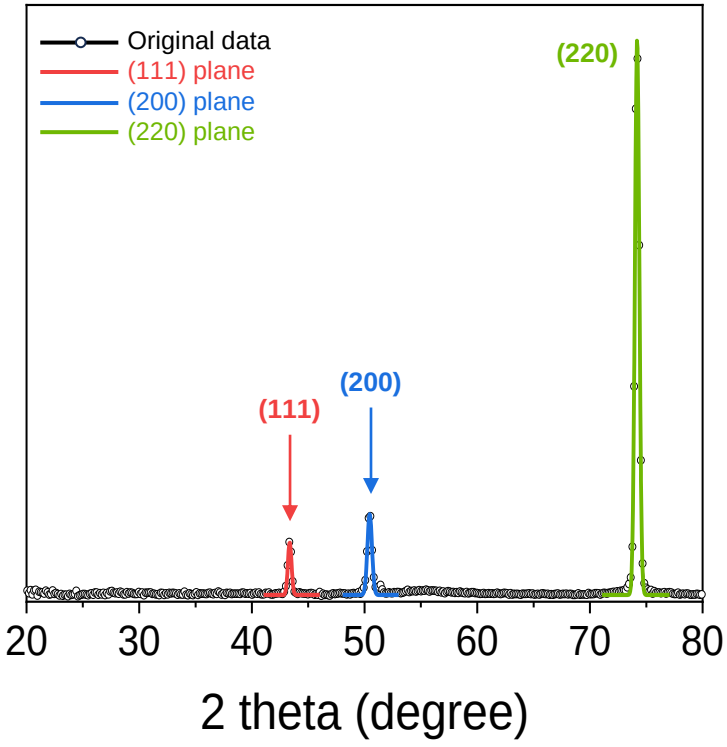
Position 2



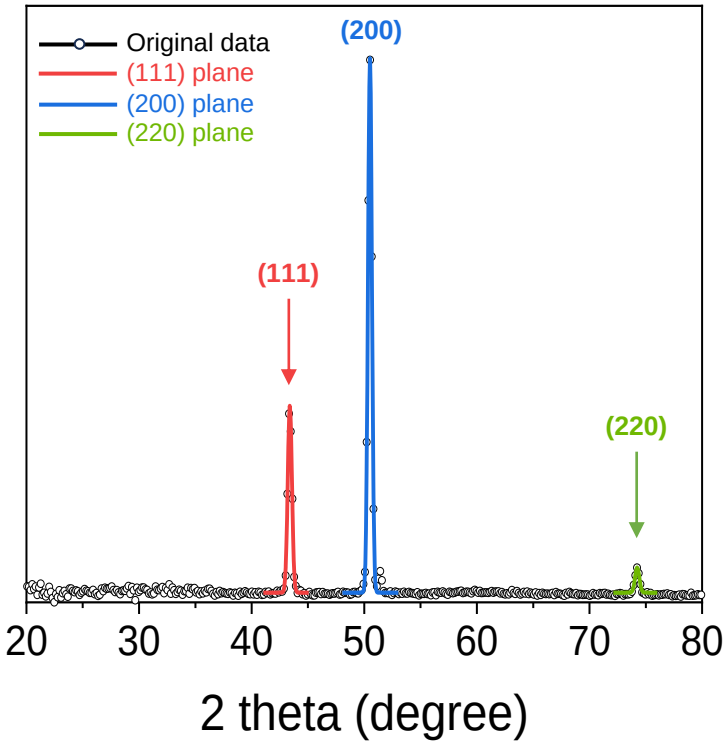
FWHM: 0.42996 ± 0.00783
Height: 1

Original data XRD_S2

Position 1

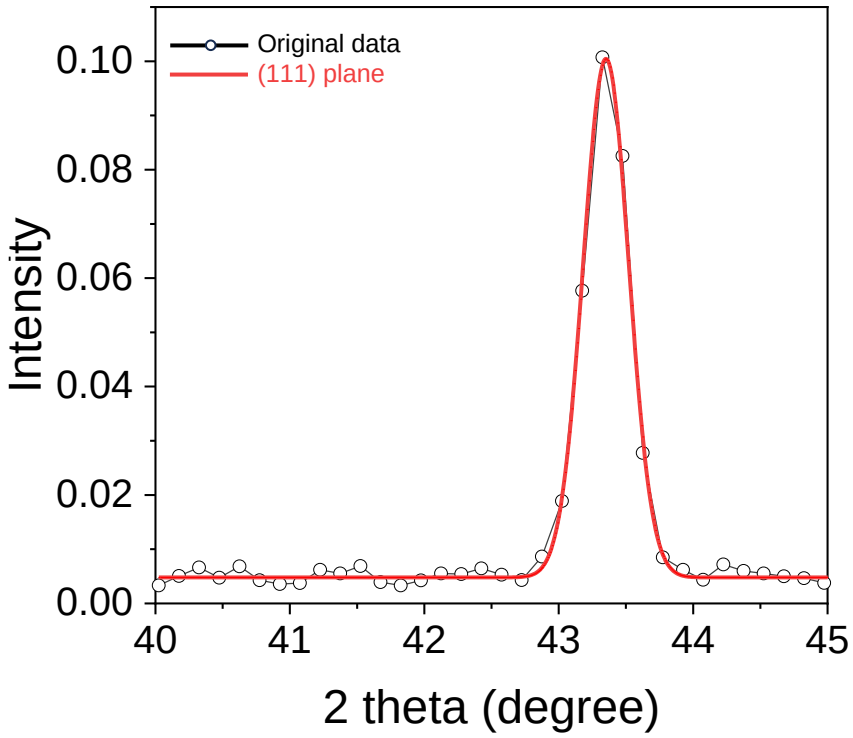


Position 2



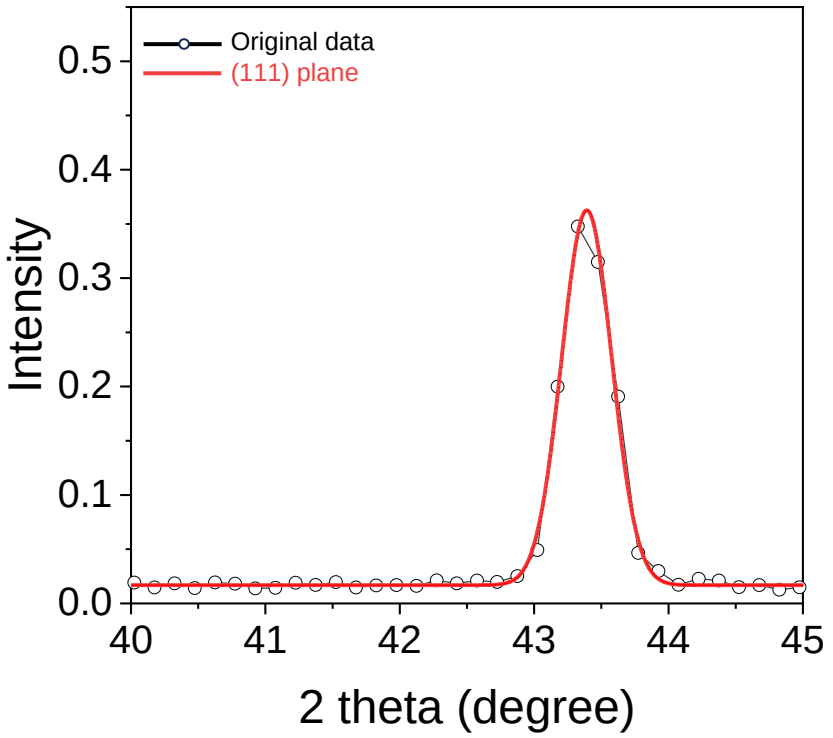
Peak Fitting Results (111) plane _S2

Position 1



FWHM:0.39848 ± 0.00711
Height: 0.099407

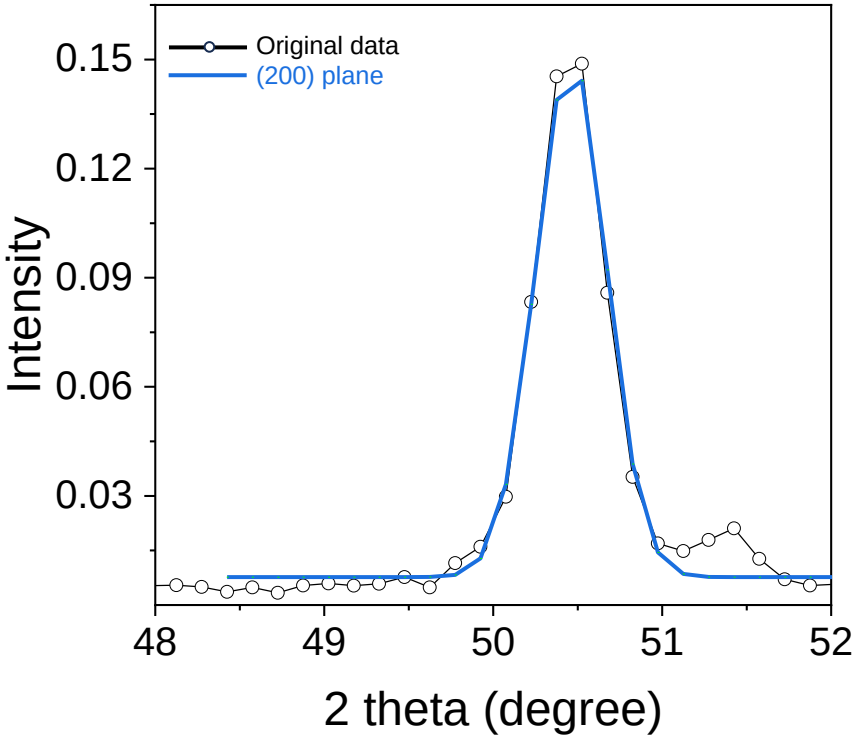
Position 2



FWHM:0.43857 ± 0.0061
Height: 0.34568

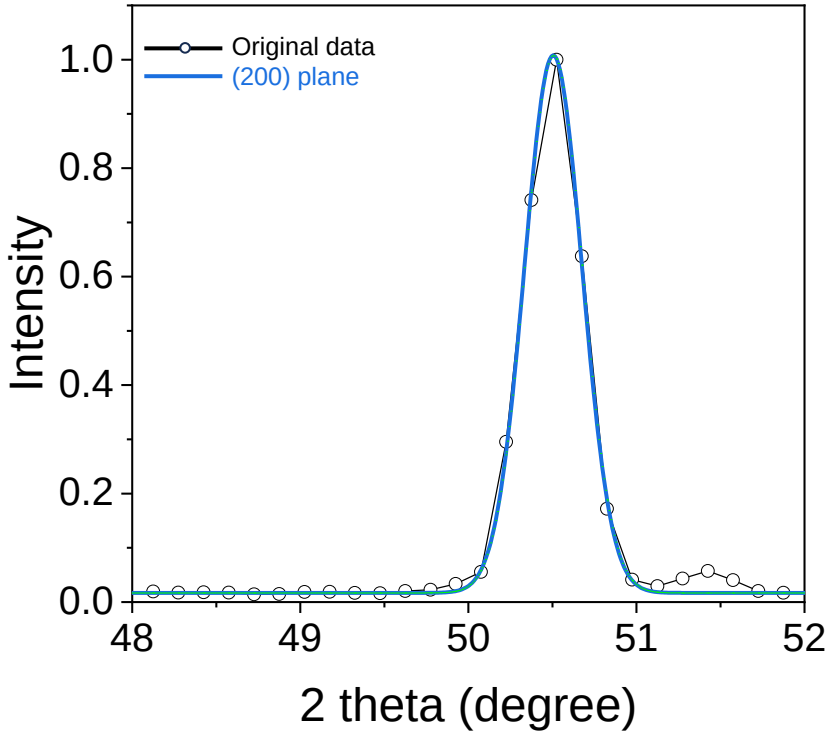
Peak Fitting Results (200) plane _S2

Position 1



FWHM: 0.49032 ± 0.01434
Height: 0.14304

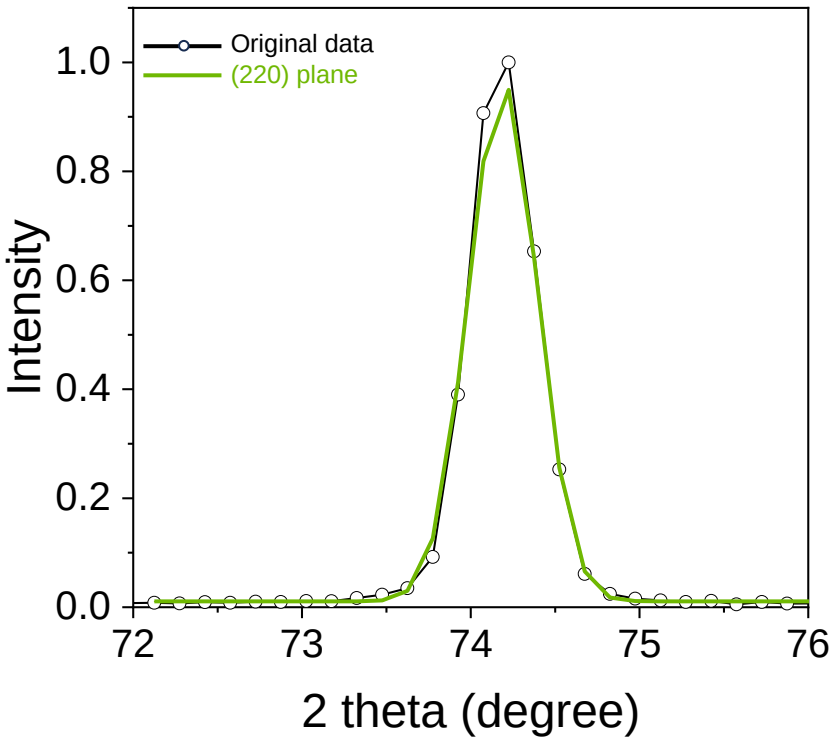
Position 2



FWHM: 0.403 ± 0.0214
Height: 1

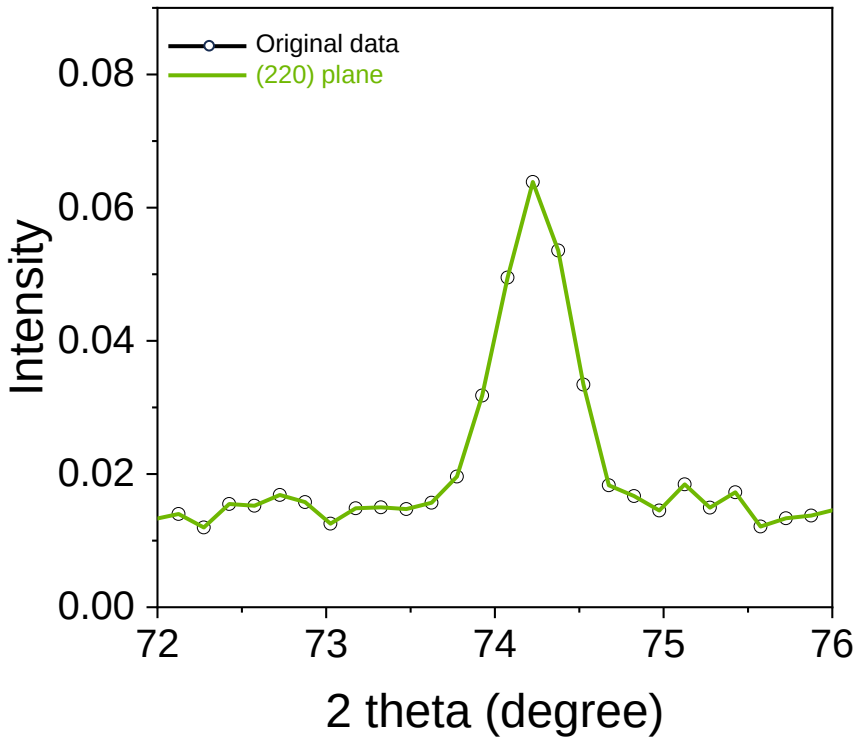
Peak Fitting Results (220) plane_S2

Position 1



FWHM: 0.47711 ± 0.0083
Height: 1

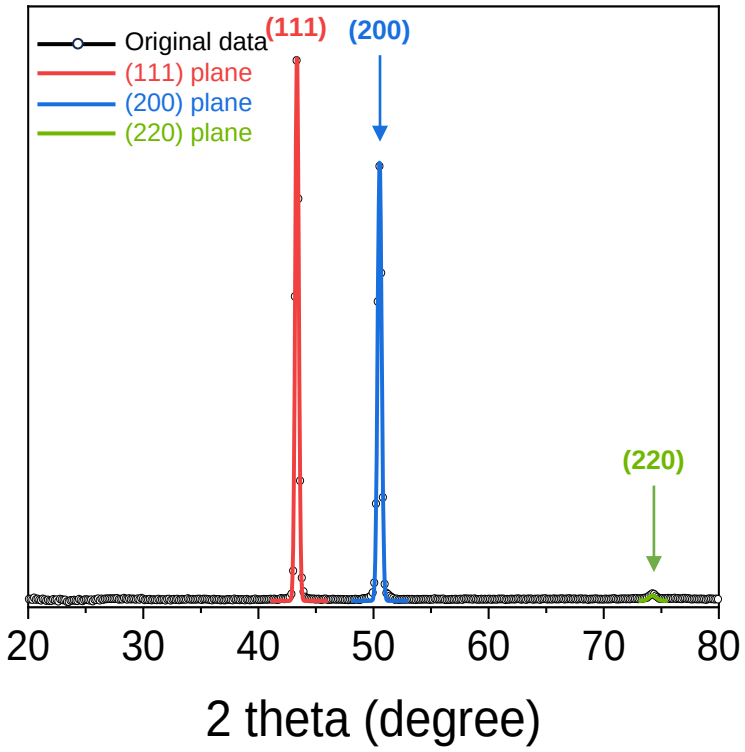
Position 2



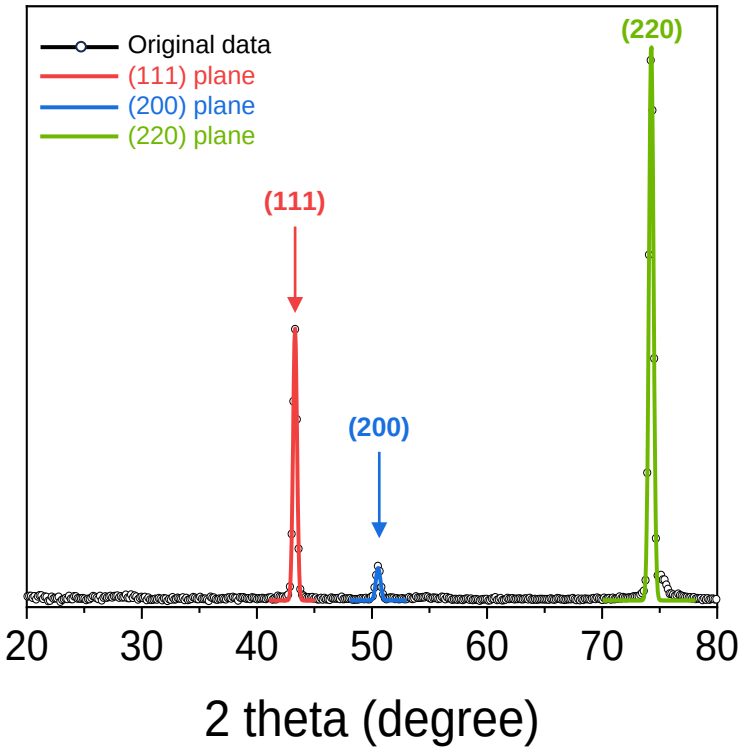
FWHM: 0.52314 ± 0.01736
Height: 0.063

Original data XRD_S3

Position 1

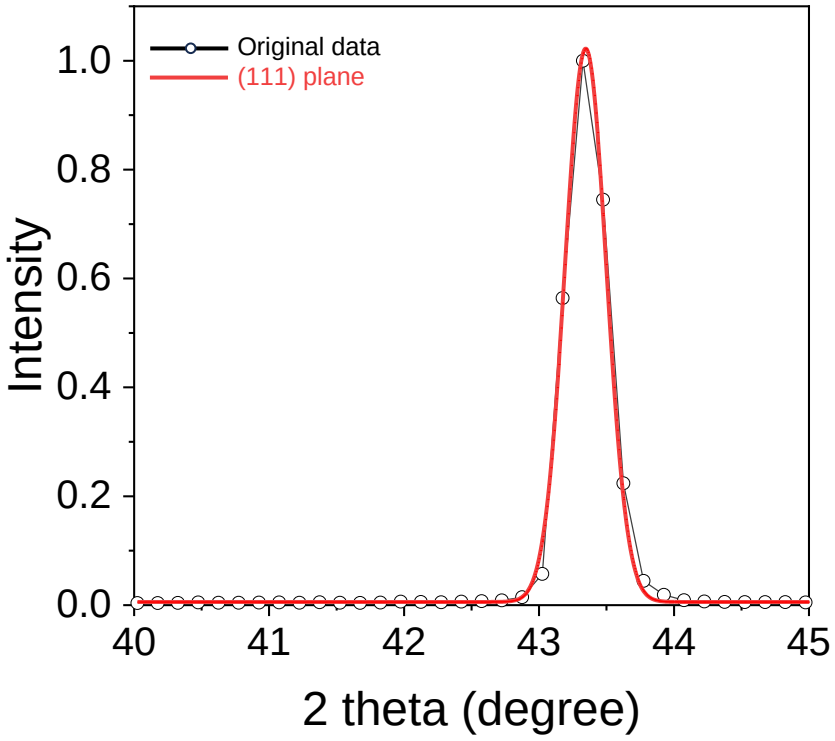


Position 2



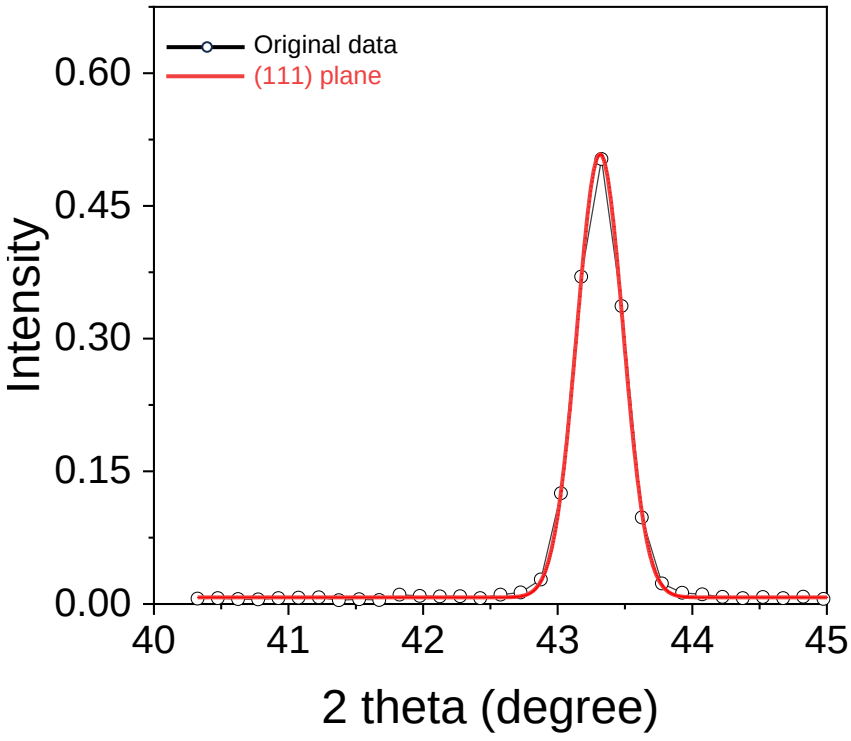
Peak Fitting Results (111) plane _S3

Position 1



FWHM:0.36056 ± 0.00503
Height: 1

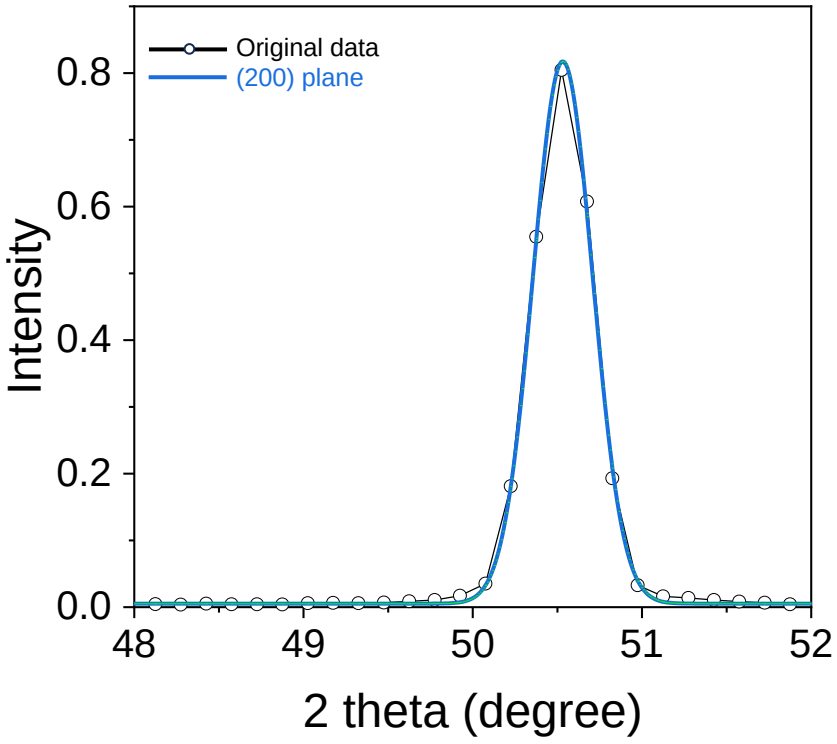
Position 2



FWHM:0.40352 ± 0.00239
Height: 0.500193

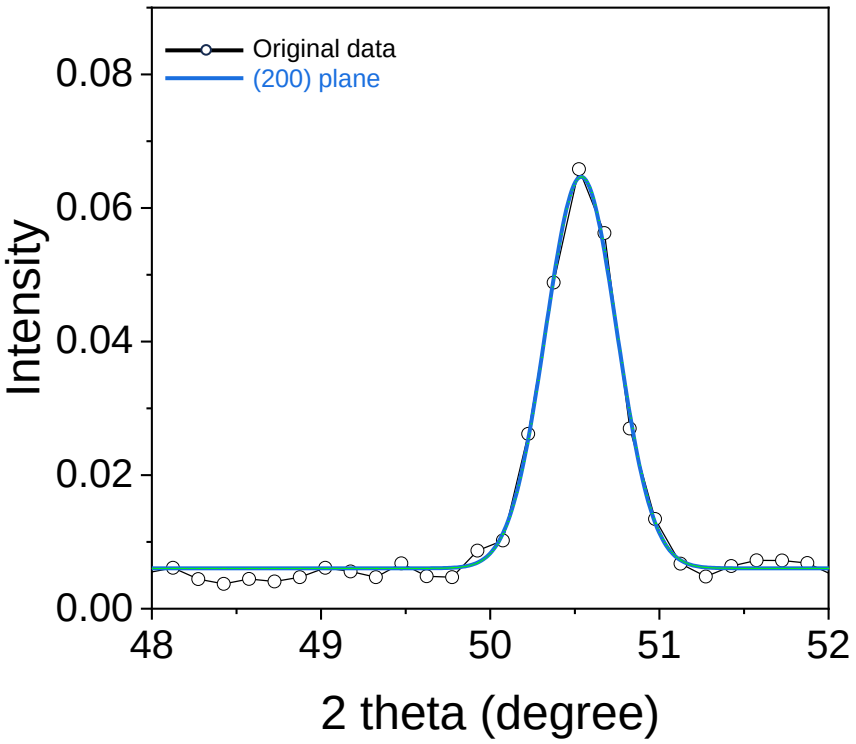
Peak Fitting Results (200) plane _S3

Position 1



FWHM: 0.41383 ± 0.00211
Height: 0.81243

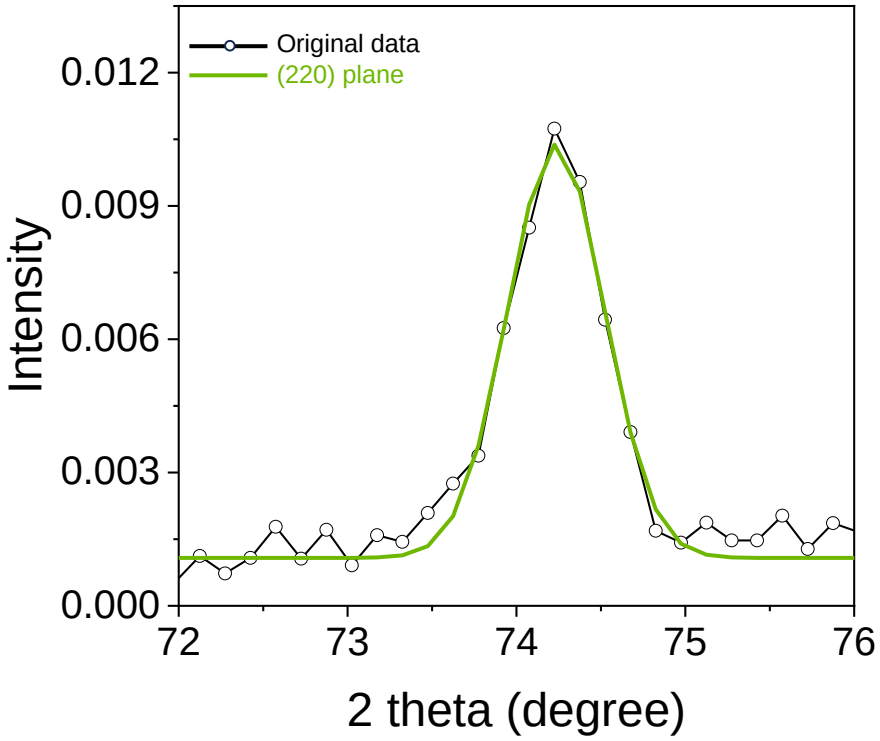
Position 2



FWHM: 0.49578 ± 0.01137
Height: 0.05863

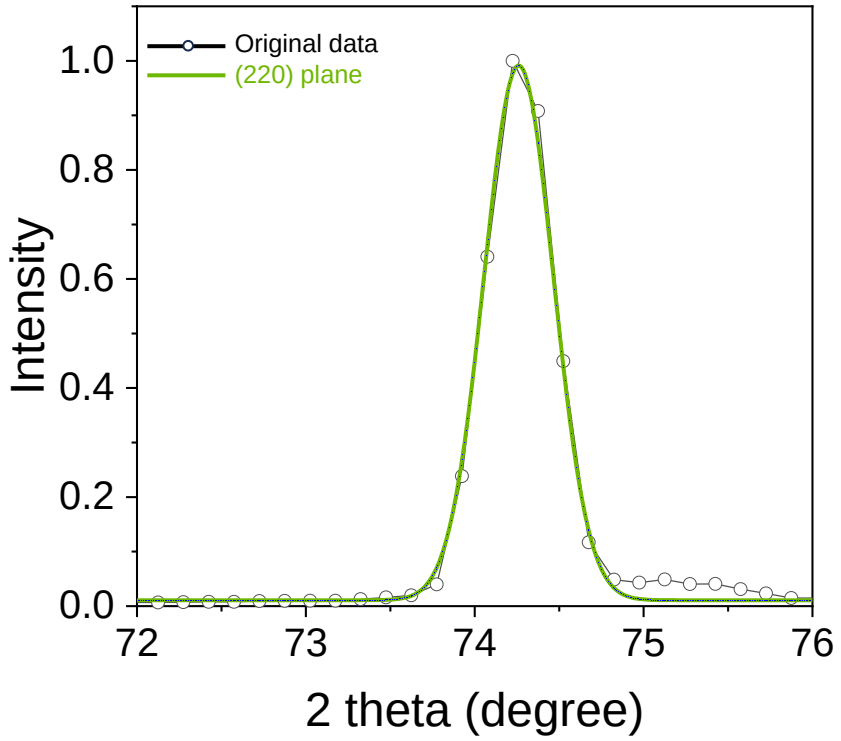
Peak Fitting Results (220) plane_S3

Position 1



FWHM: 0.67071 ± 0.0316
Height: 0.00931

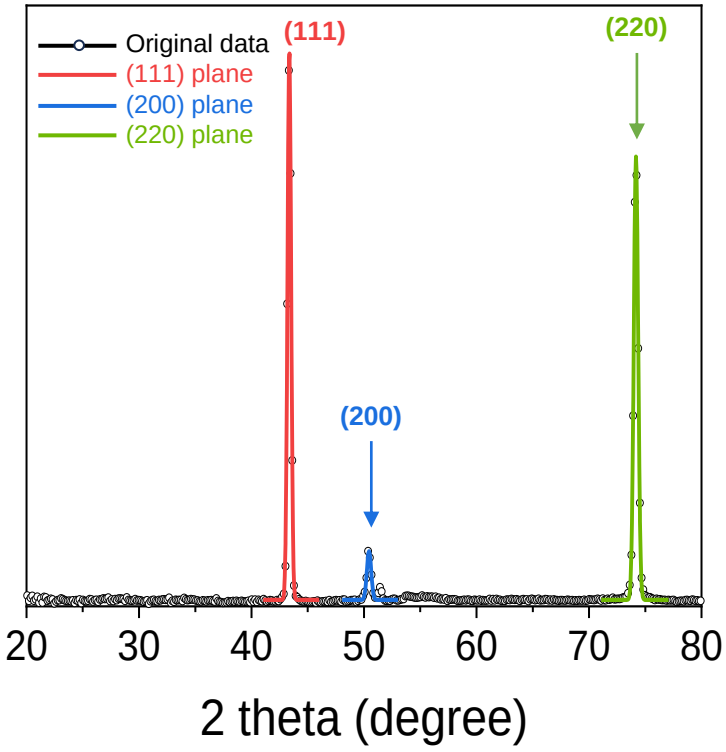
Position 2



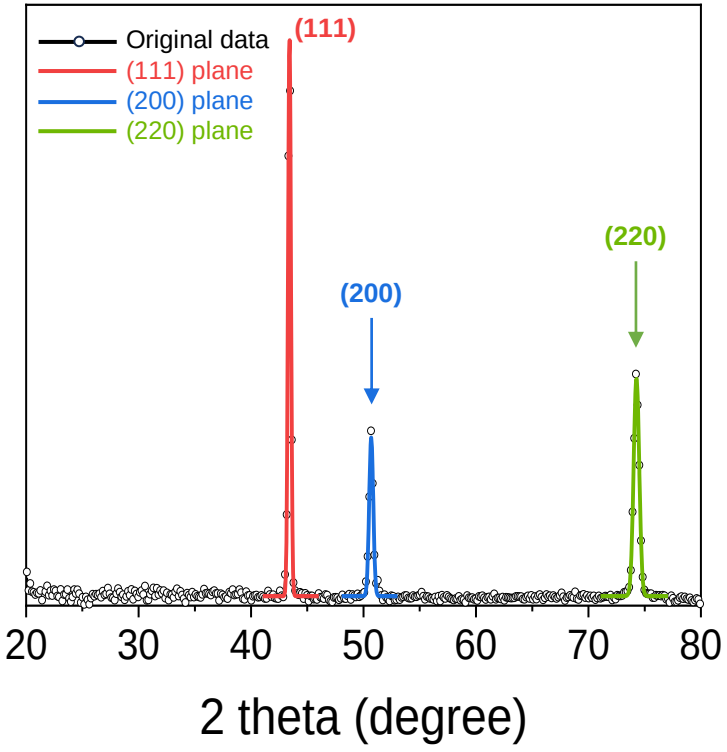
FWHM: 0.4834± 0.00649
Height: 1

Original data XRD_S4

Position 1

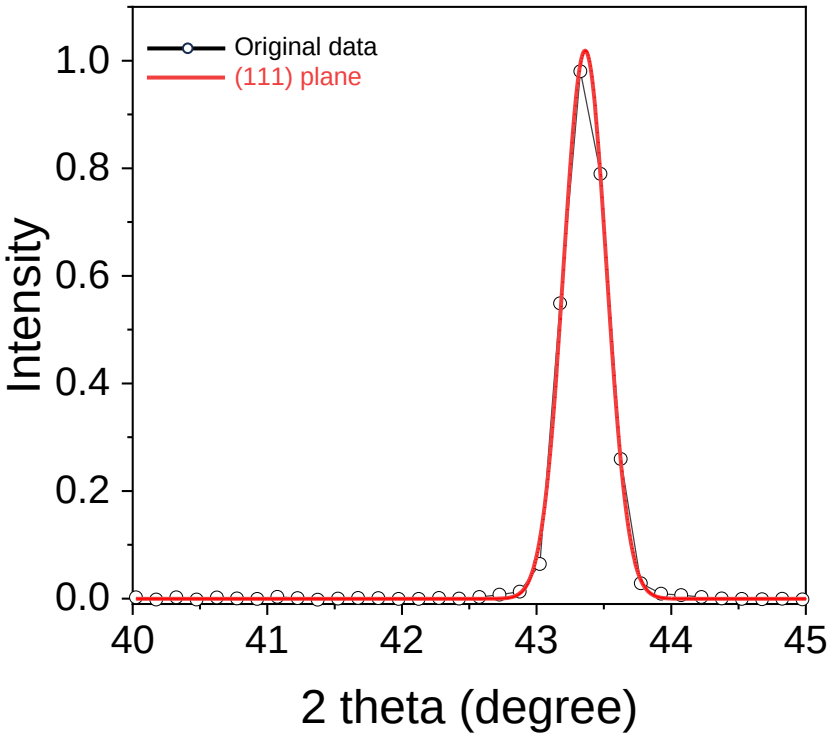


Position 2



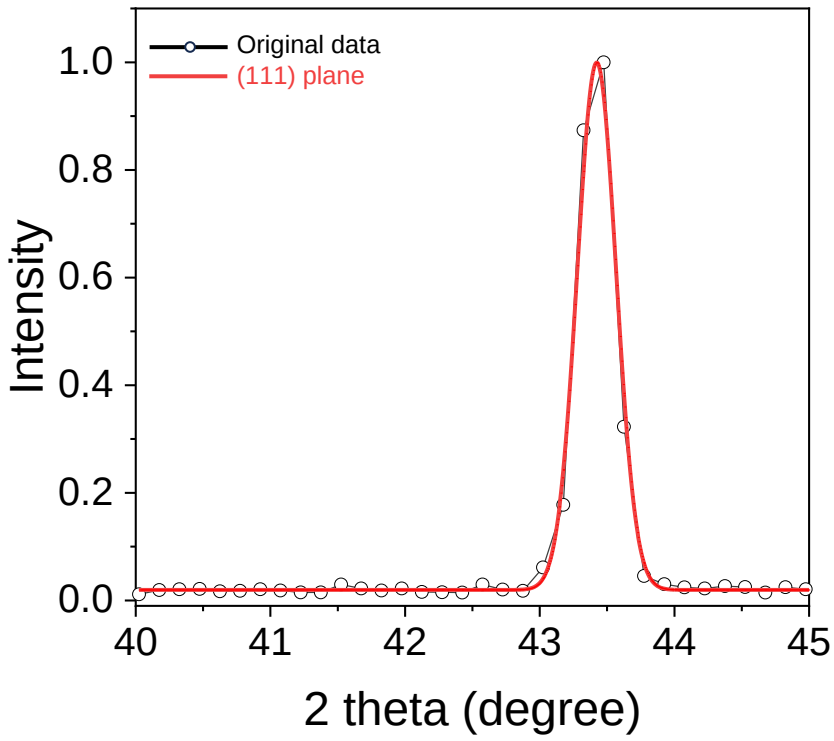
Peak Fitting Results (111) plane _S4

Position 1



FWHM:0.37541 ± 0.00393
Height: 1

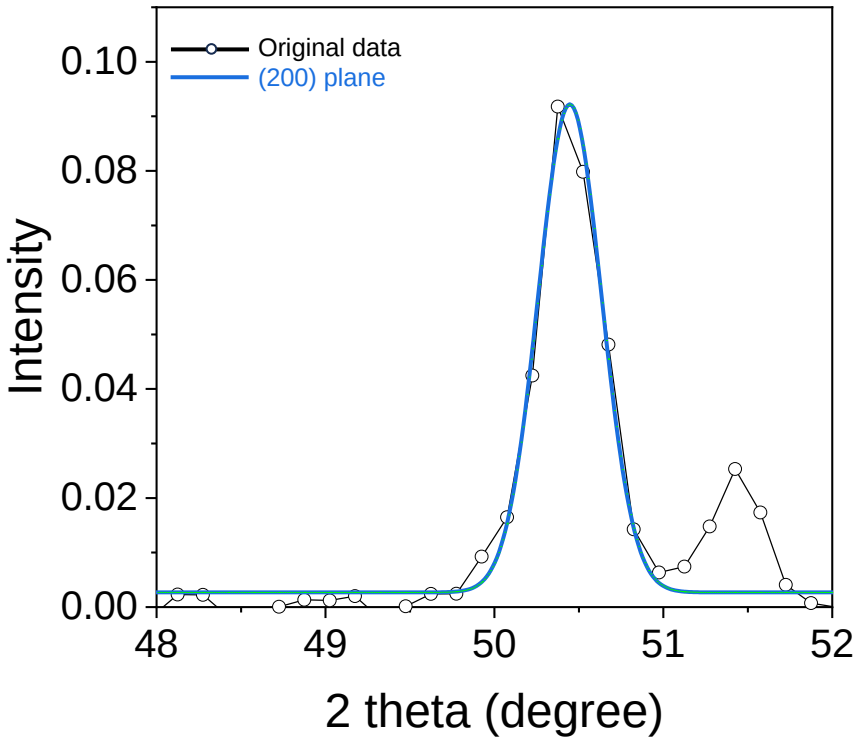
Position 2



FWHM:0.33892 ± 0.00904
Height: 1

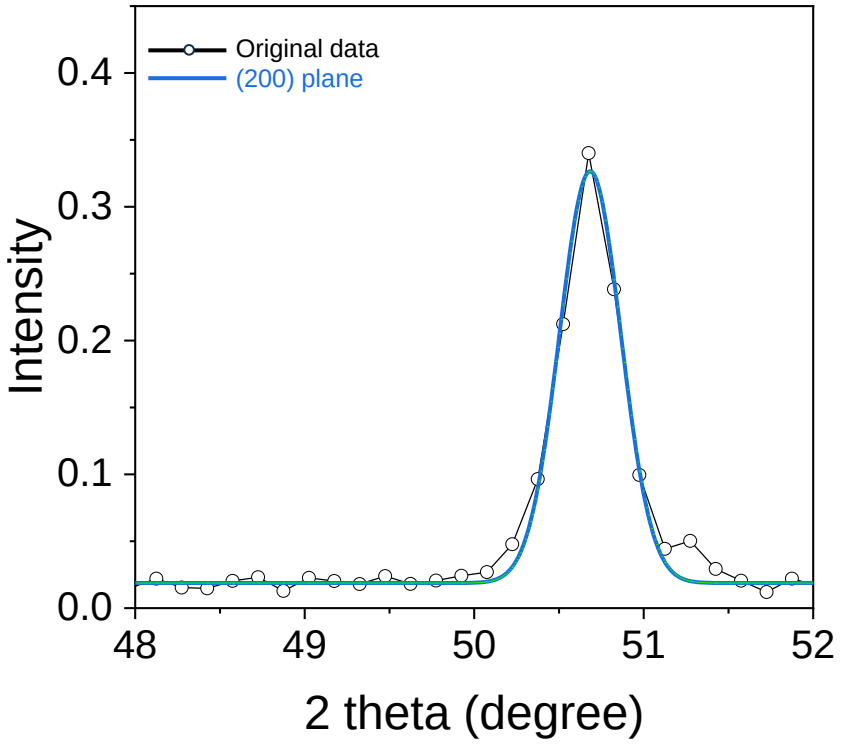
Peak Fitting Results (200) plane _S4

Position 1



FWHM: 0.34466 ± 0.06001
Height: 0.02265

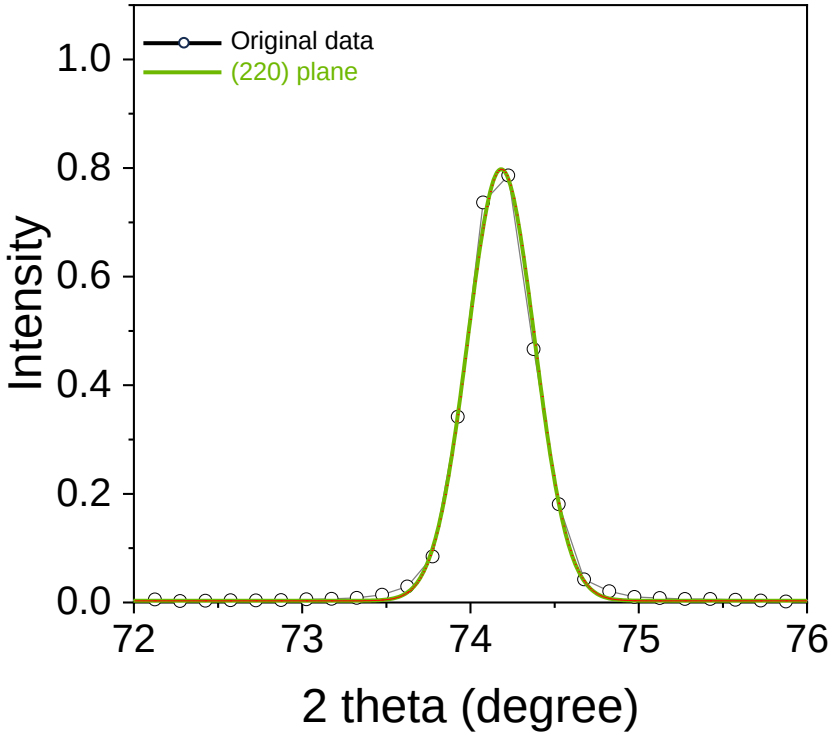
Position 2



FWHM: 0.4249 ± 0.00932
Height: 0.3914

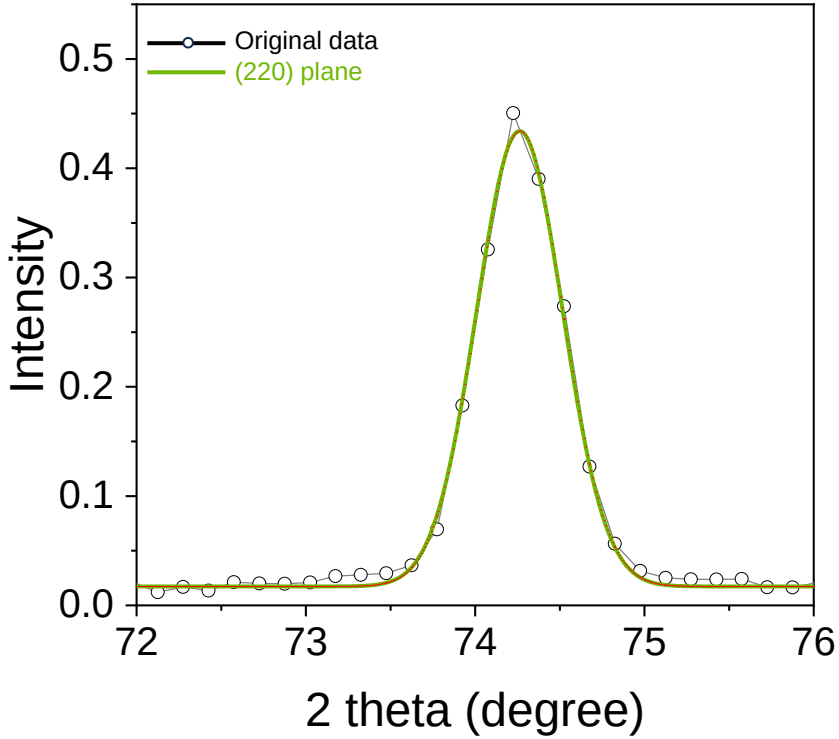
Peak Fitting Results (220) plane_S4

Position 1



FWHM: 0.4672 ± 0.00457
Height: 0.79478

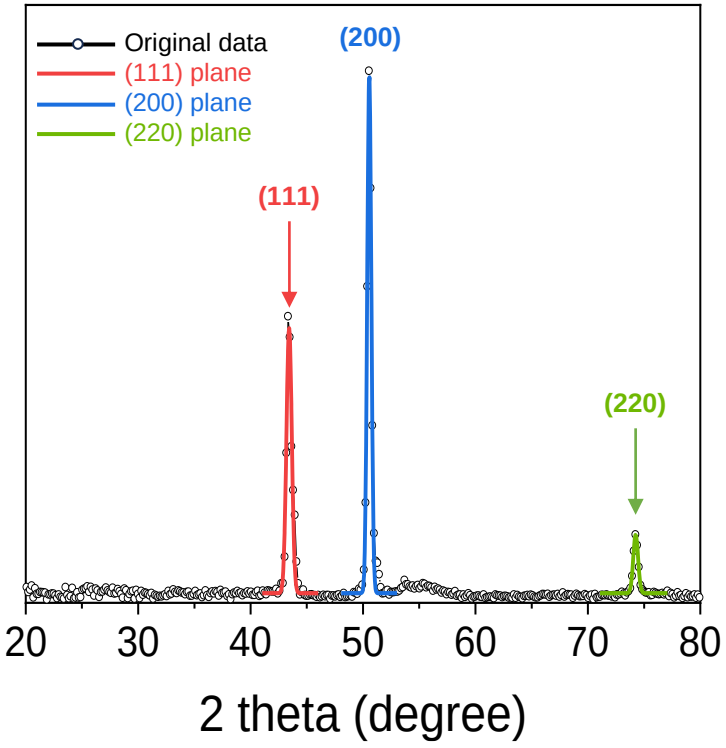
Position 2



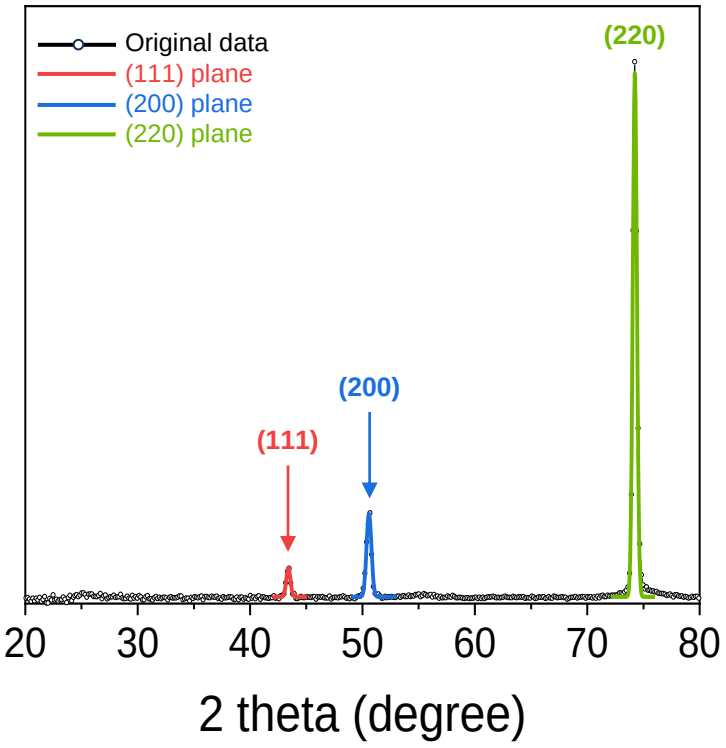
FWHM: 0.59733 ± 0.00729
Height: 0.41673

Original data XRD_S5

Position 1

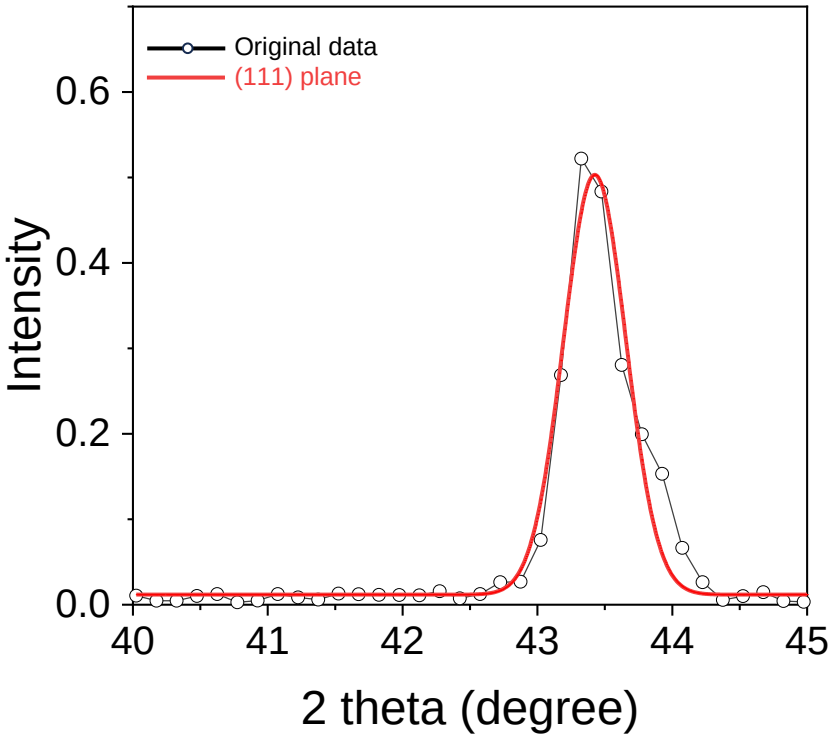


Position 2



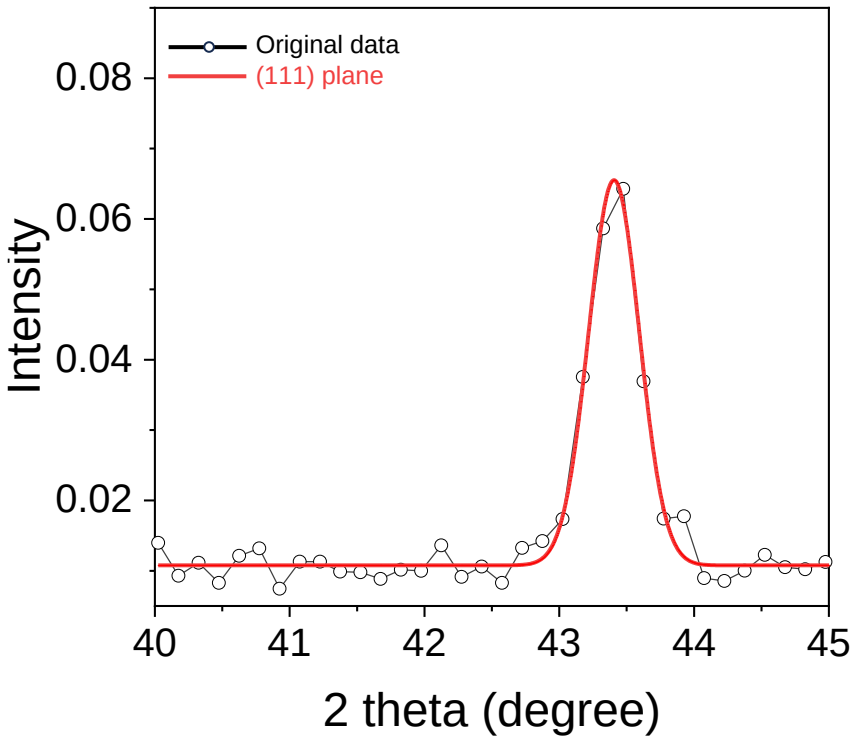
Peak Fitting Results (111) plane _S5

Position 1



FWHM: 0.54541 + 0.02609
Height: 0.49132

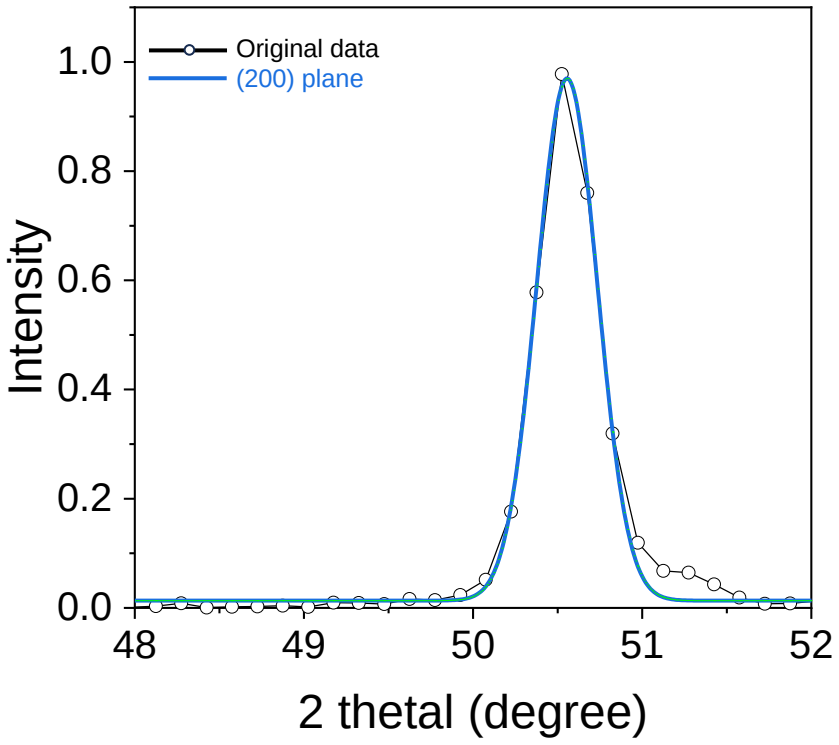
Position 2



FWHM:0.44222 + 0.01601
Height: 0.05474

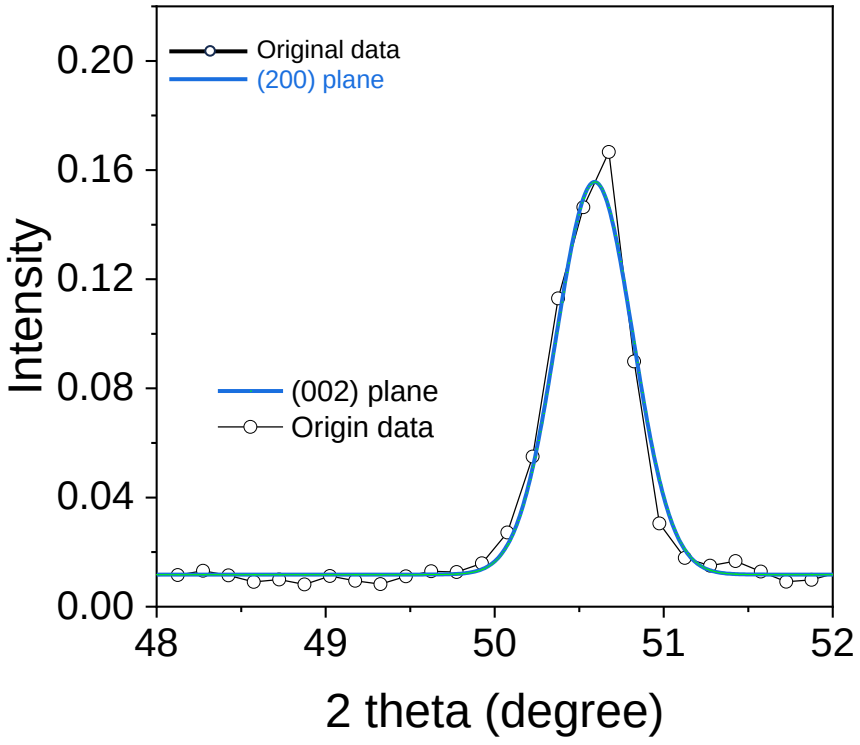
Peak Fitting Results (200) plane _S5

Position 1



FWHM: 0.42178 ± 0.00637
Height: 1

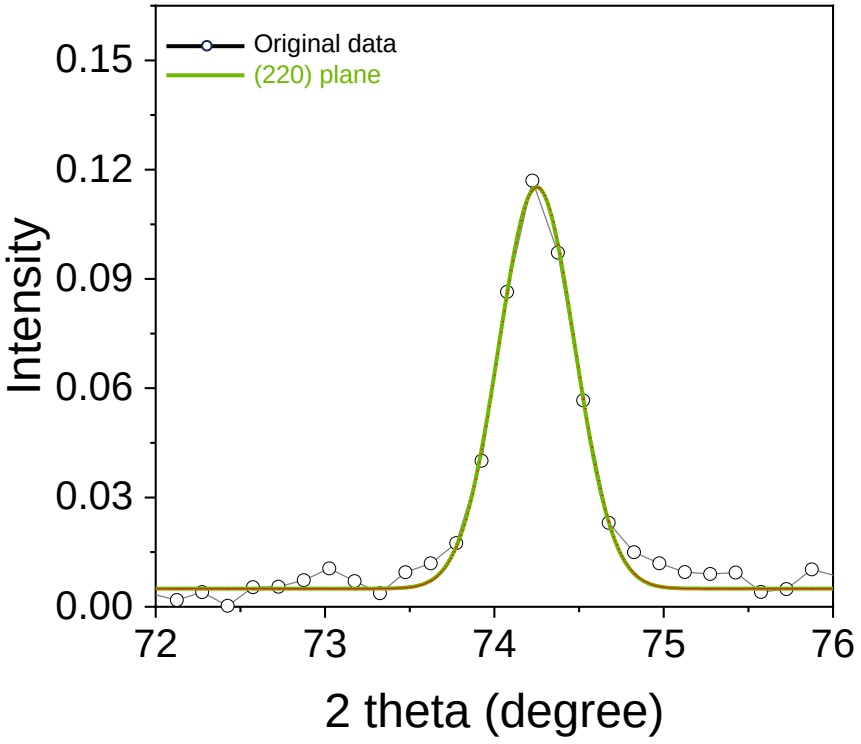
Position 2



FWHM: 0.53468 ± 0.01367
Height: 0.14391

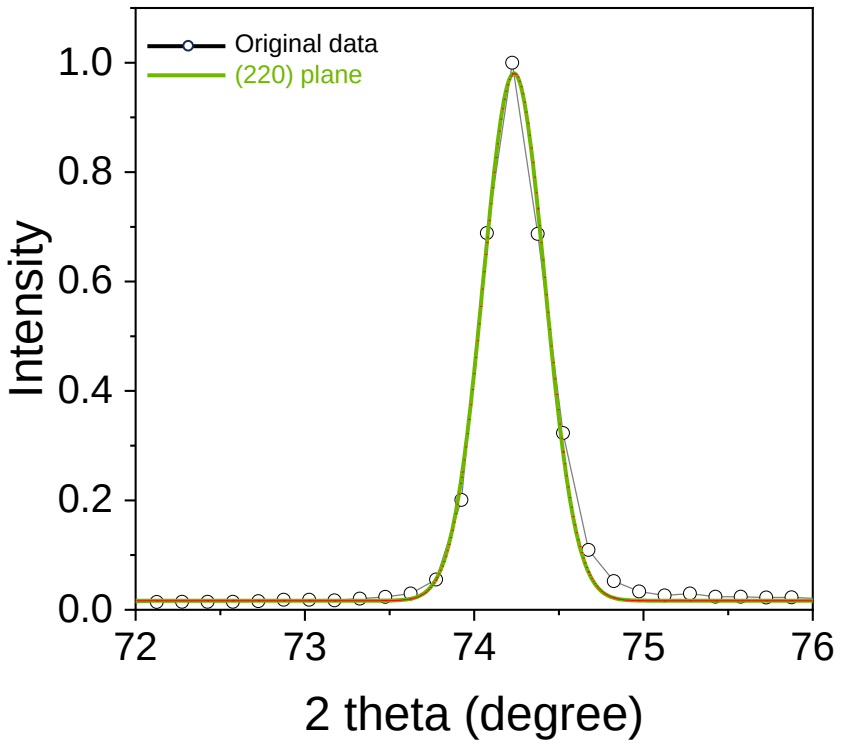
Peak Fitting Results (220) plane_S5

Position 1



FWHM: 0.532726 ± 0.01465
Height: 0.11022

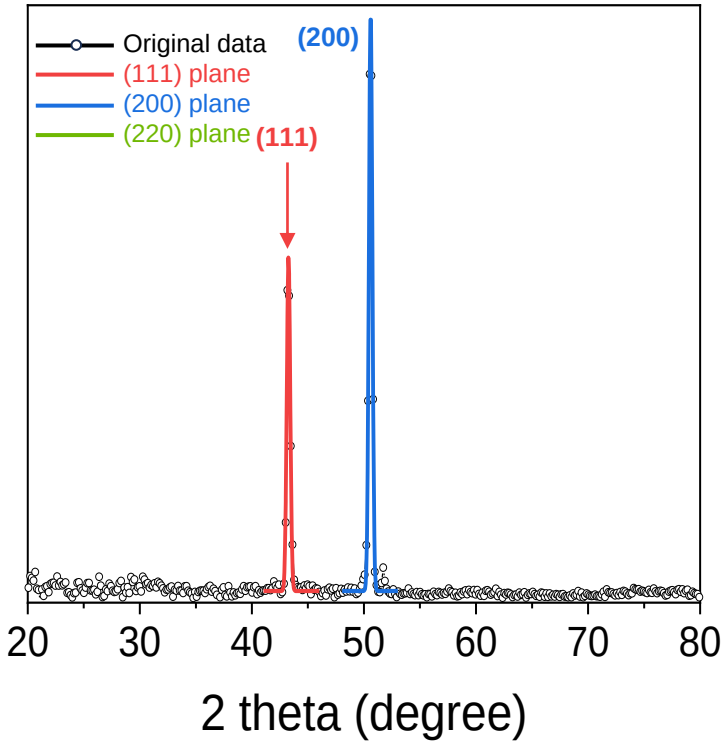
Position 2



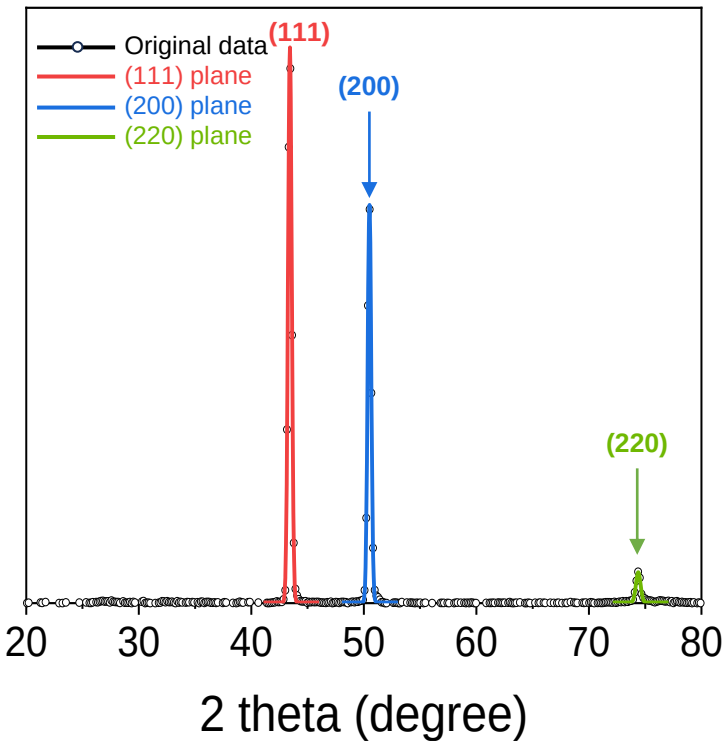
FWHM: 0.42854 ± 0.00555
Height: 1

Original data XRD_S6

Position 1

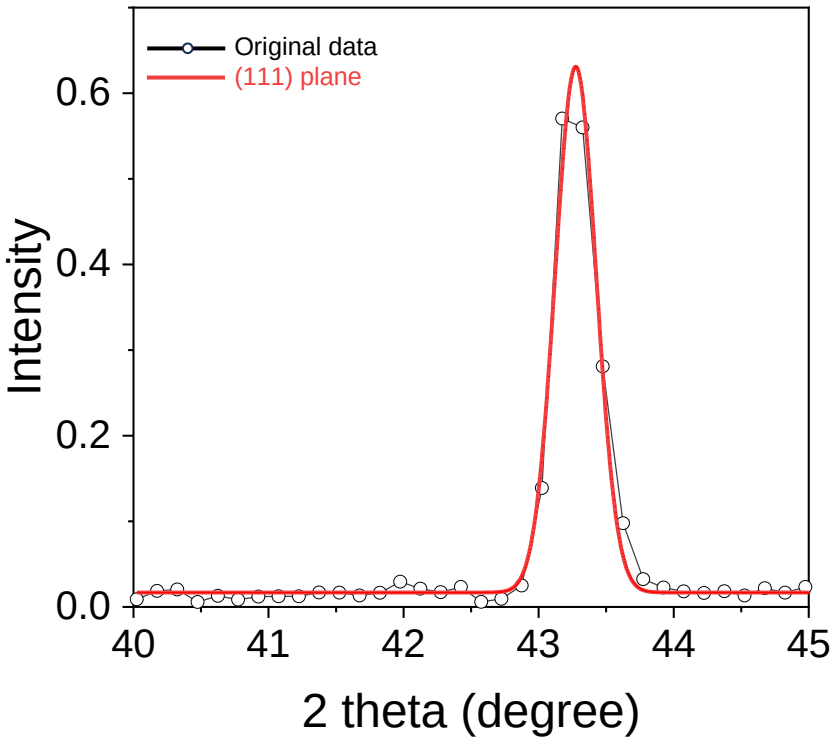


Position 2



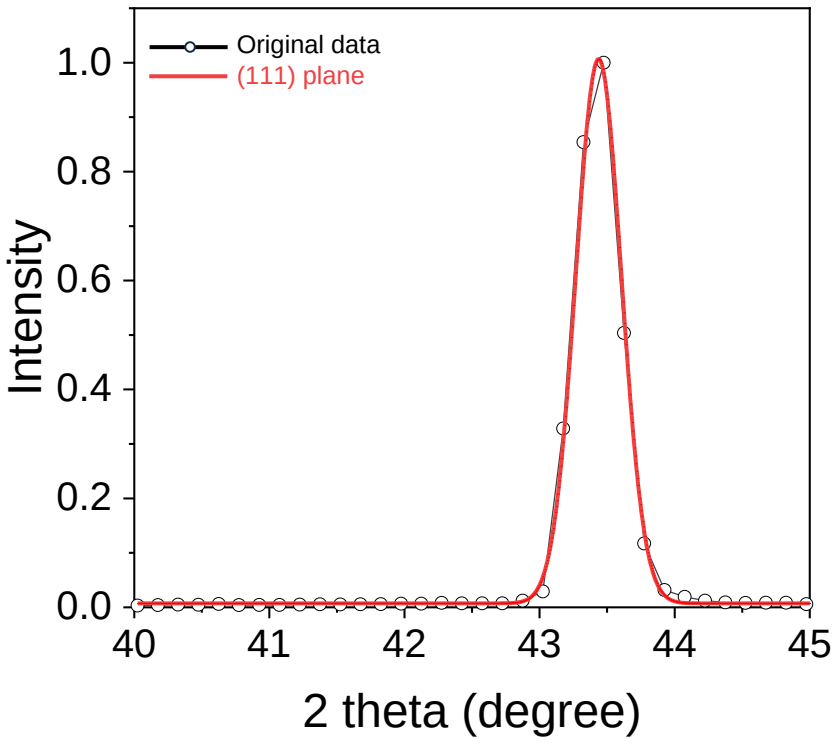
Peak Fitting Results (111) plane _S6

Position 1



FWHM: 0.35712 ± 0.01015
Height: 0.6141

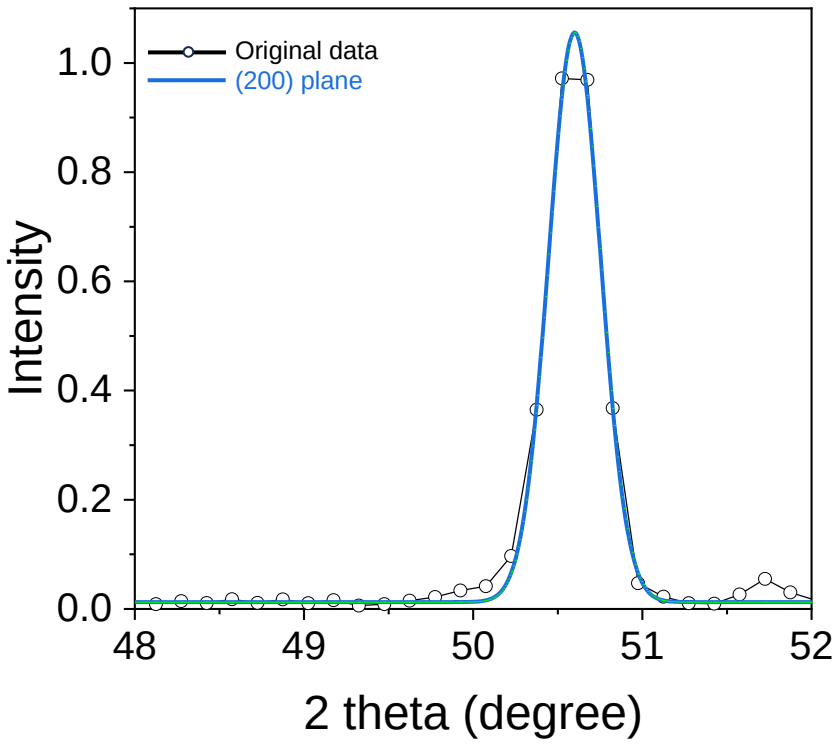
Position 2



FWHM: 0.39527 ± 0.0055
Height: 1

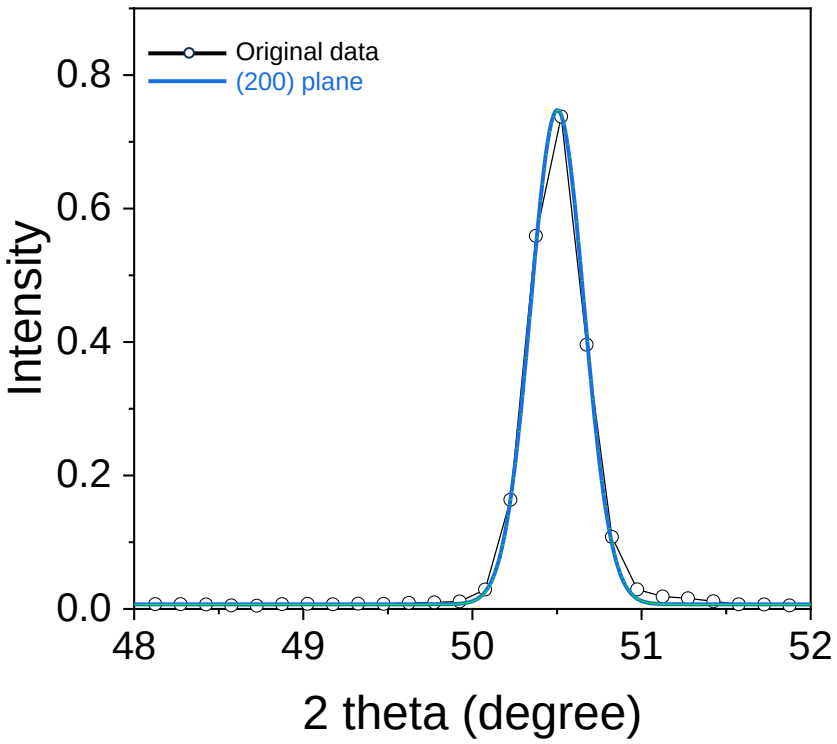
Peak Fitting Results (200) plane _S6

Position 1



FWHM: 0.36194 ± 0.00389
Height: 1

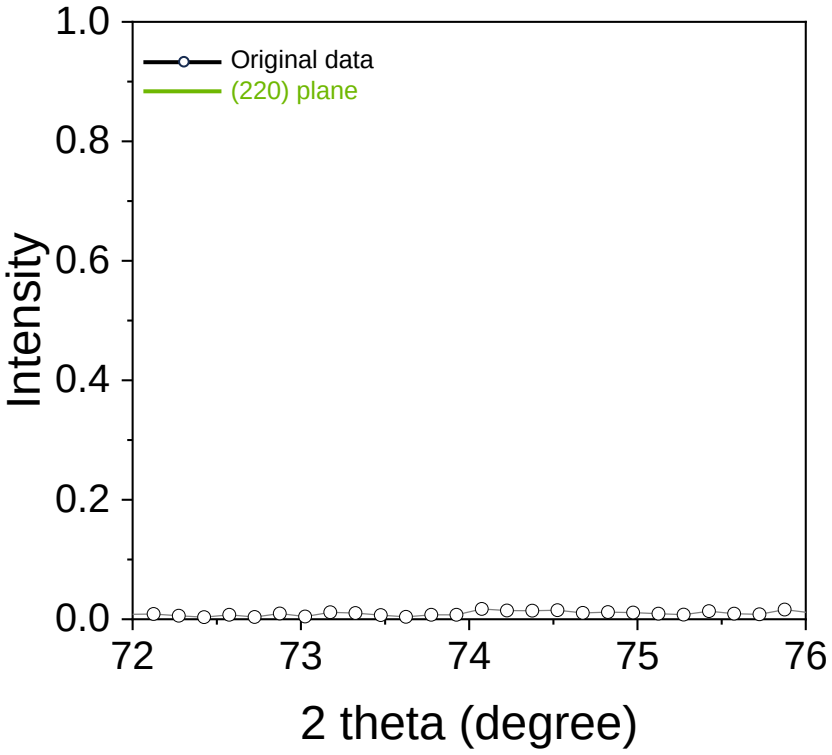
Position 2



FWHM: 0.37075 ± 0.0023
Height: 0.74075

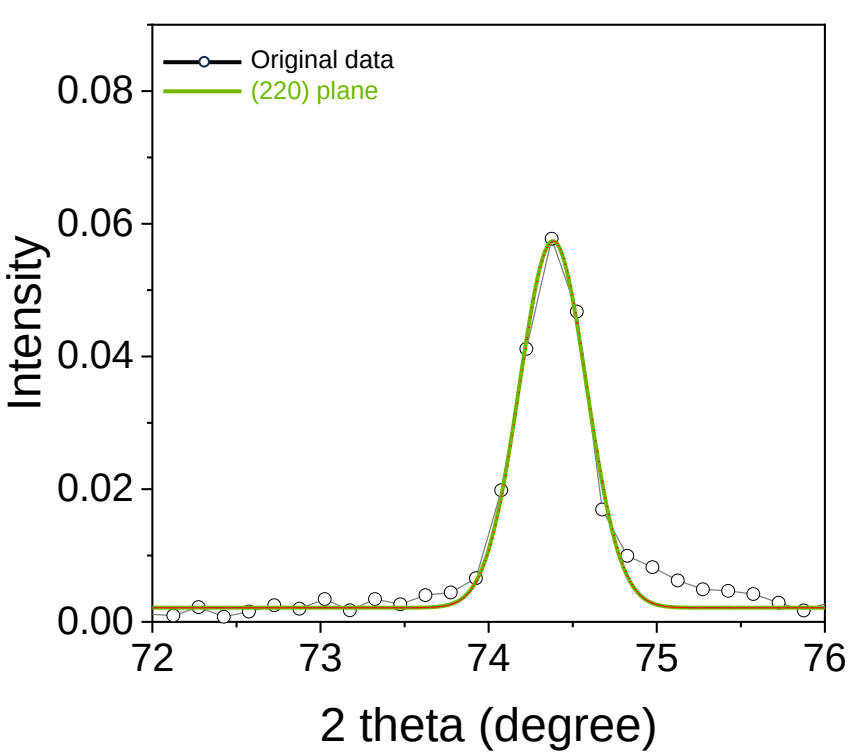
Peak Fitting Results (220) plane_S6

Position 1



No peak found

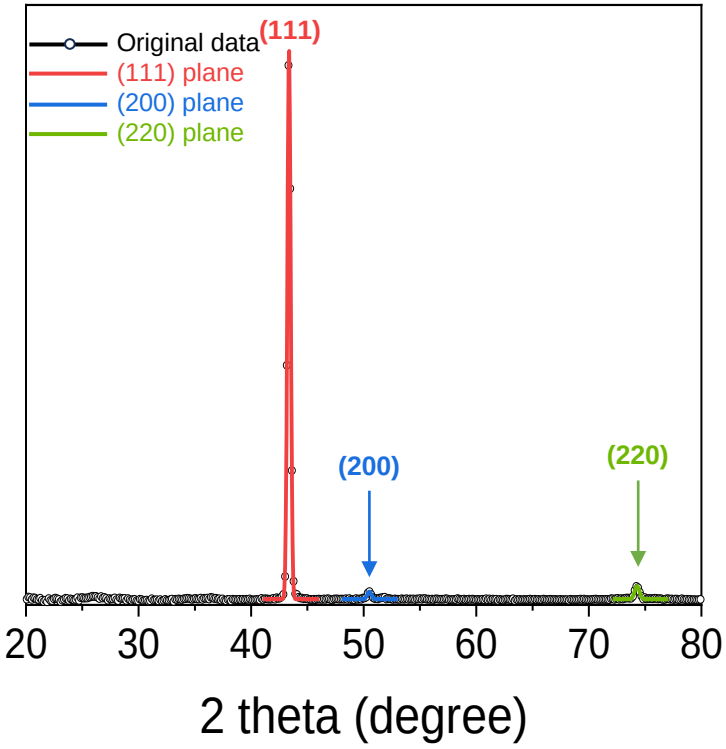
Position 2



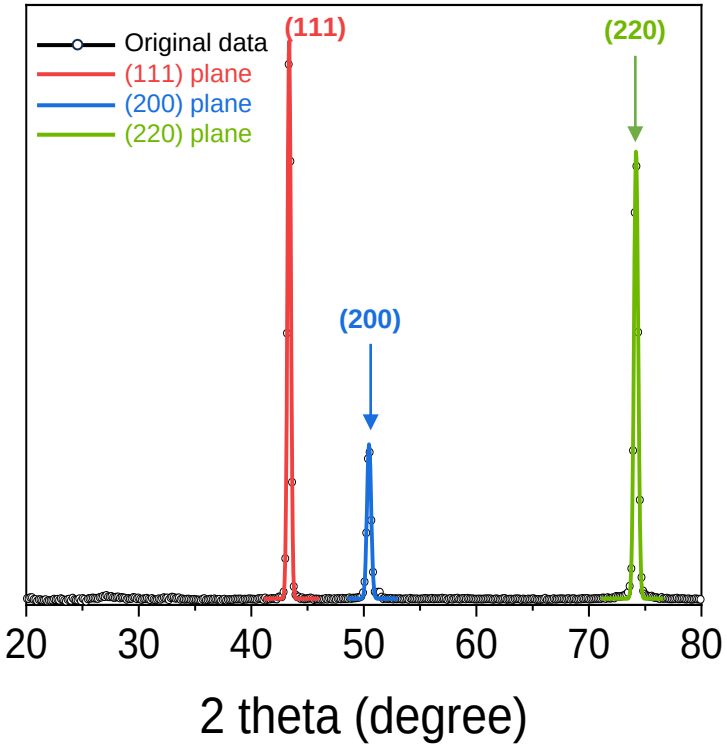
FWHM: 0.46961 ± 0.01426
Height: 0.05526

Original data XRD_S7

Position 1

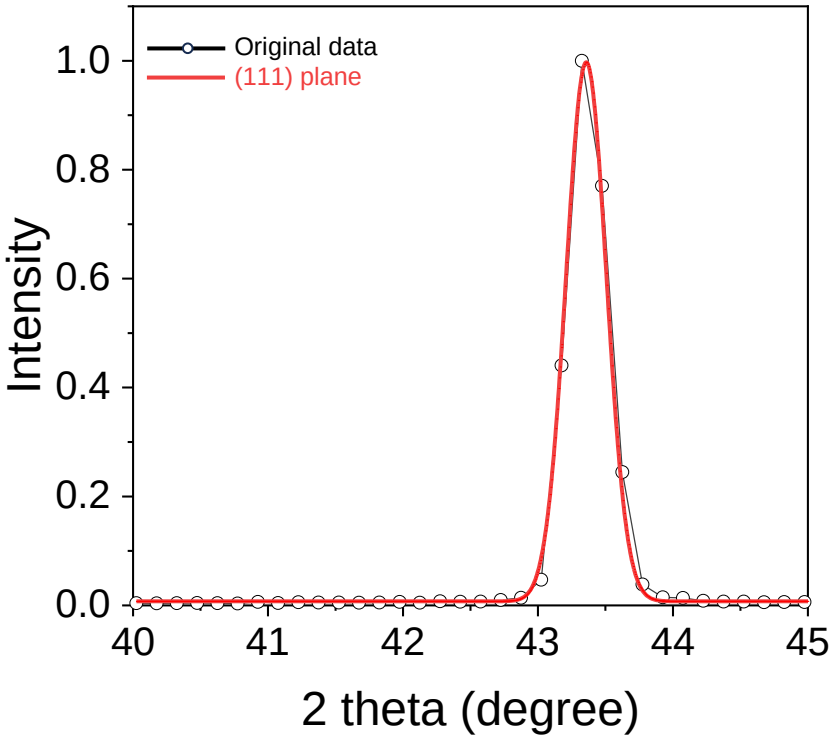


Position 2



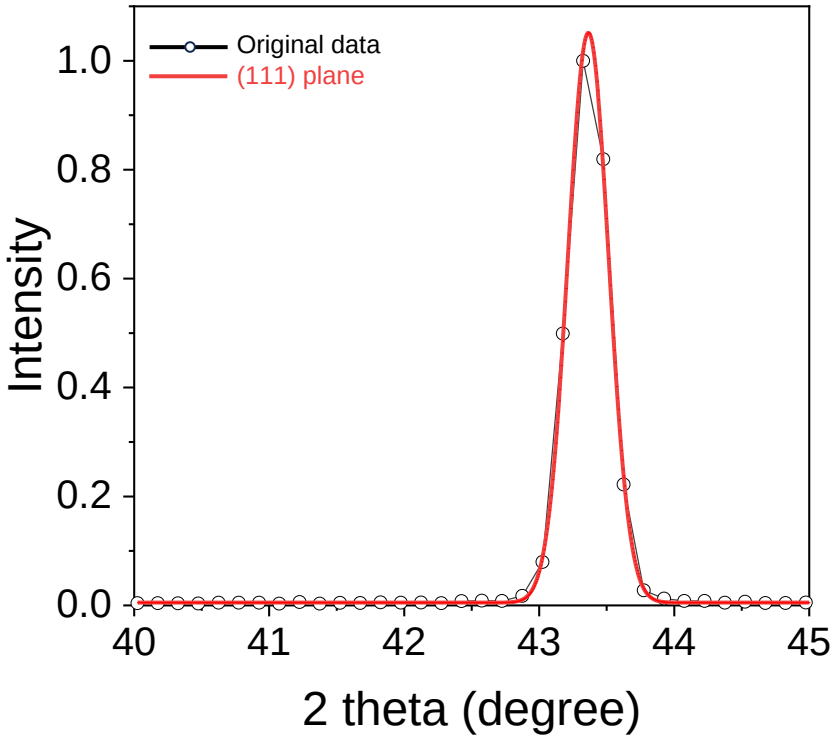
Peak Fitting Results (111) plane _S7

Position 1



FWHM: 0.35064 ± 0.00558
Height: 1

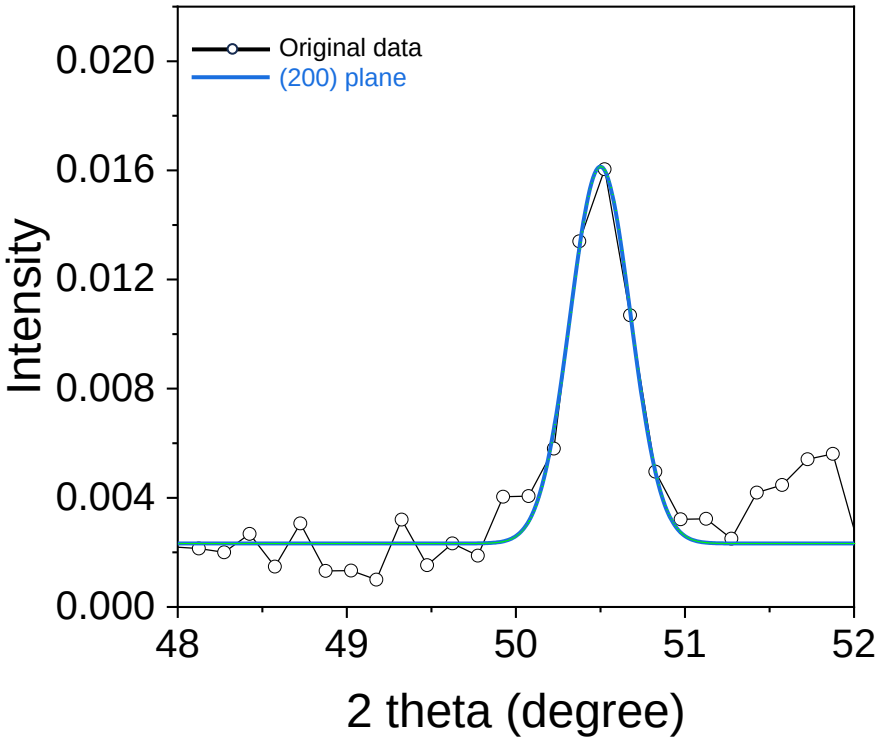
Position 2



FWHM:0.35534 ± 0.00229
Height: 1

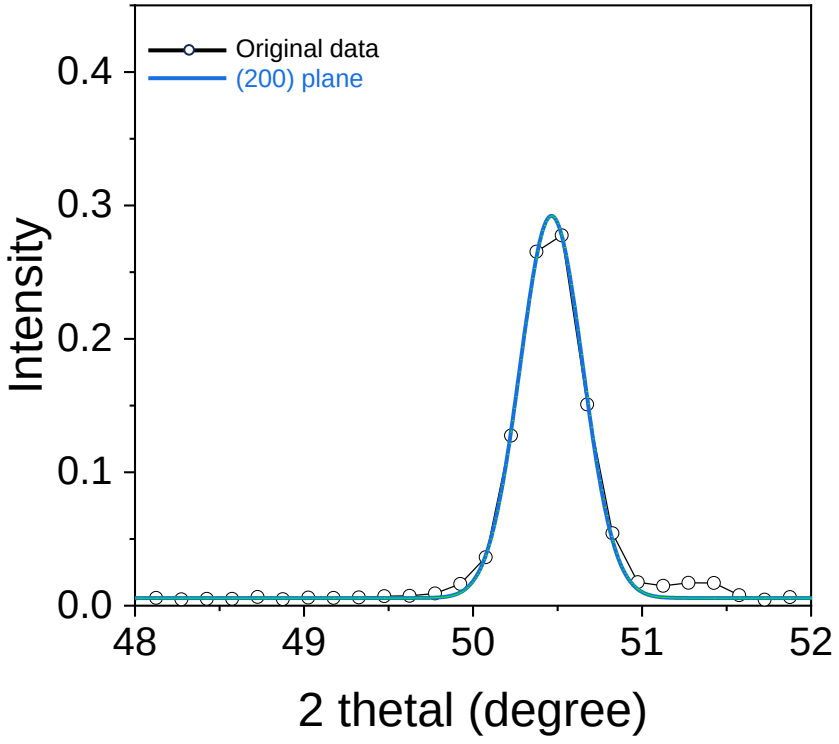
Peak Fitting Results (200) plane _S7

Position 1



FWHM: 0.405 ± 0.00397
Height: 0.01606

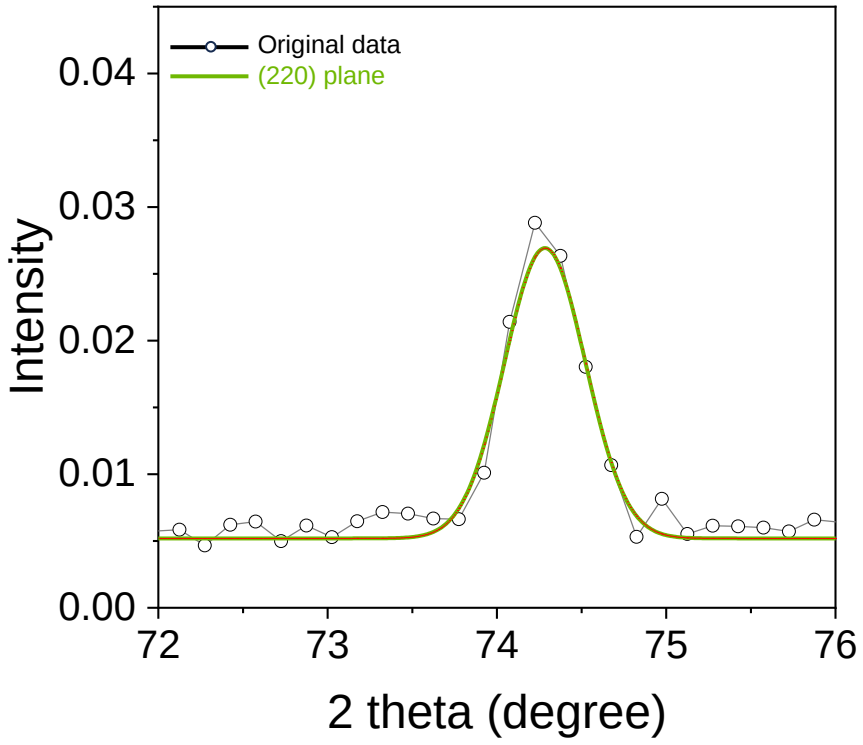
Position 2



FWHM: 0.43748 ± 0.0045
Height: 0.28651

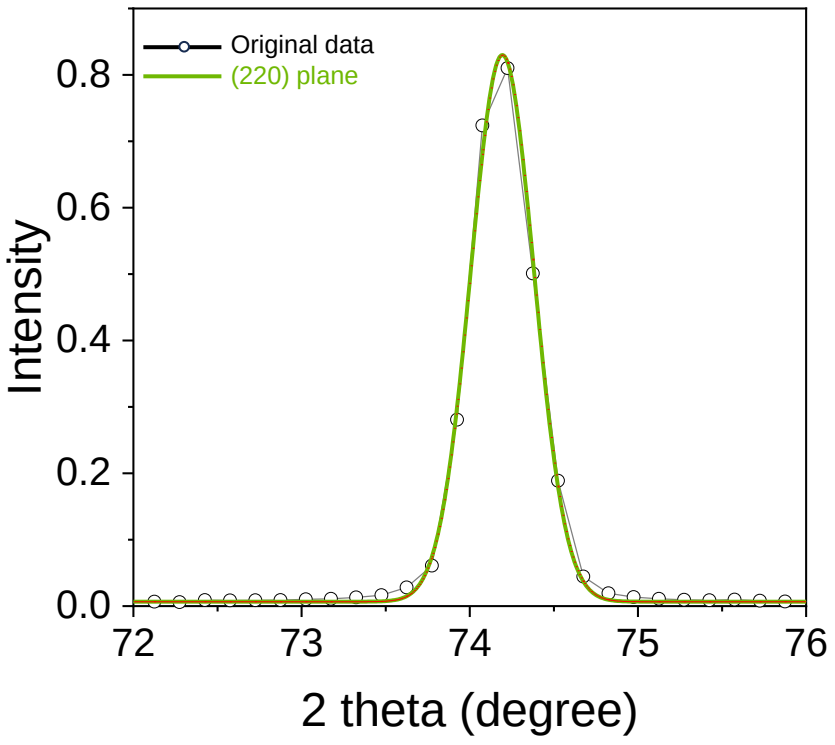
Peak Fitting Results (220) plane_S7

Position 1



FWHM: 0.5625 ± 0.02391
Height: 0.02714

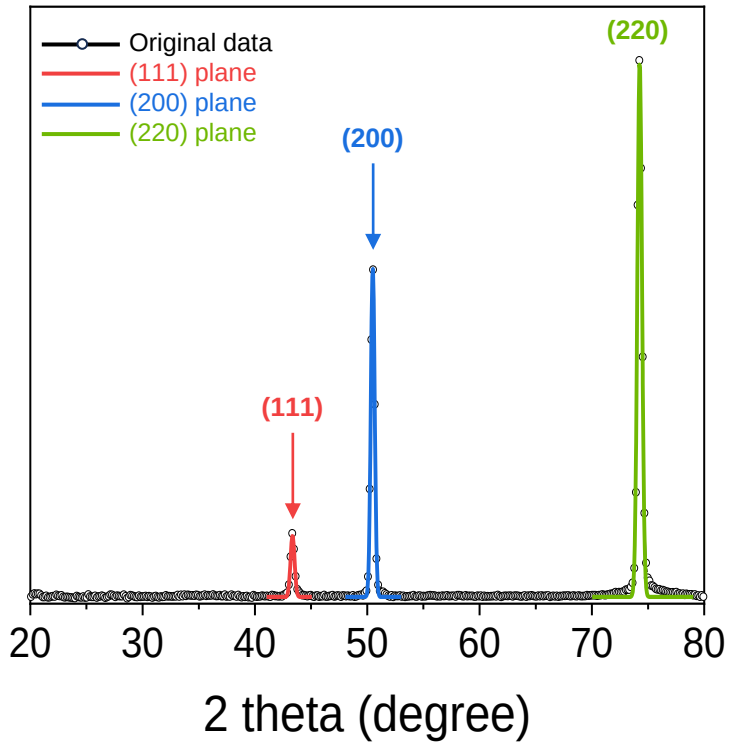
Position 2



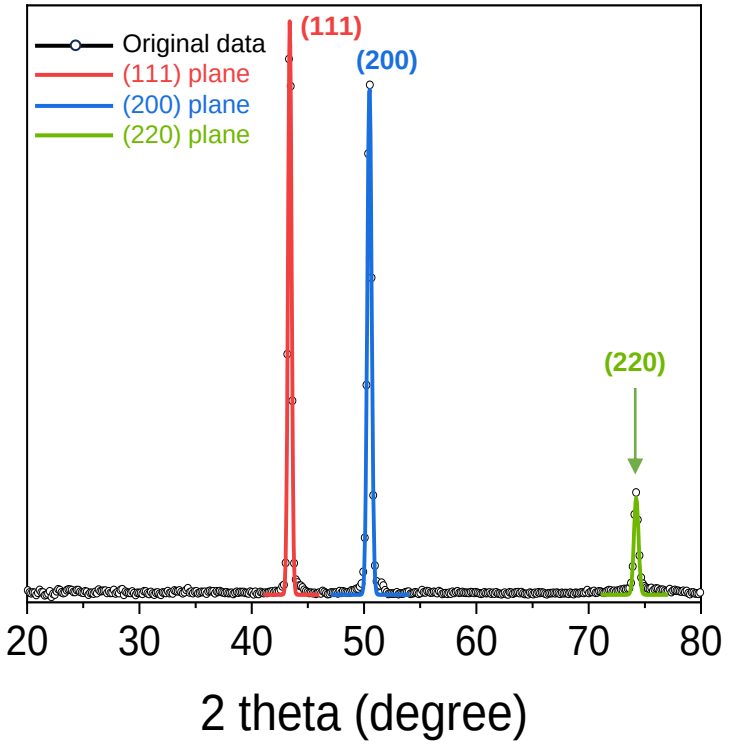
FWHM: 0.44139 ± 0.00406
Height: 0.82237

Original data XRD_ S8

Position 1

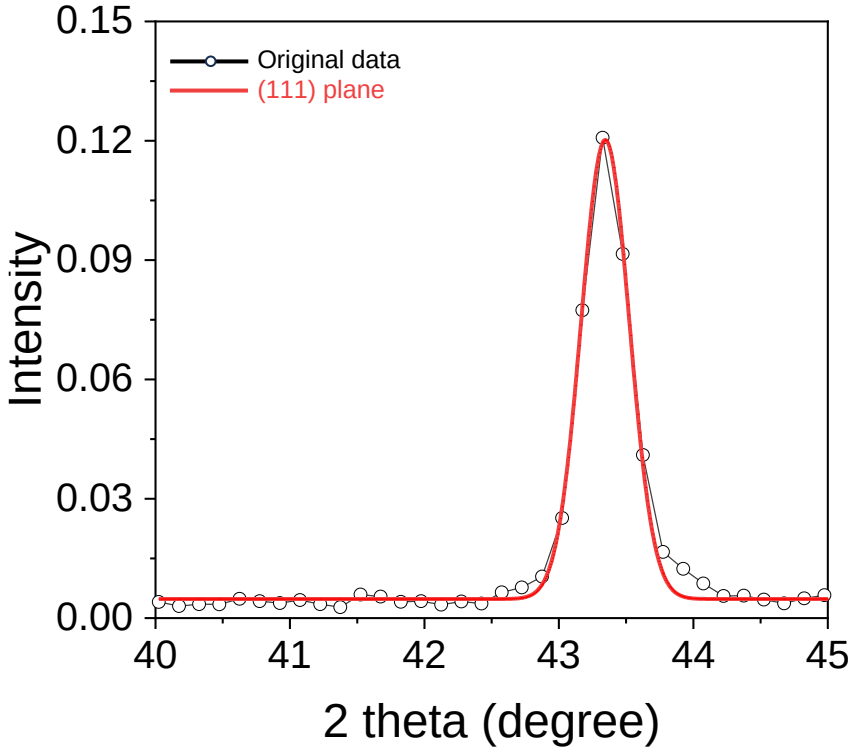


Position 2



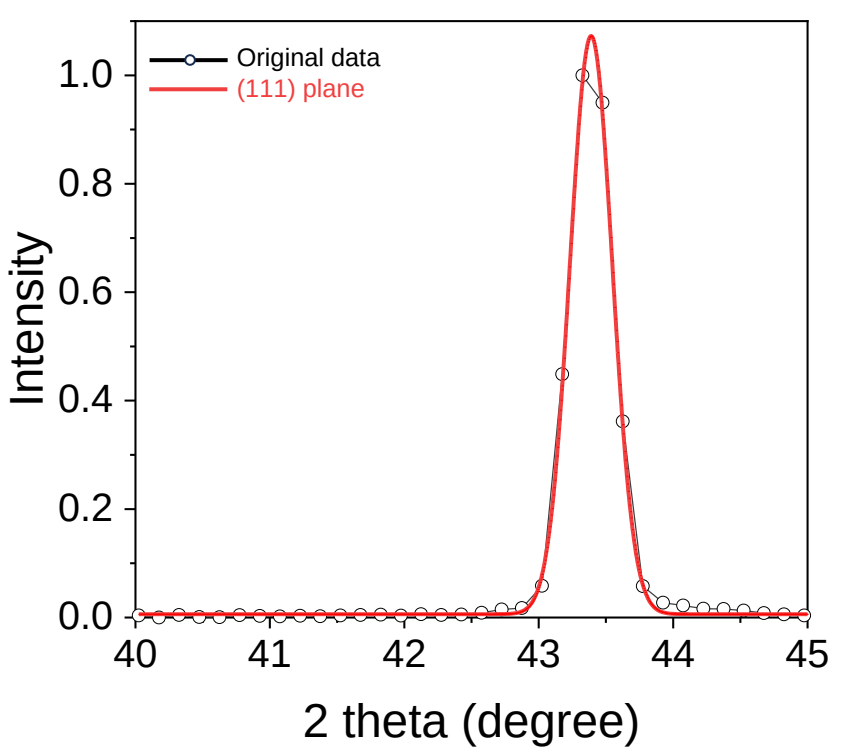
Peak Fitting Results (111) plane _S8

Position 1



FWHM: 0.42096 ± 0.00737
Height: 0.12087

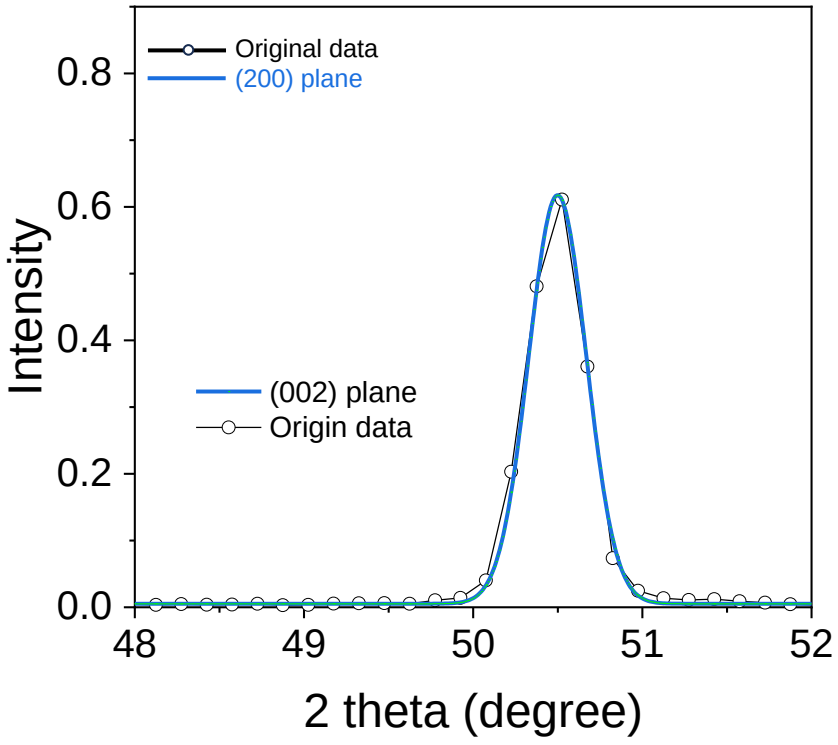
Position 2



FWHM: 0.37092 ± 0.00278
Height: 1

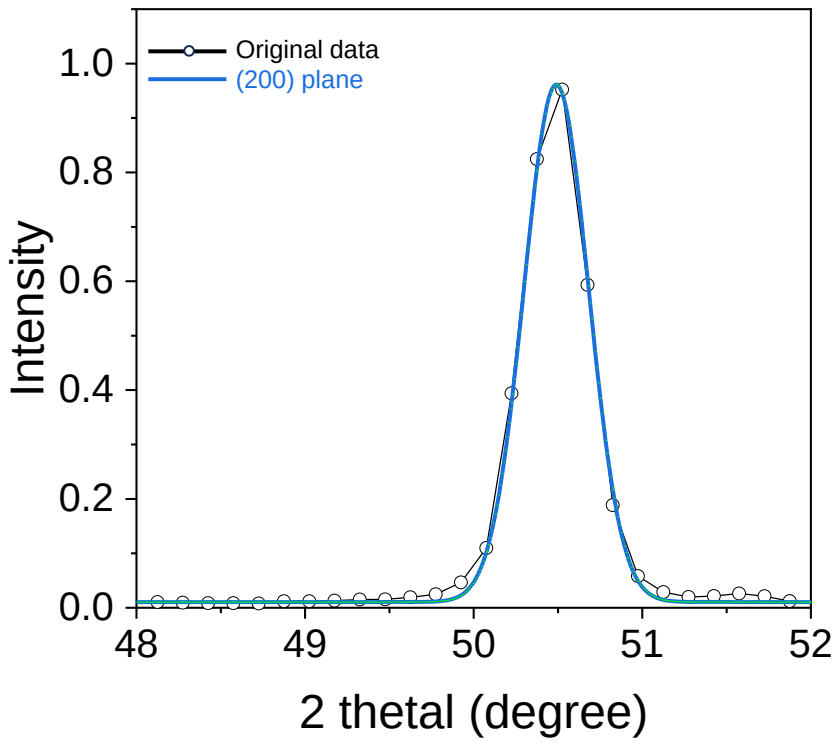
Peak Fitting Results (200) plane _S8

Position 1



FWHM: 0.405 ± 0.00397
Height: 0.614

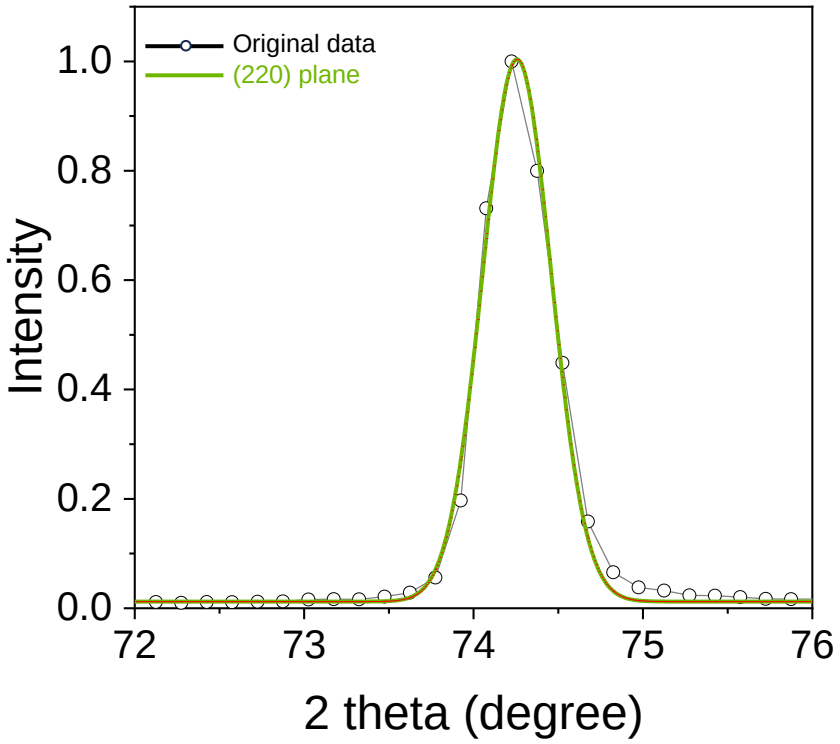
Position 2



FWHM: 0.45142 ± 0.00365
Height: 0.0.95061

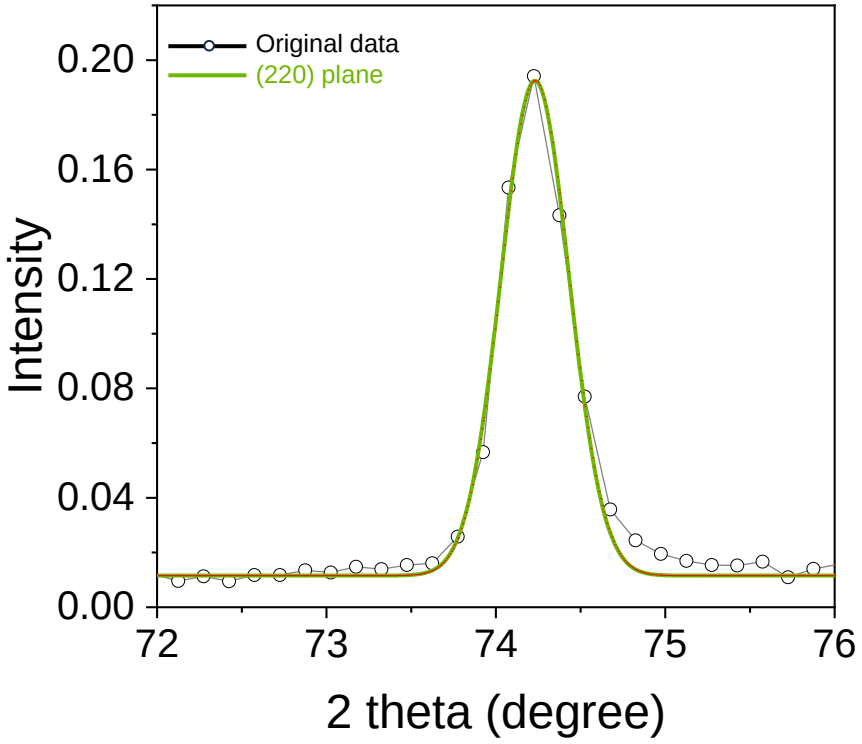
Peak Fitting Results (220) plane_S8

Position 1



FWHM: 0.447618 ± 0.00765
Height: 1

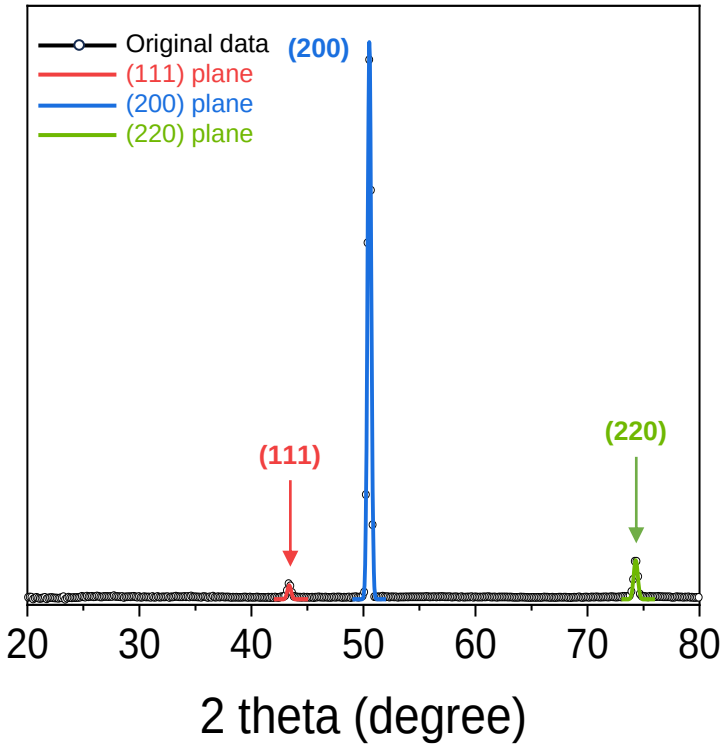
Position 2



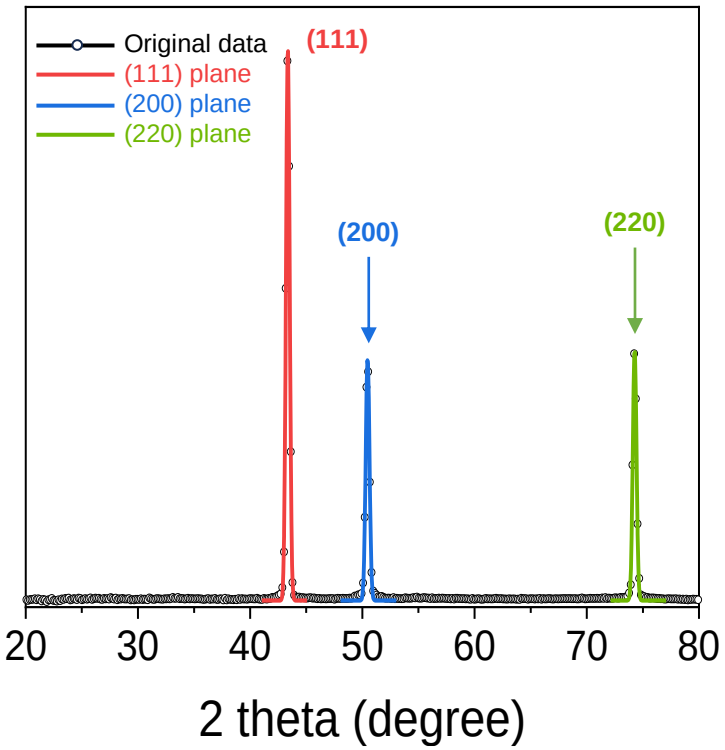
FWHM: 0.47263 ± 0.0099
Height: 0.1904

Original data XRD_ S9

Position 1

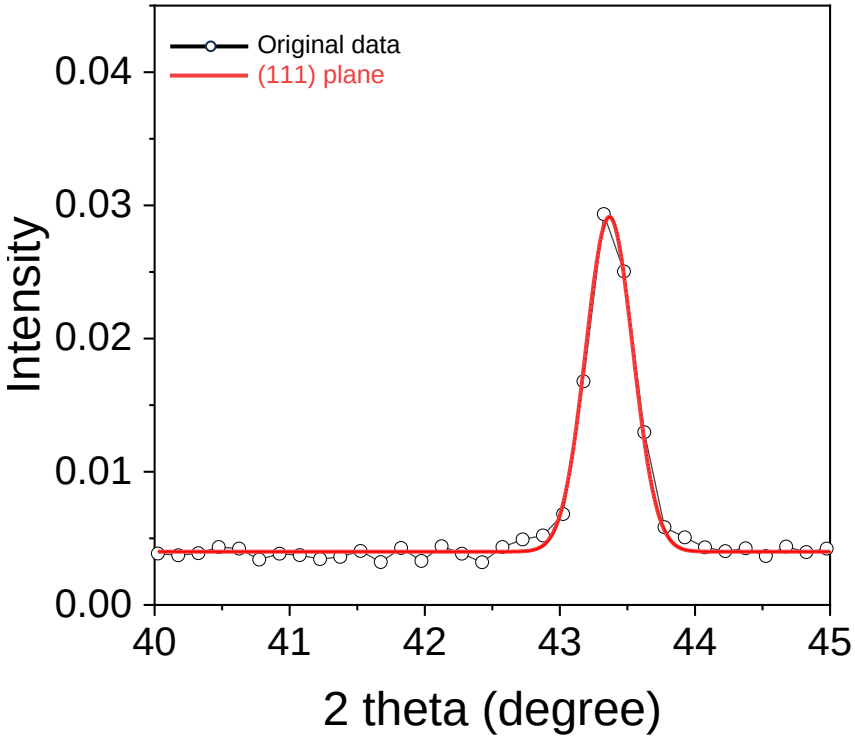


Position 2



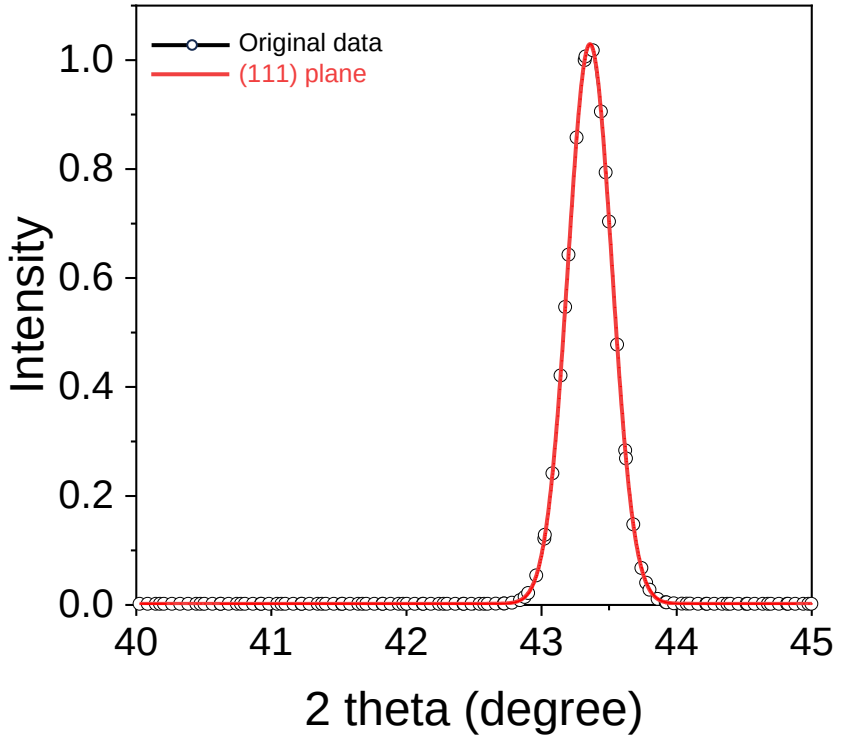
Peak Fitting Results (111) plane _S9

Position 1



FWHM: 0.40953 ± 0.00892
Height: 0.2926

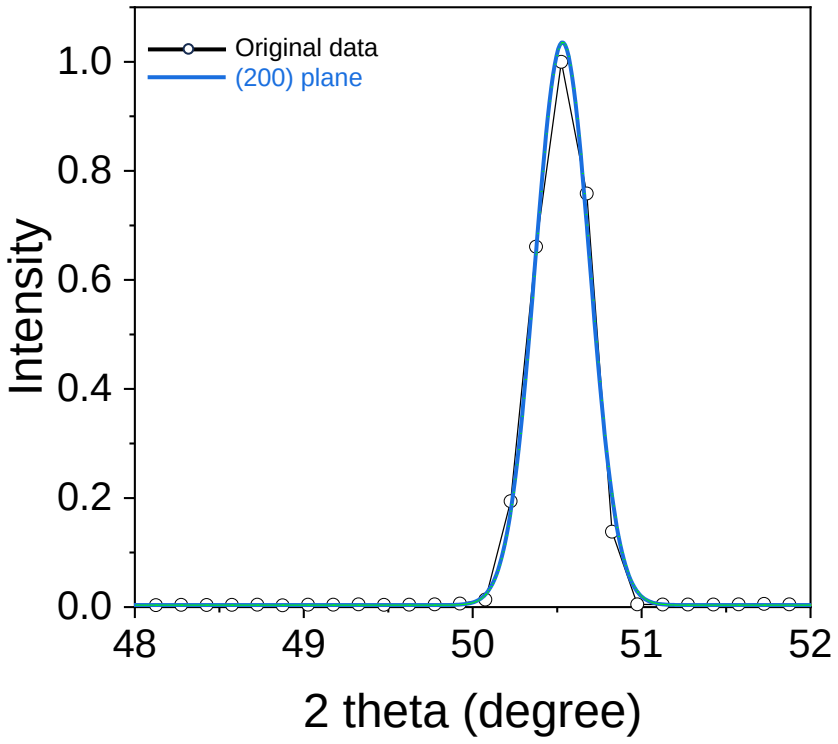
Position 2



FWHM:0.38243 ± 9.2696E-5
Height: 1

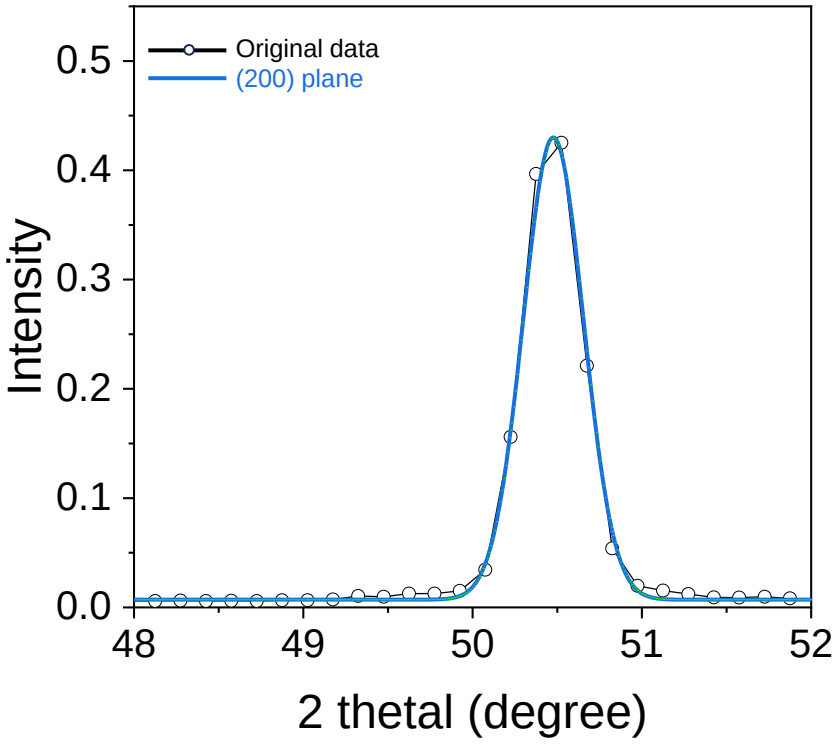
Peak Fitting Results (200) plane _S9

Position 1



FWHM: 0.38146 ± 0.00473
Height: 1

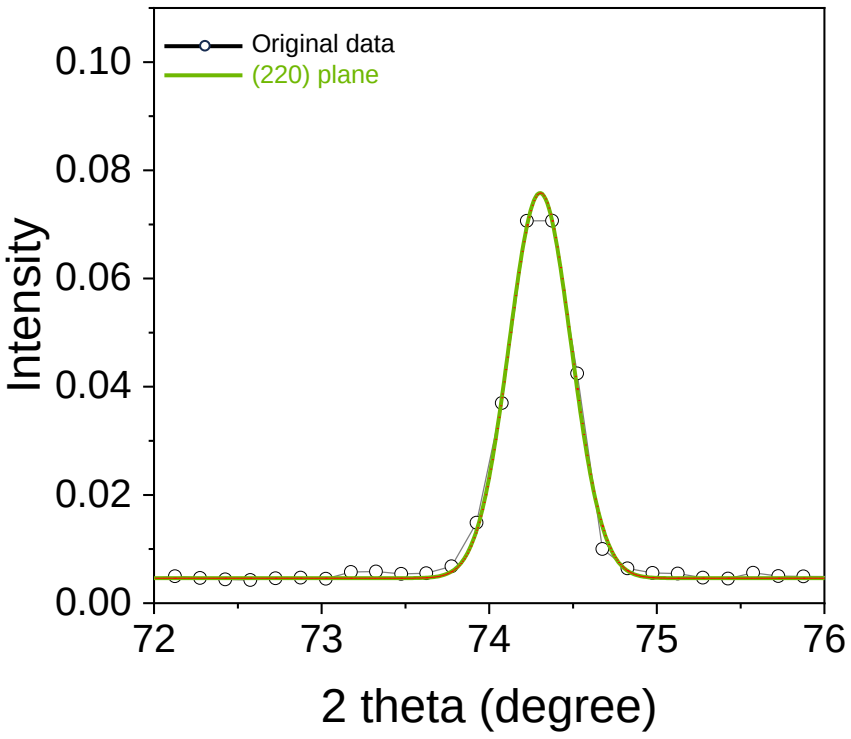
Position 2



FWHM: 0.41899 ± 0.00499
Height: 0.42871

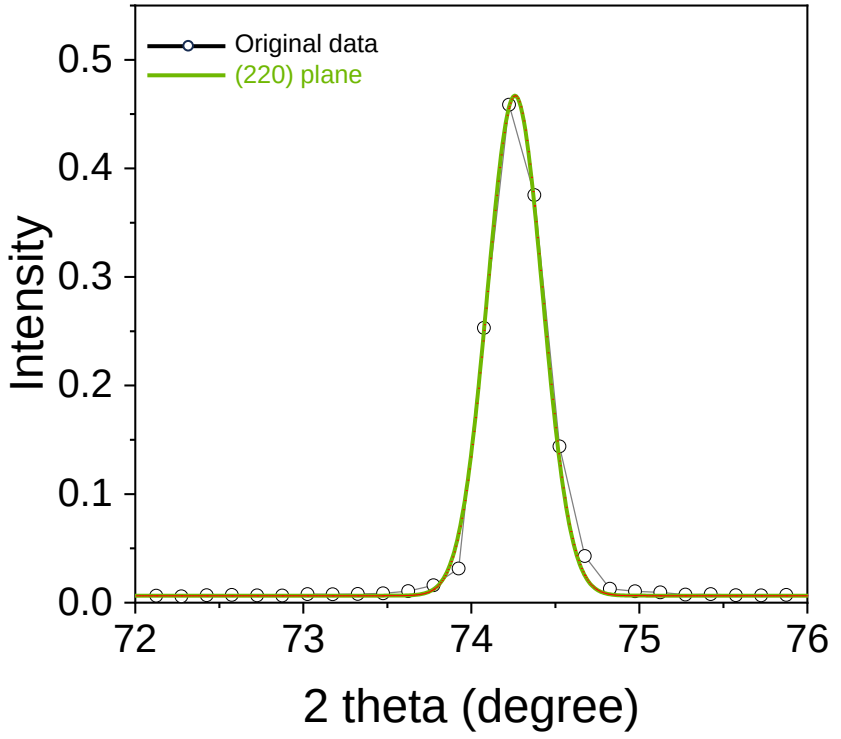
Peak Fitting Results (220) plane_S9

Position 1



FWHM: 0.43671 ± 0.0053
Height: 0.07119

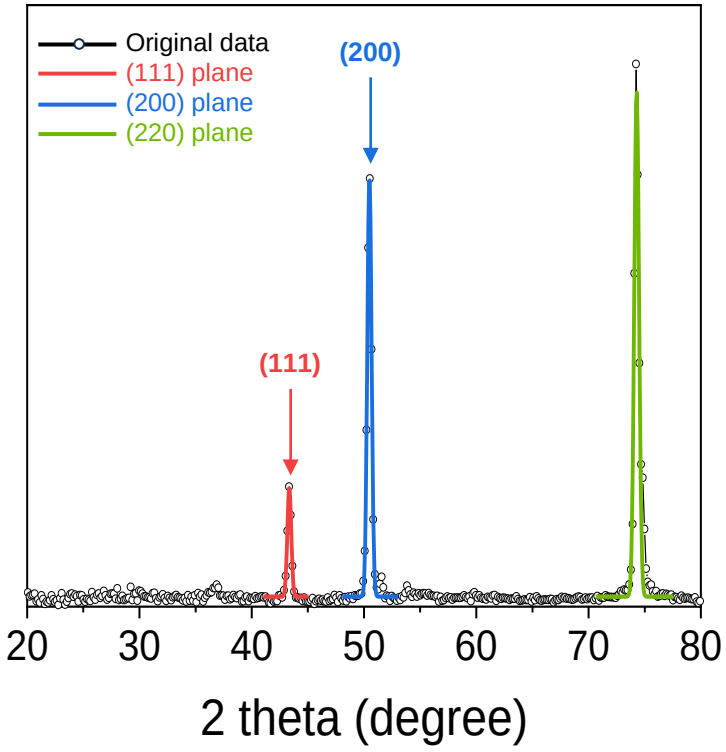
Position 2



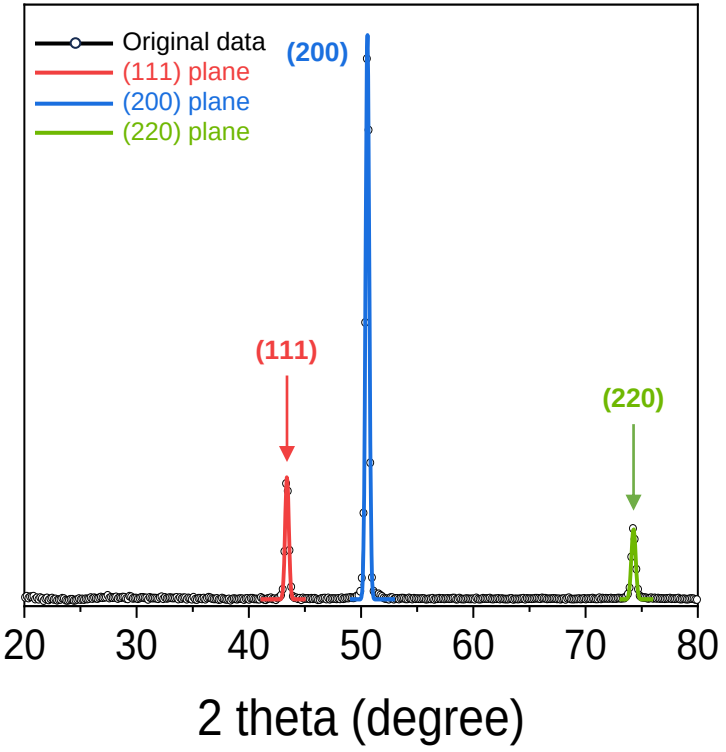
FWHM: 0.38595 ± 0.0049
Height: 0.47058

Original data XRD_S10

Position 1

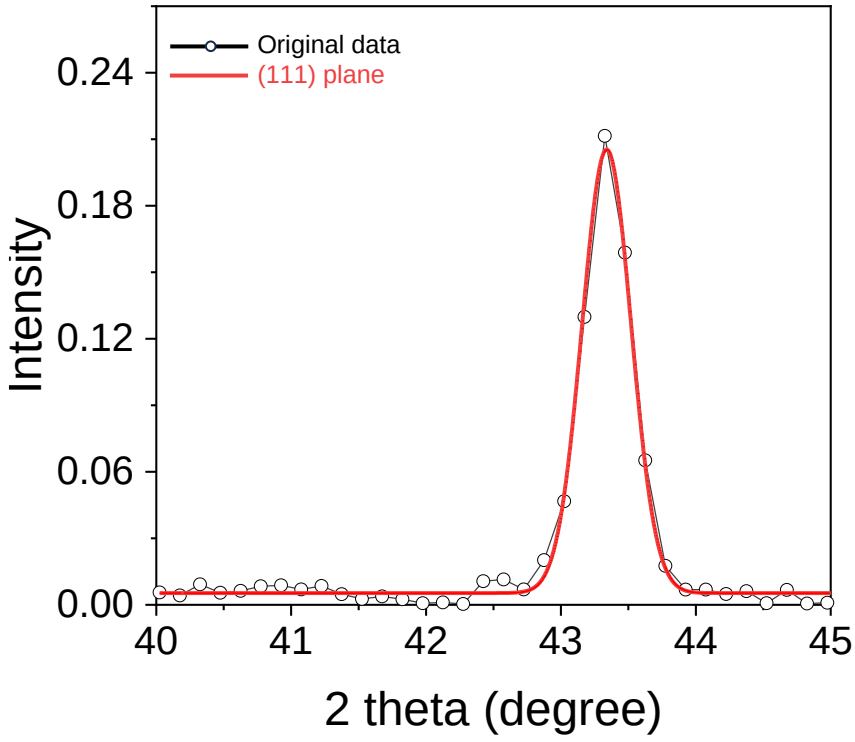


Position 2



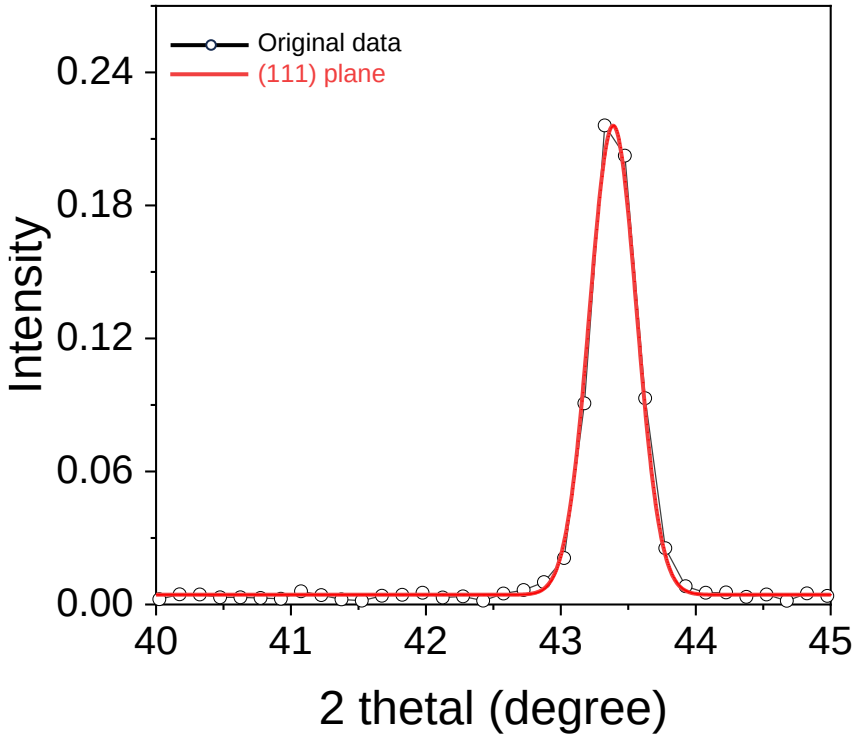
Peak Fitting Results (111) plane _S10

Position 1



FWHM: 0.42715 ± 0.00846
Height: 0.2102

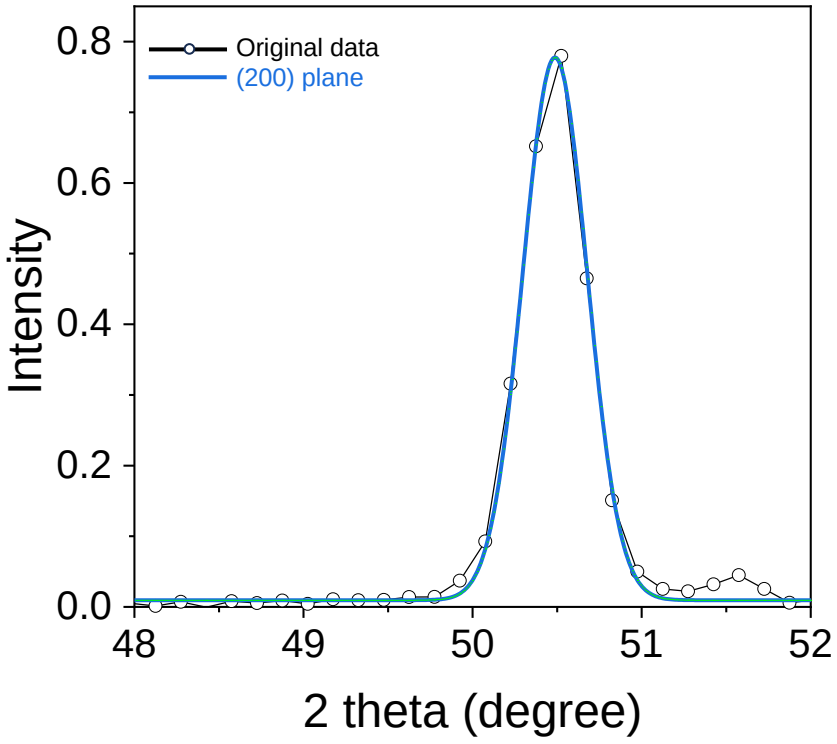
Position 2



FWHM: 0.40993 ± 0.00867
Height: 0.2162

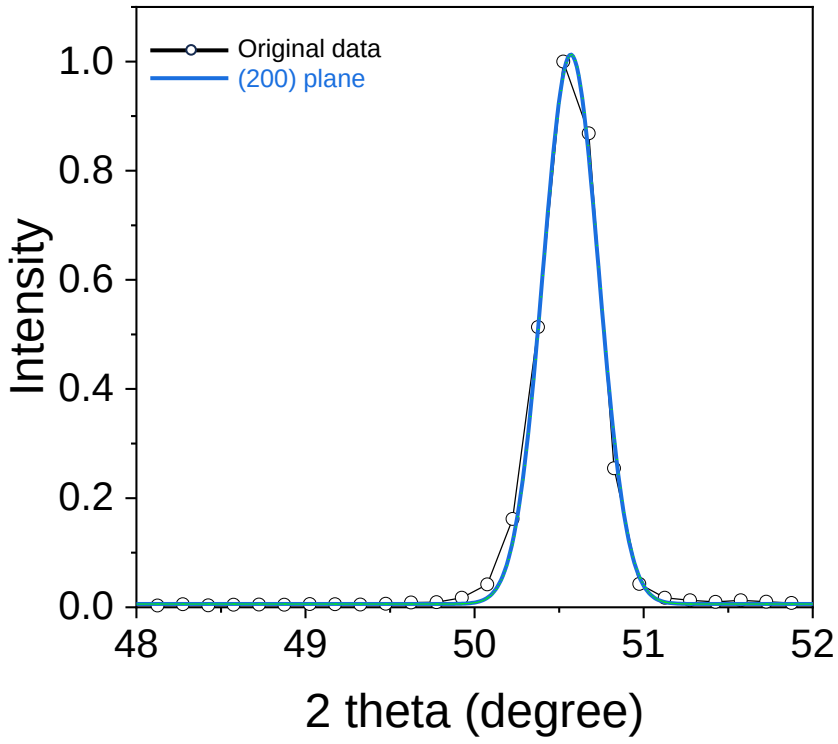
Peak Fitting Results (200) plane _S10

Position 1



FWHM: 0.44685 ± 0.00552
Height: 0.7769

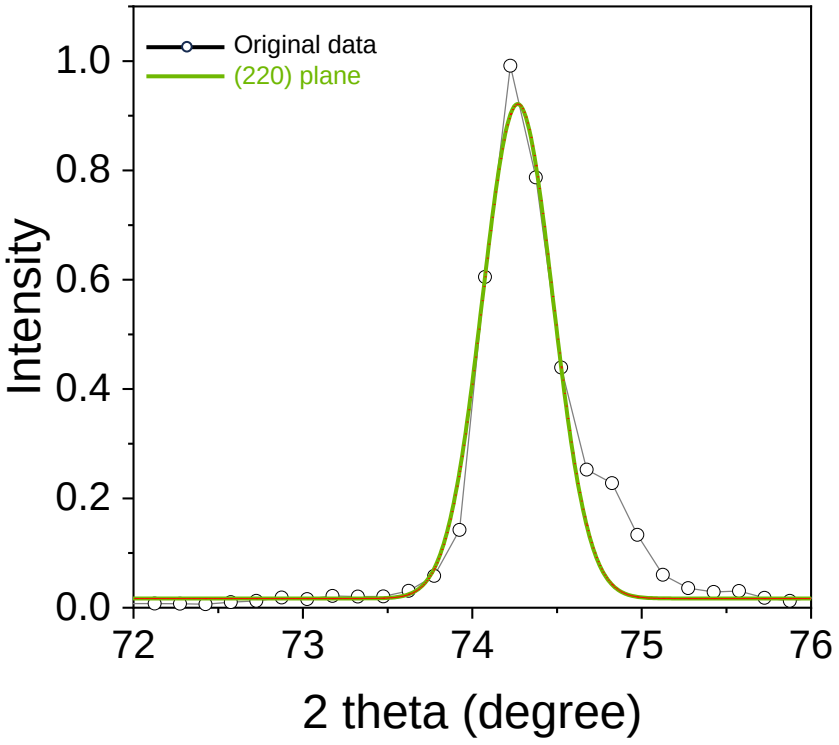
Position 2



FWHM: 0.38867 ± 0.00441
Height: 1

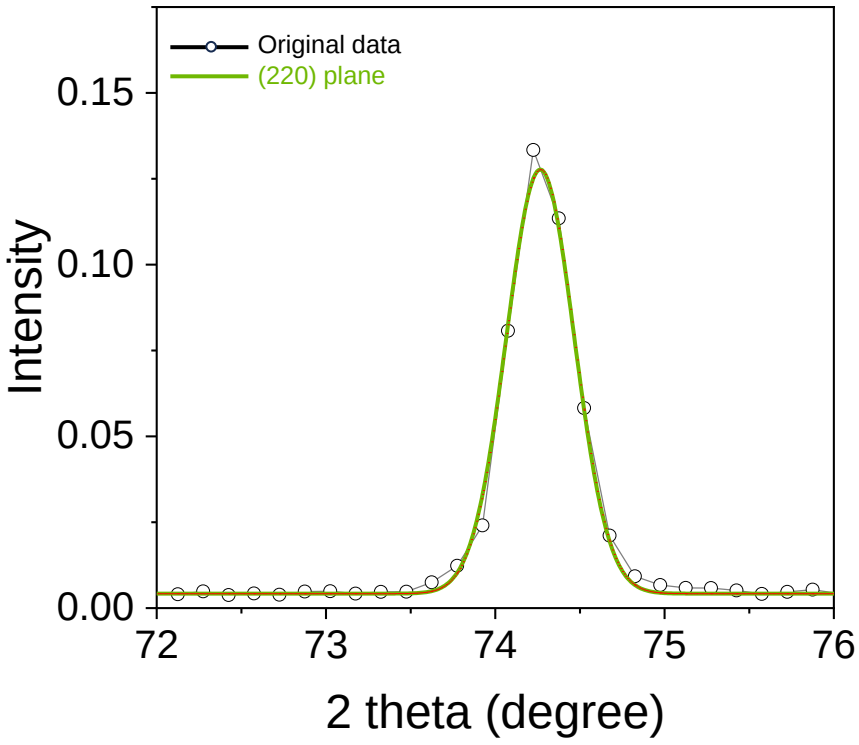
Peak Fitting Results (220) plane_S10

Position 1



FWHM: 0.4901 ± 0.01839
Height: 1

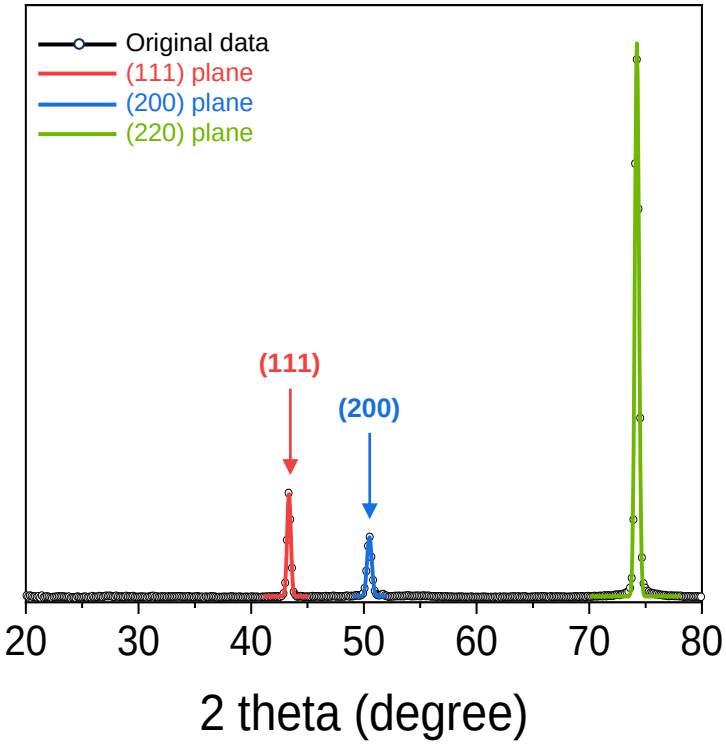
Position 2



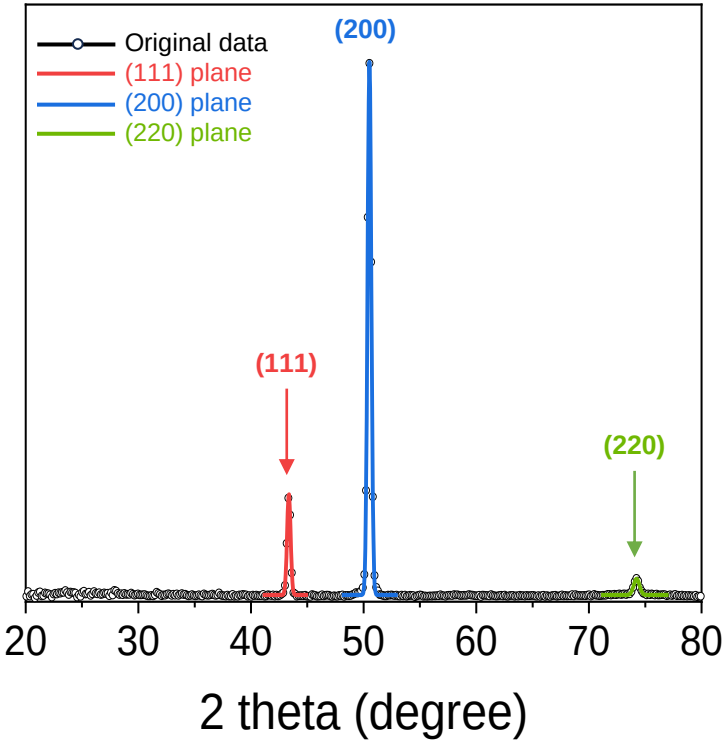
FWHM: 0.47126 ± 0.00681
Height: 0.12915

Original data XRD_S11

Position 1

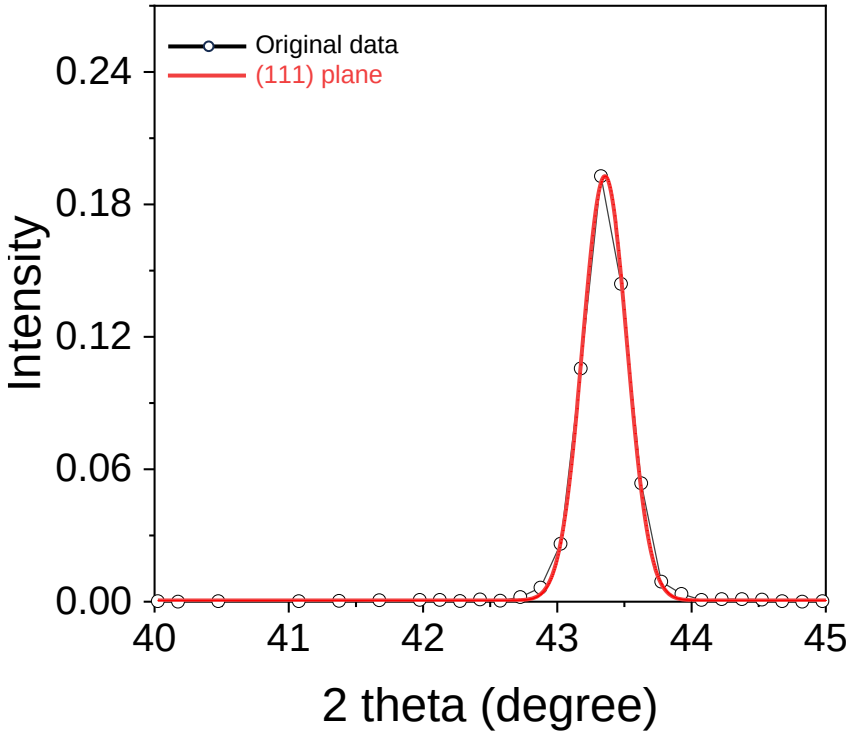


Position 2



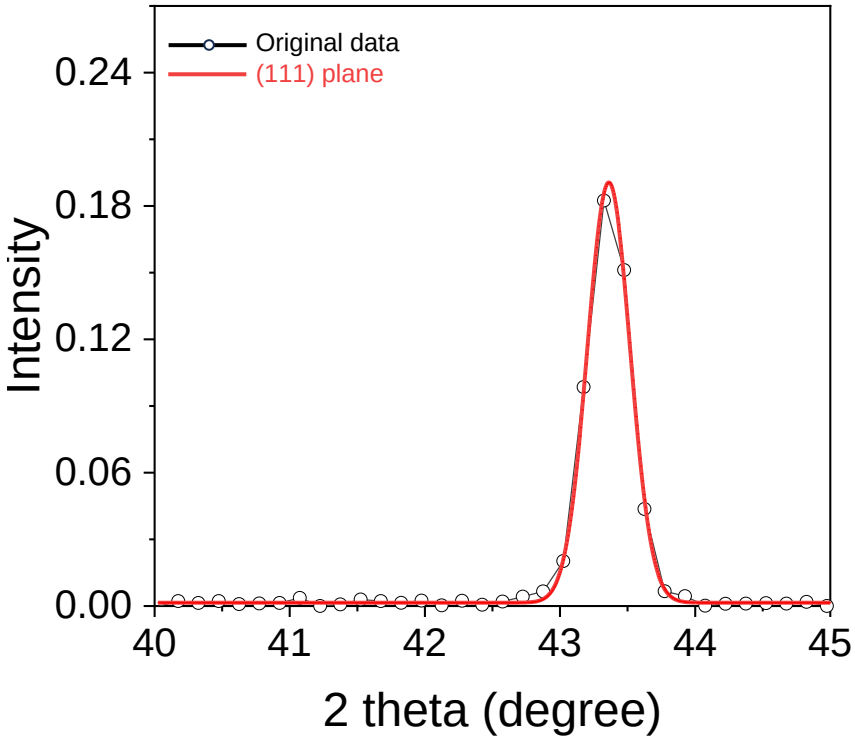
Peak Fitting Results (111) plane _S11

Position 1



FWHM:0.38862 ± 0.00277
Height: 0.19226

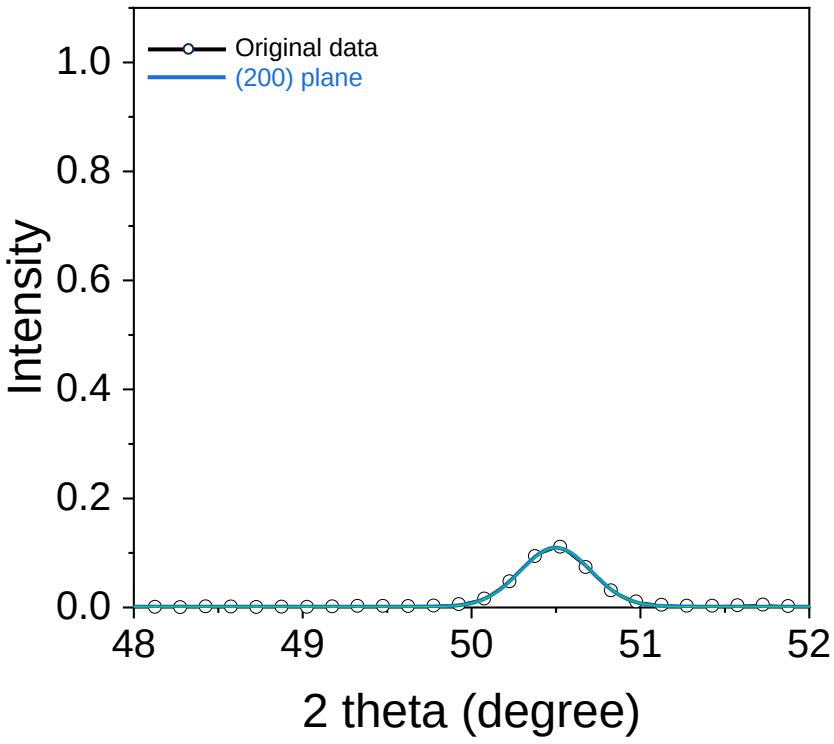
Position 2



FWHM:0.37145 ± 0.00374
Height: 0.18909

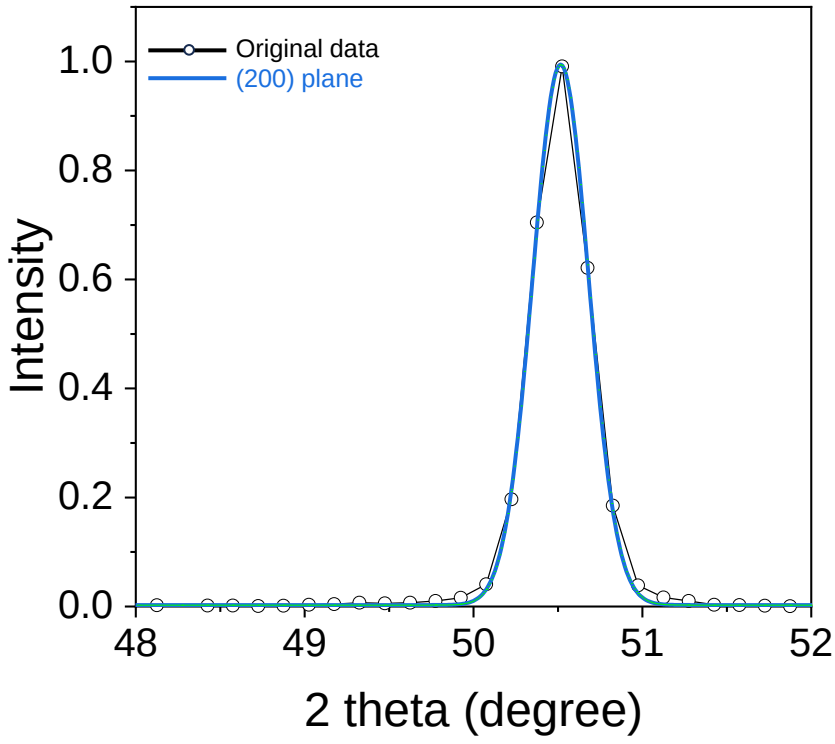
Peak Fitting Results (200) plane _S11

Position 1



FWHM: 0.49436 ± 0.00468
Height: 0.10804

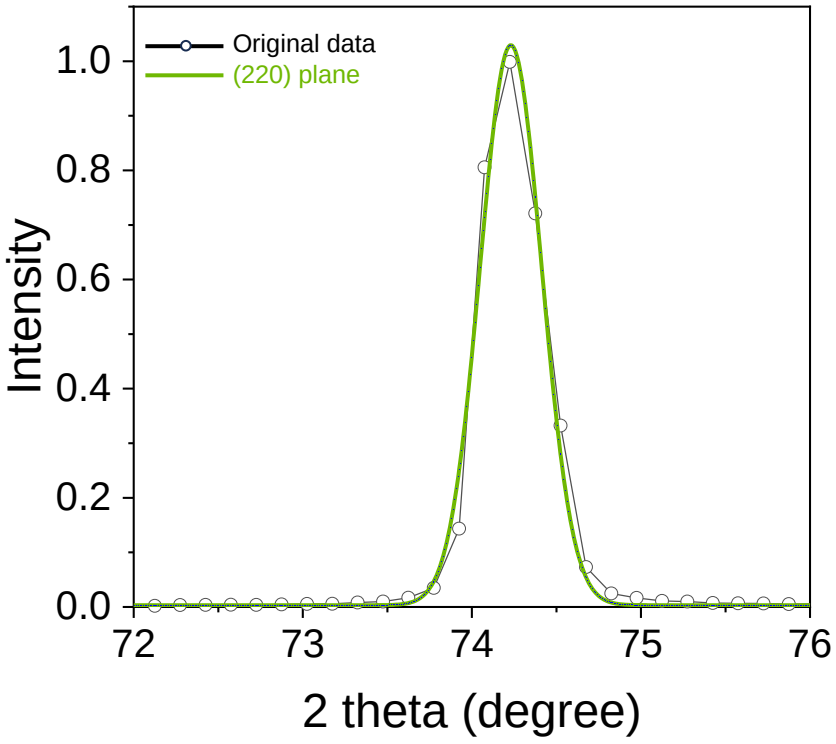
Position 2



FWHM: 0.38995 + 0.00196
Height: 0.0428

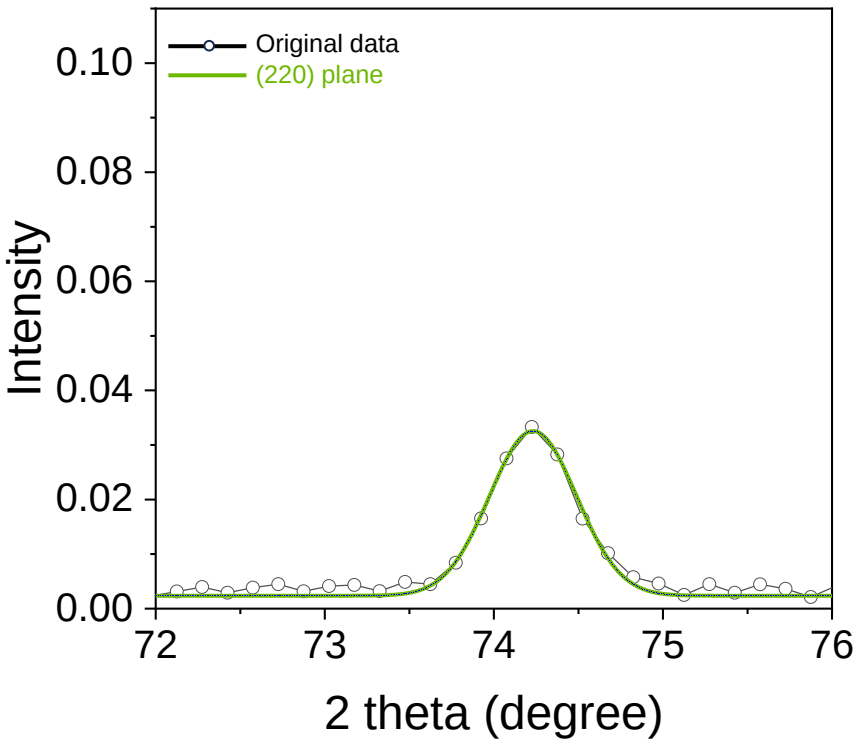
Peak Fitting Results (220) plane_S11

Position 1



FWHM: 0.4273 ± 0.00836
Height: 1

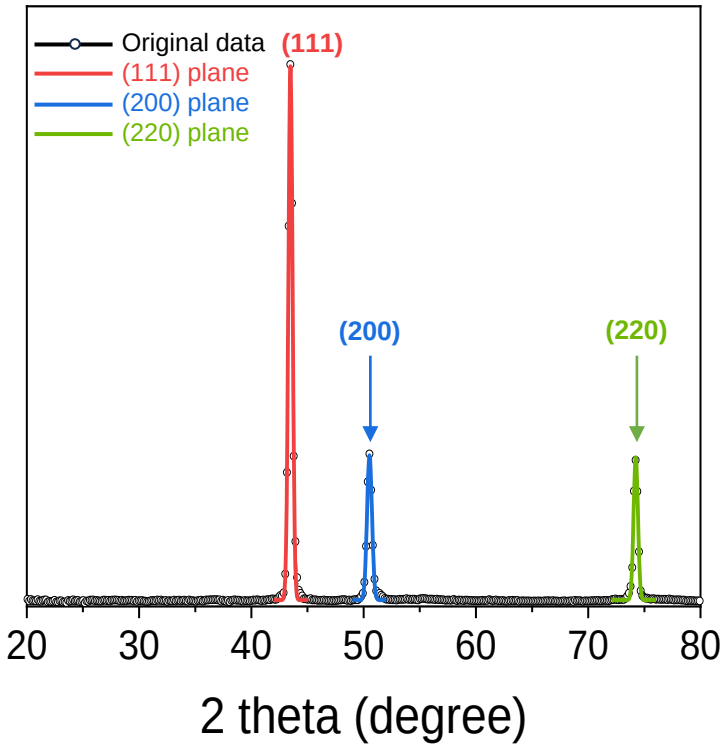
Position 2



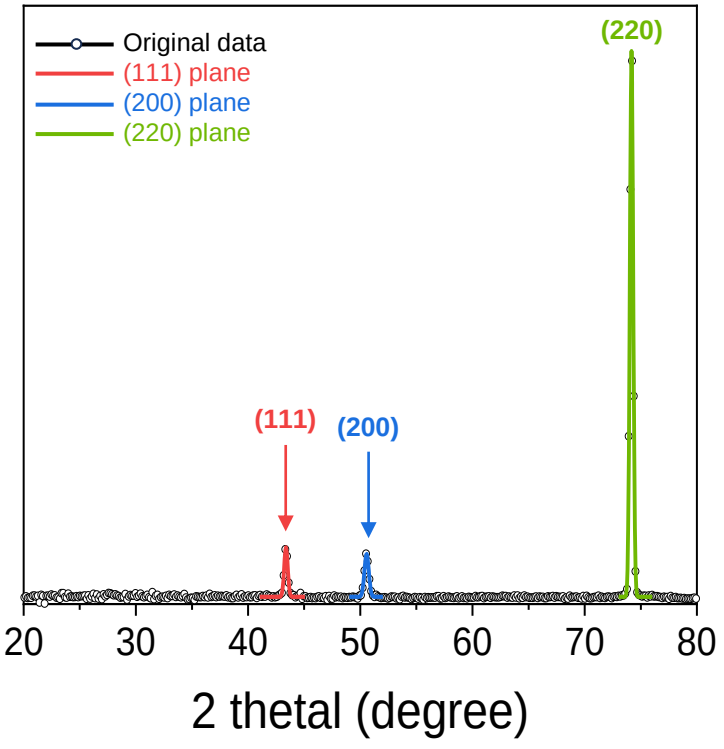
FWHM: 0.60607 ± 0.02156
Height: 0.03018

Original data XRD_S12

Position 1

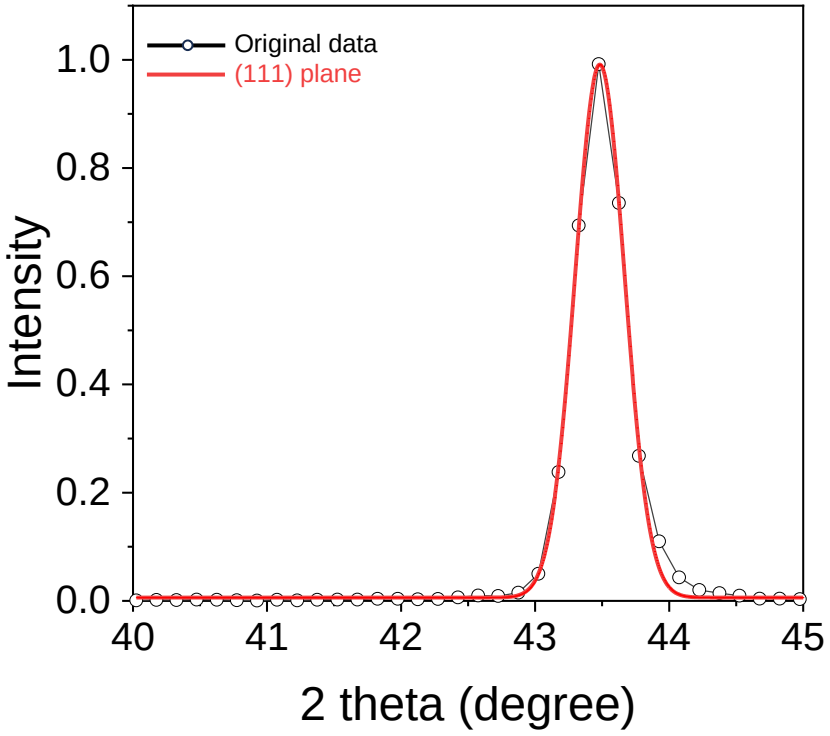


Position 2



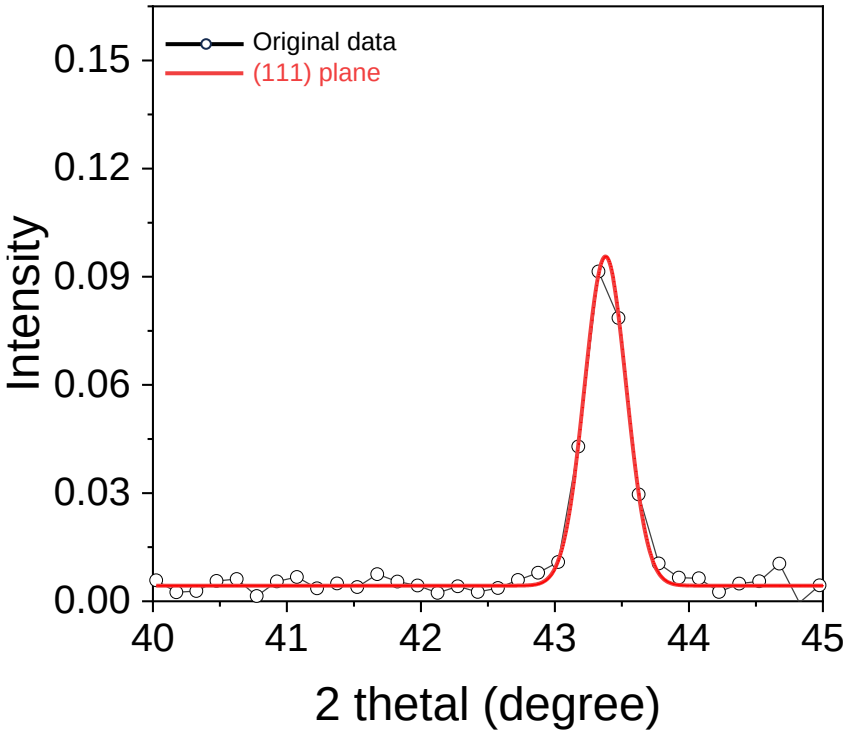
Peak Fitting Results (111) plane _S12

Position 1



FWHM:0.43314 ± 0.00489
Height: 1

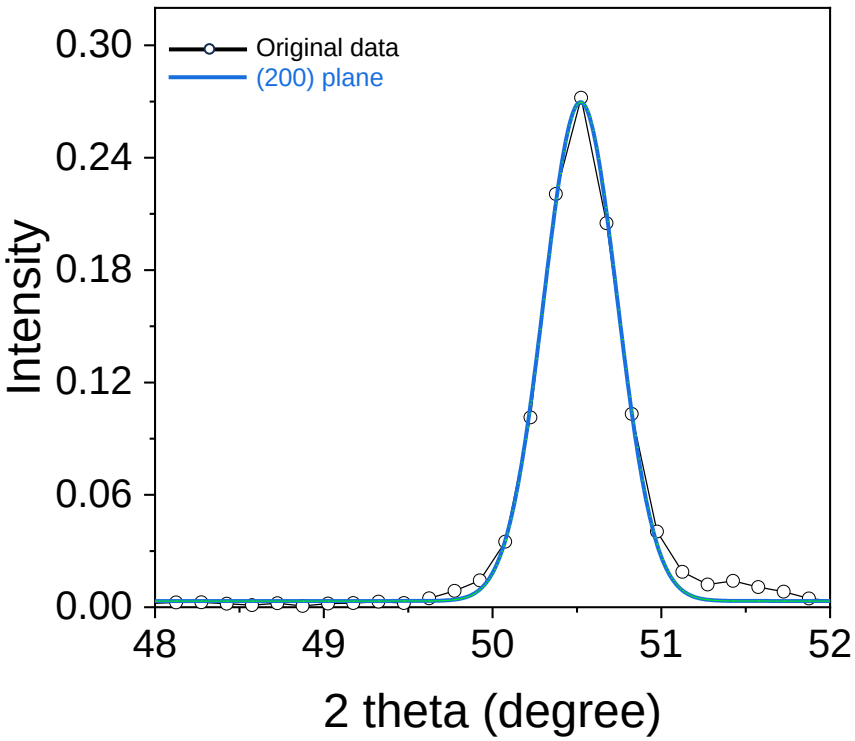
Position 2



FWHM:0.36584 ± 0.00952
Height: 0.09131

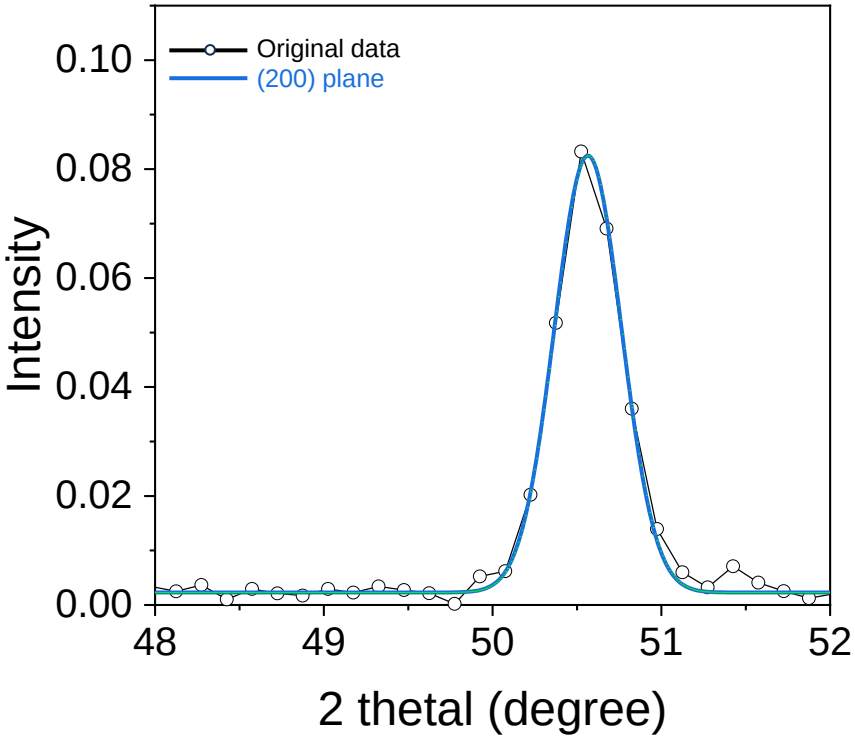
Peak Fitting Results (002) plane _S12

Position 1



FWHM: 0.51169 ± 0.00546
Height: 0.26634

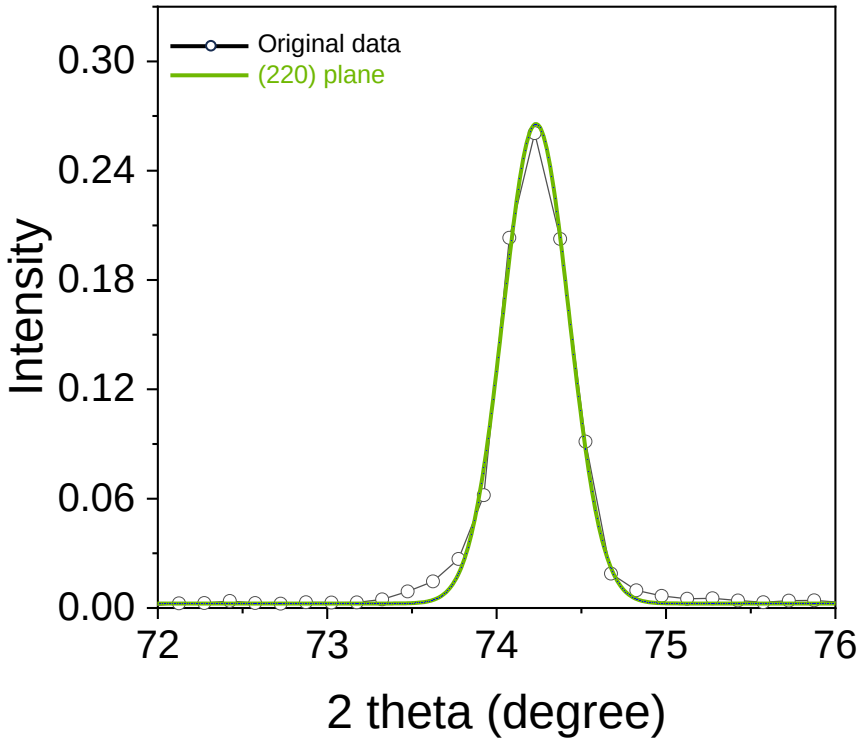
Position 2



FWHM: 0.467 ± 0.0094
Height: 0.08019

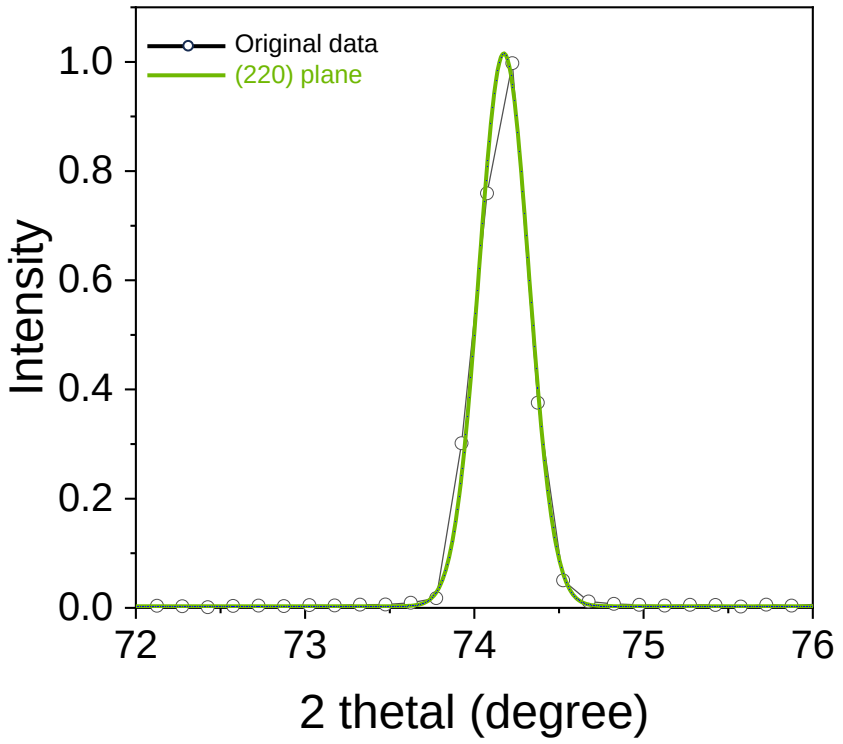
Peak Fitting Results (022) plane_S12

Position 1



FWHM: 0.45683 ± 0.00589
Height: 0.2632

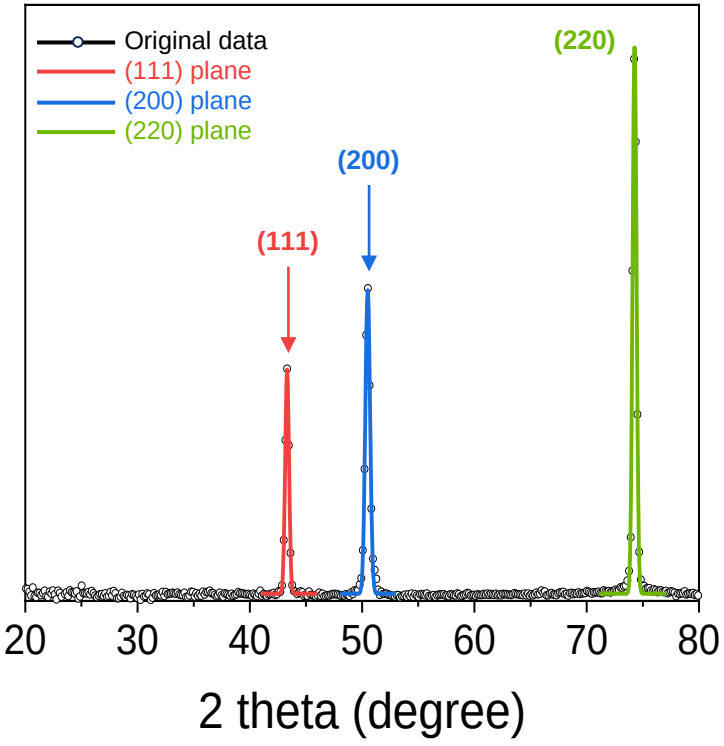
Position 2



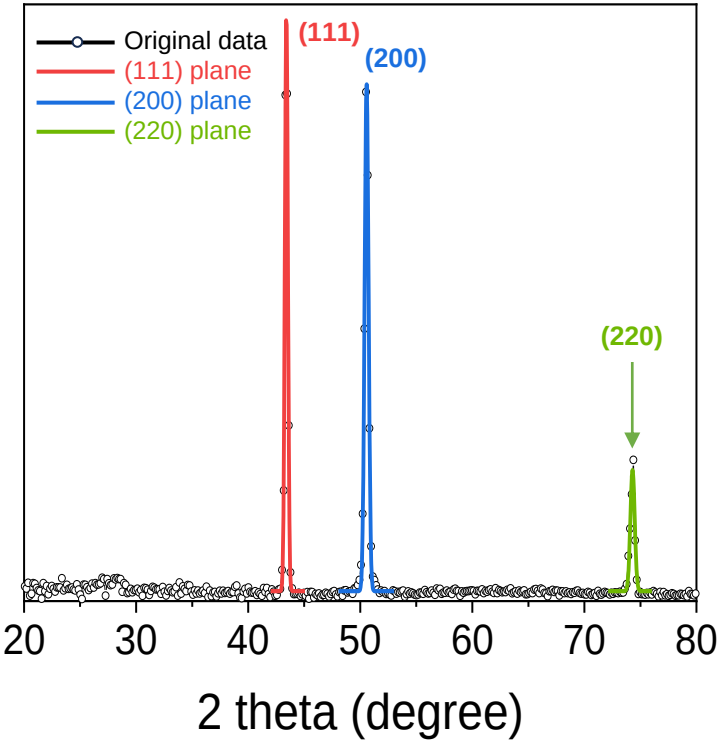
FWHM: 0.3456 ± 0.00439
Height: 1

Original data XRD_S13

Position 1

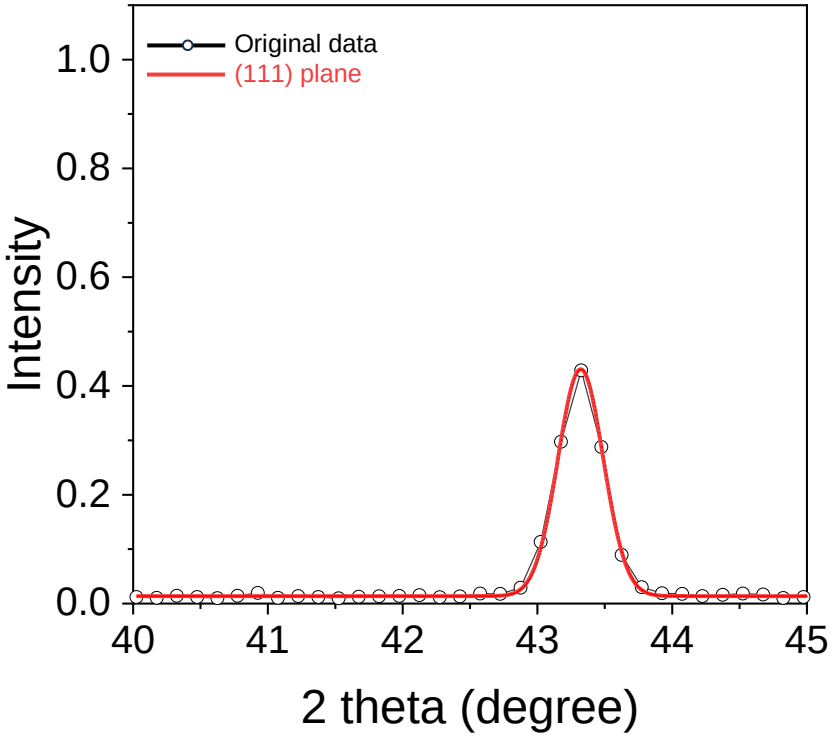


Position 2



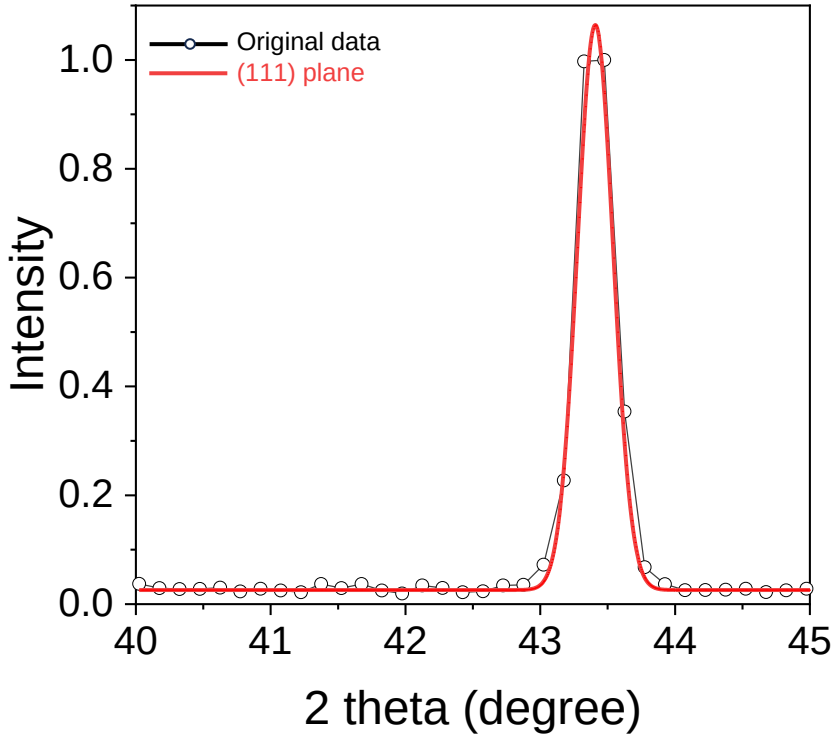
Peak Fitting Results (111) plane _S13

Position 1



FWHM: 0.39623 ± 0.00359
Height: 0.41689

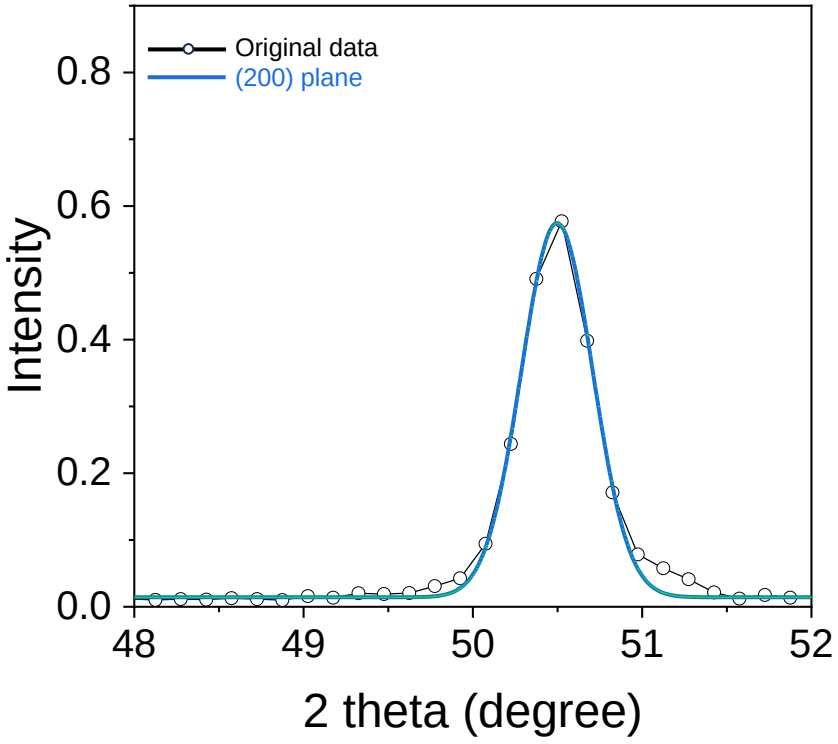
Position 2



FWHM: 0.31221 ± 0.00448
Height: 1

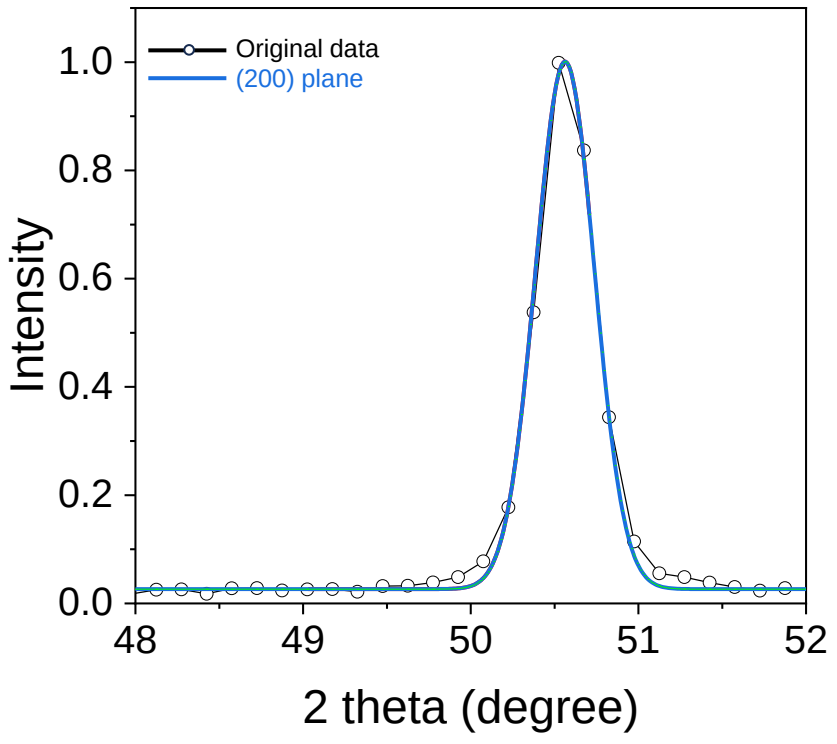
Peak Fitting Results (200) plane _S13

Position 1



FWHM: 0.49513 ± 0.00661
Height: 0.56032

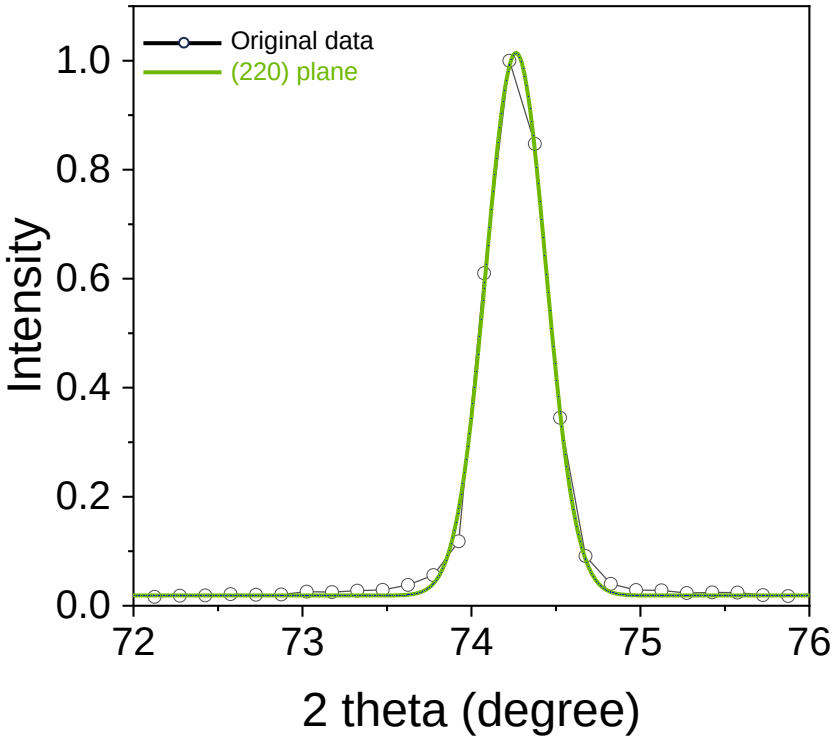
Position 2



FWHM: 0.4079 ± 0.00395
Height: 1

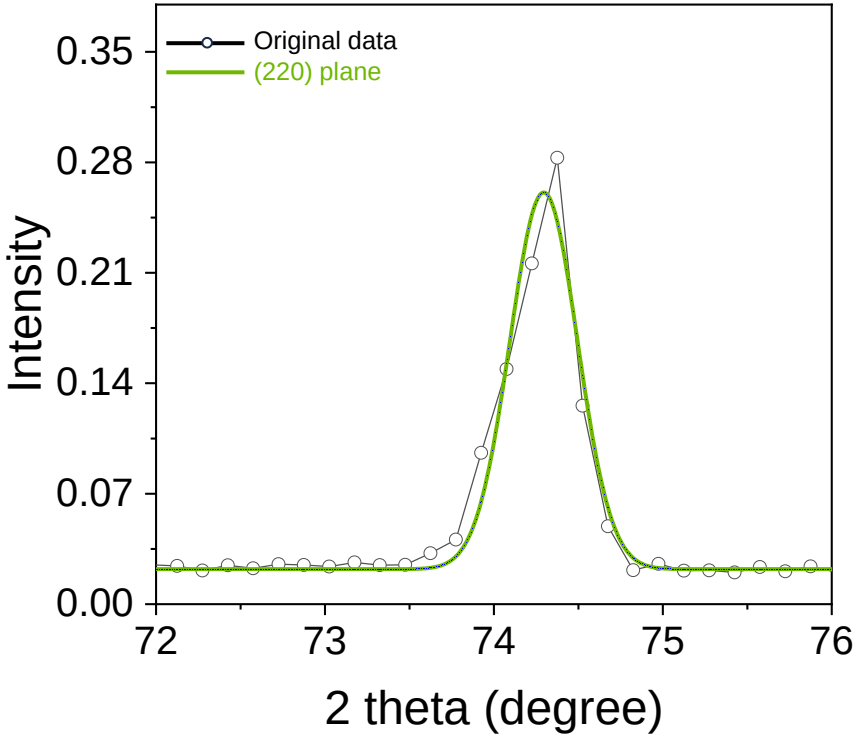
Peak Fitting Results (220) plane_S13

Position 1



FWHM: 0.41665 ± 0.00424
Height: 1

Position 2

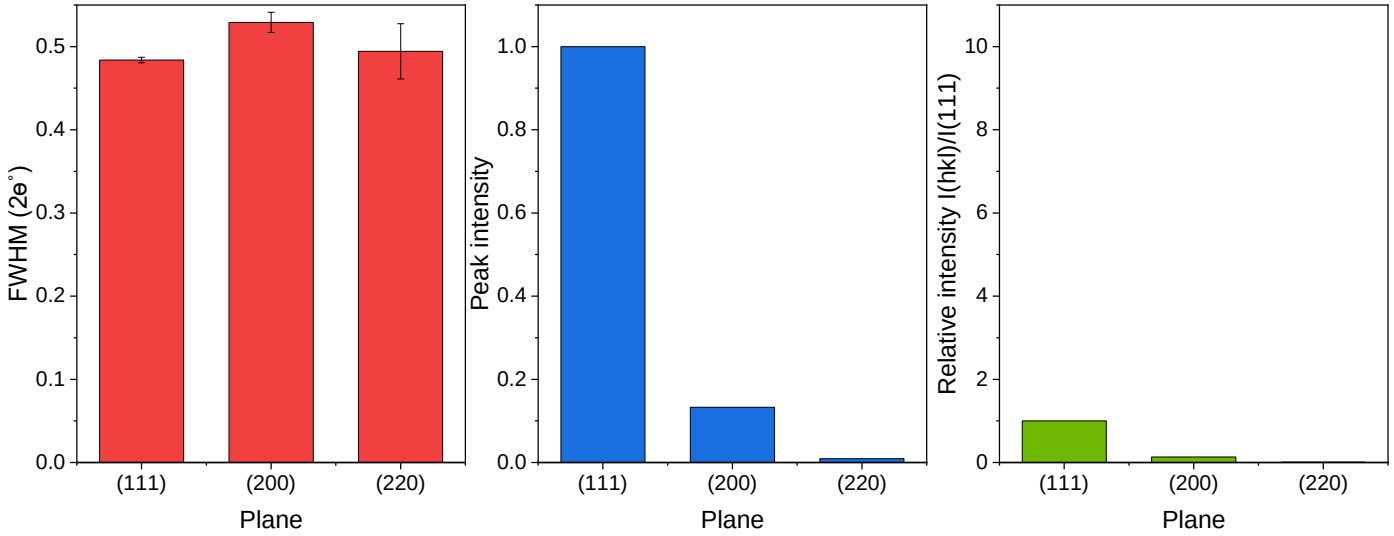


FWHM: 0.47047 ± 0.01634
Height: 0.23881

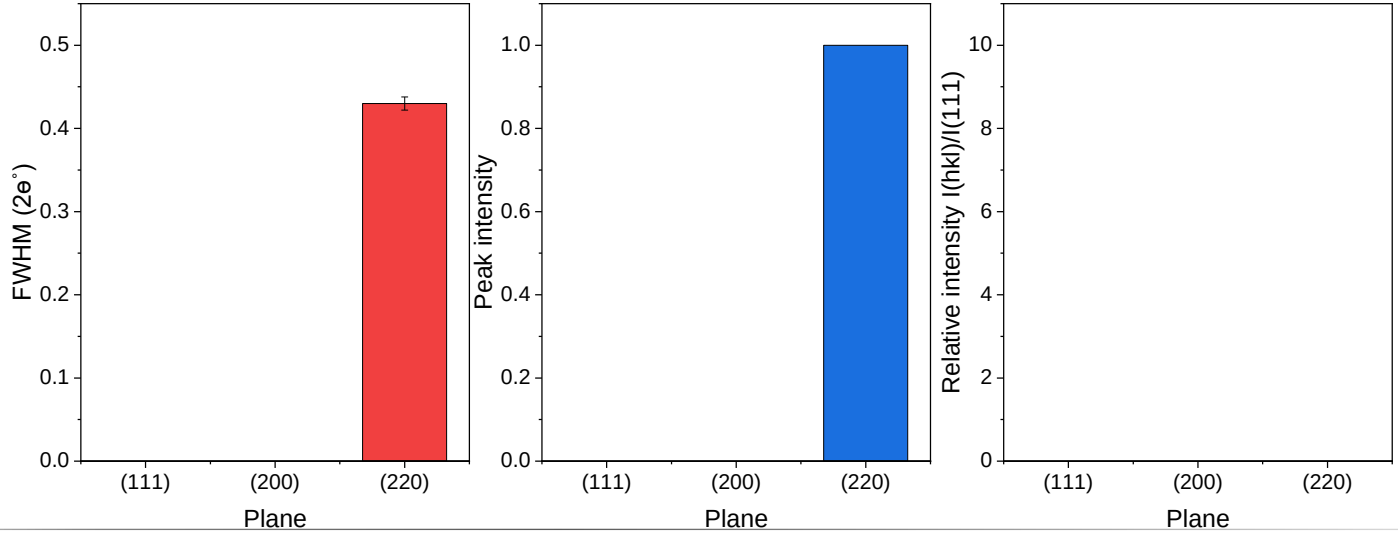
Summarized Data

Sample S1

Position 1



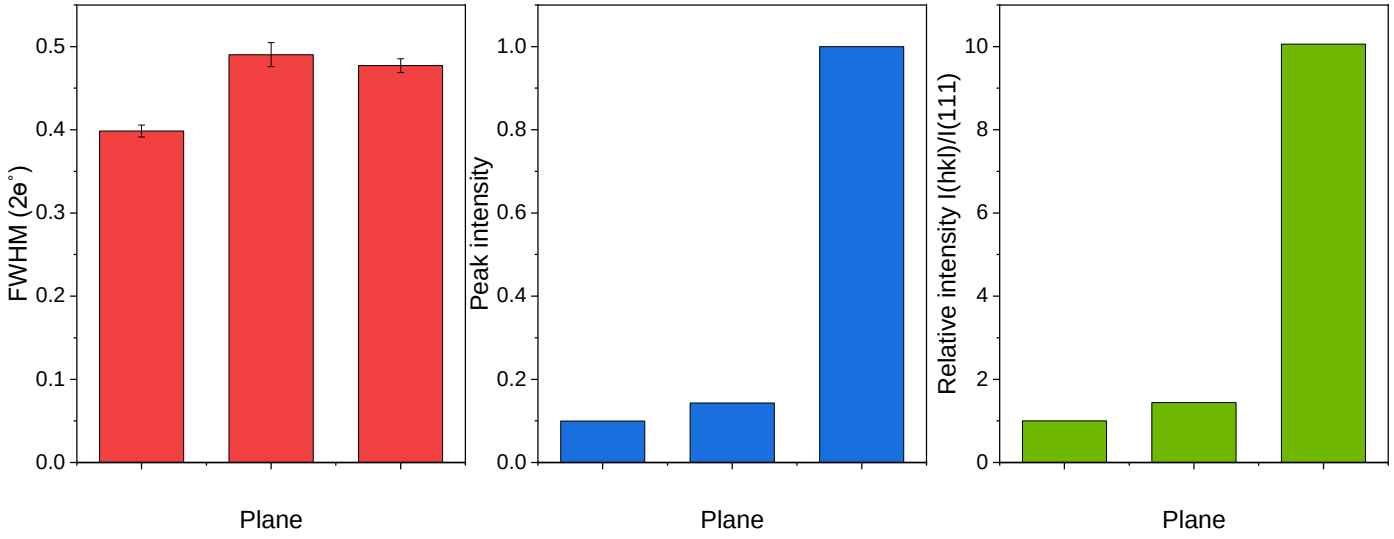
Position 2



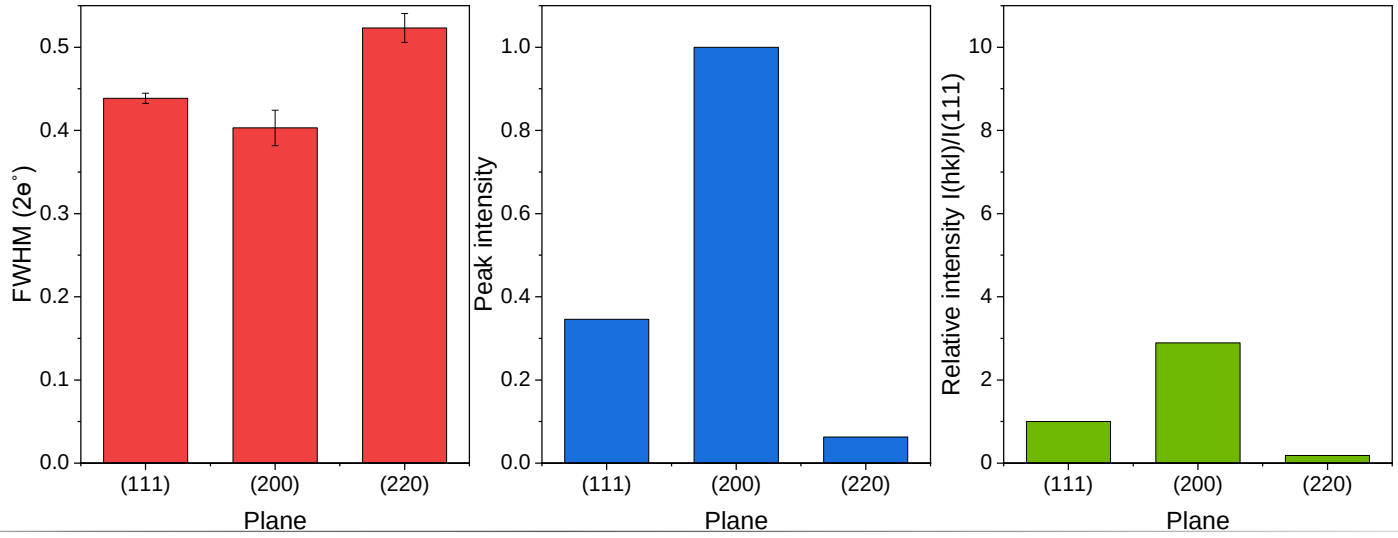
Summarized Data

Sample S2

Position 1



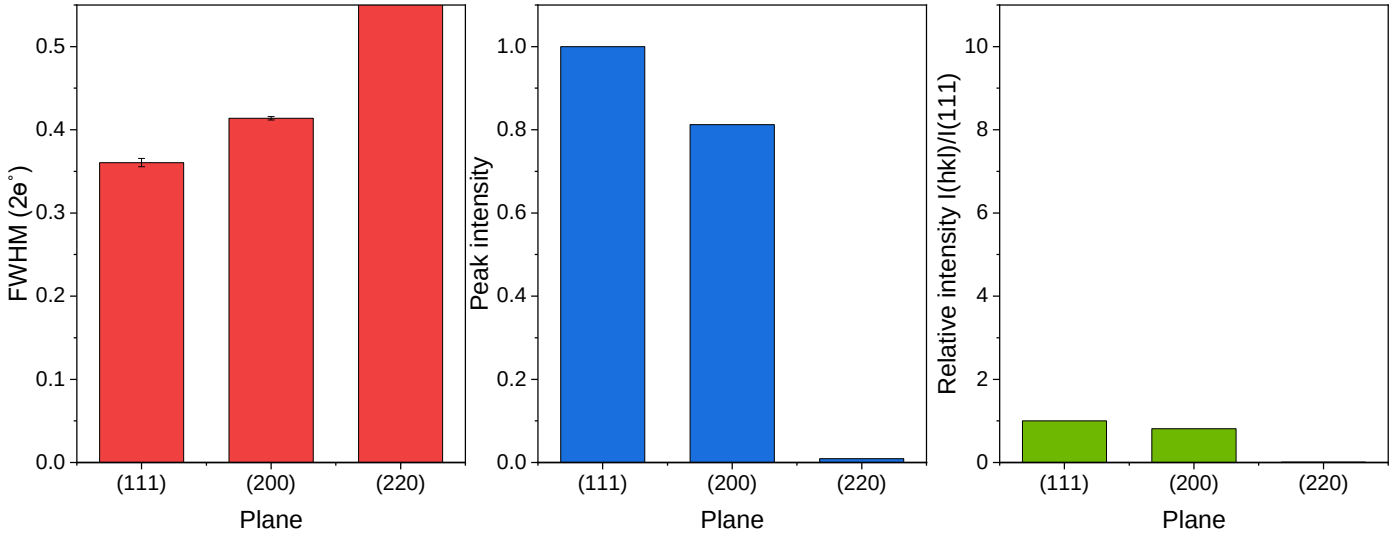
Position 2



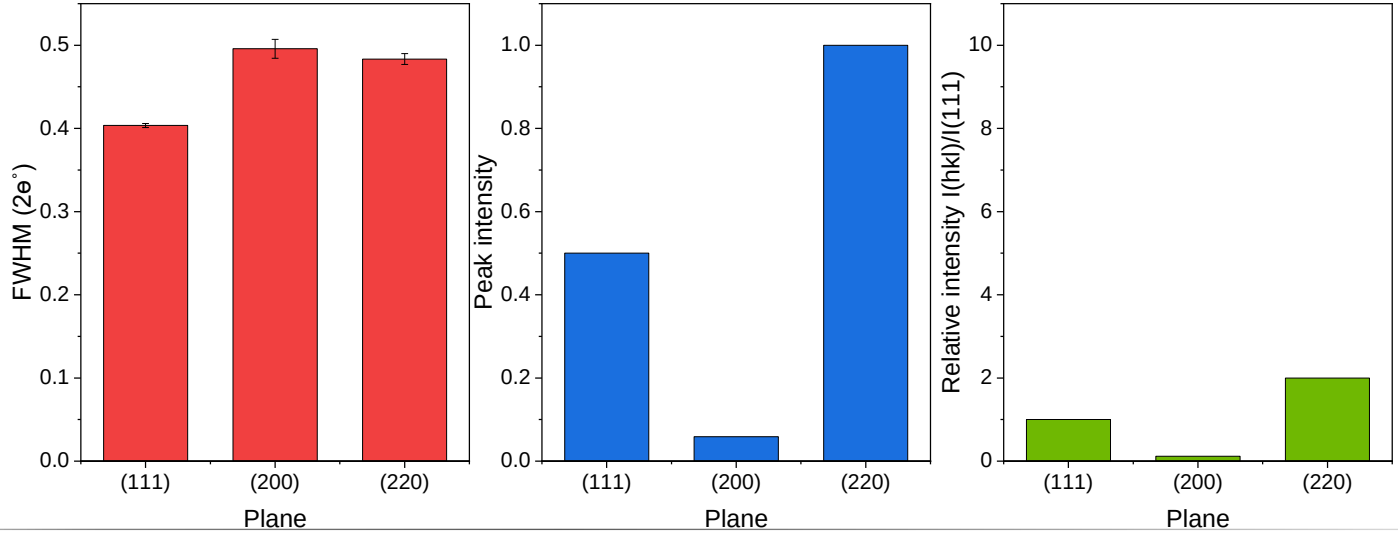
Summarized Data

Sample S3

Position 1



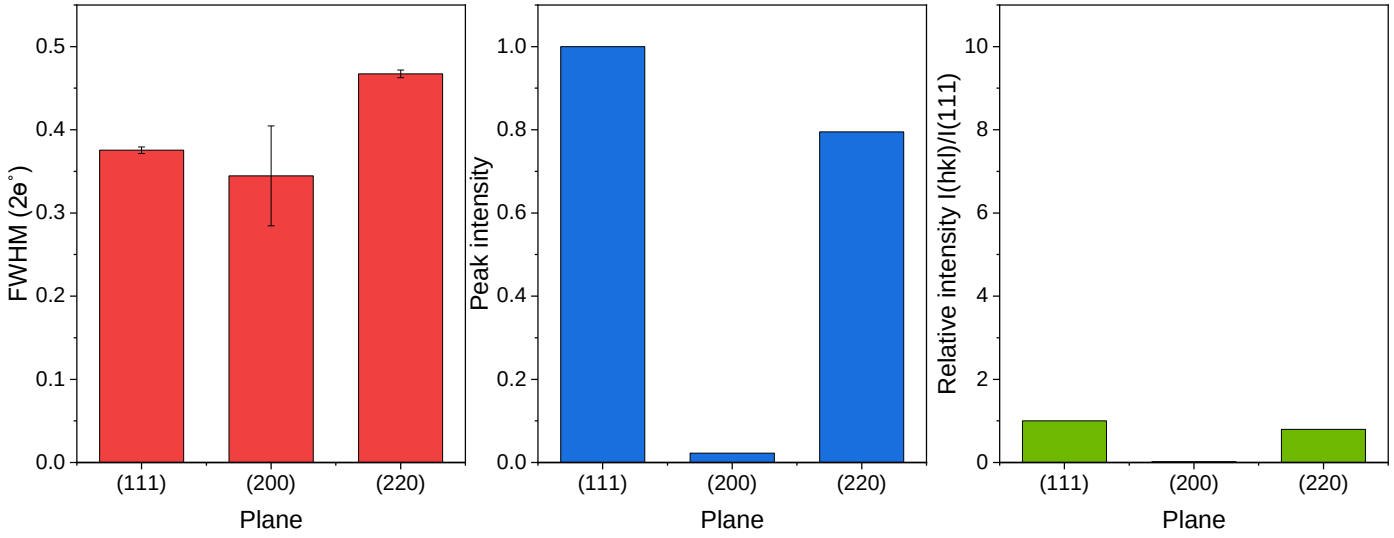
Position 2



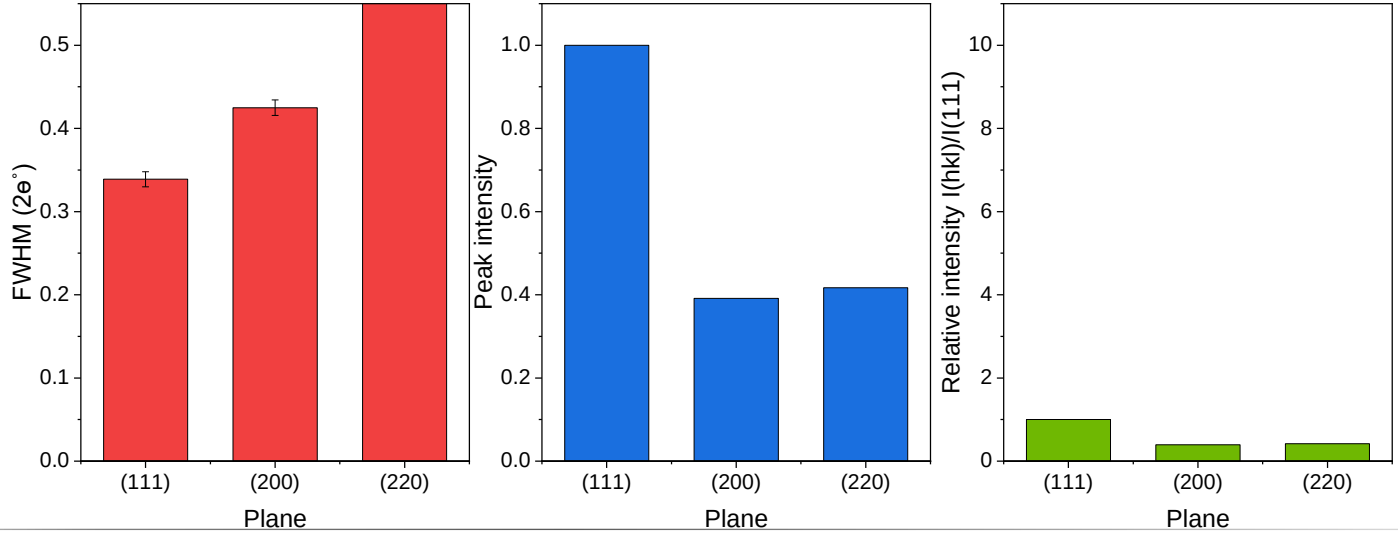
Summarized Data

Sample S4

Position 1



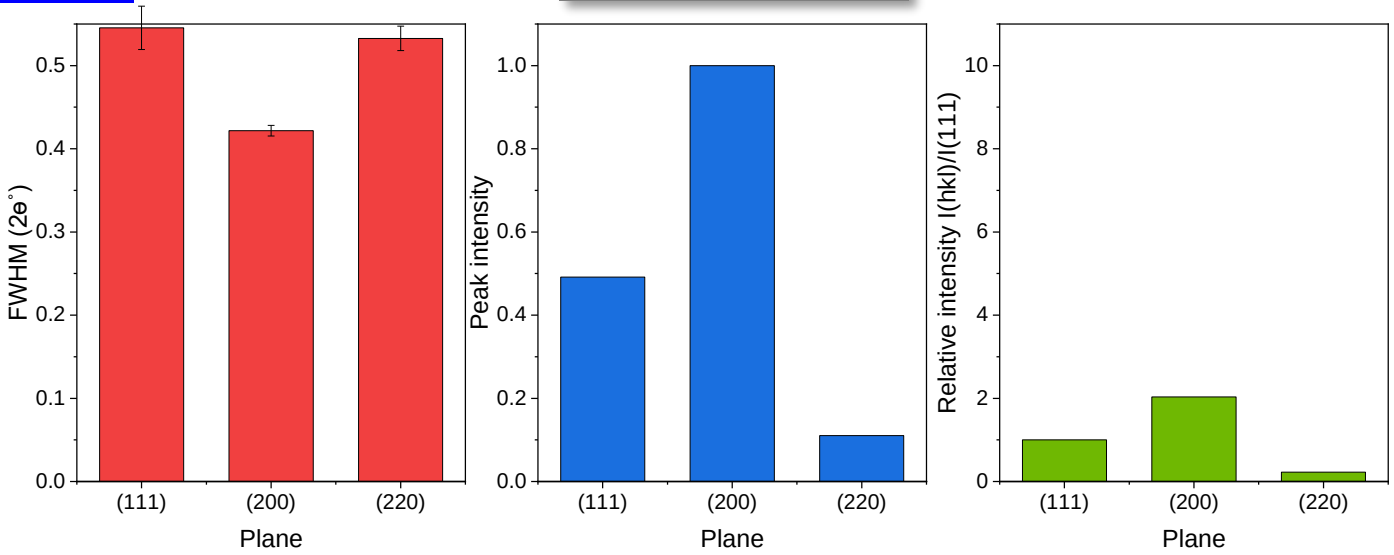
Position 2



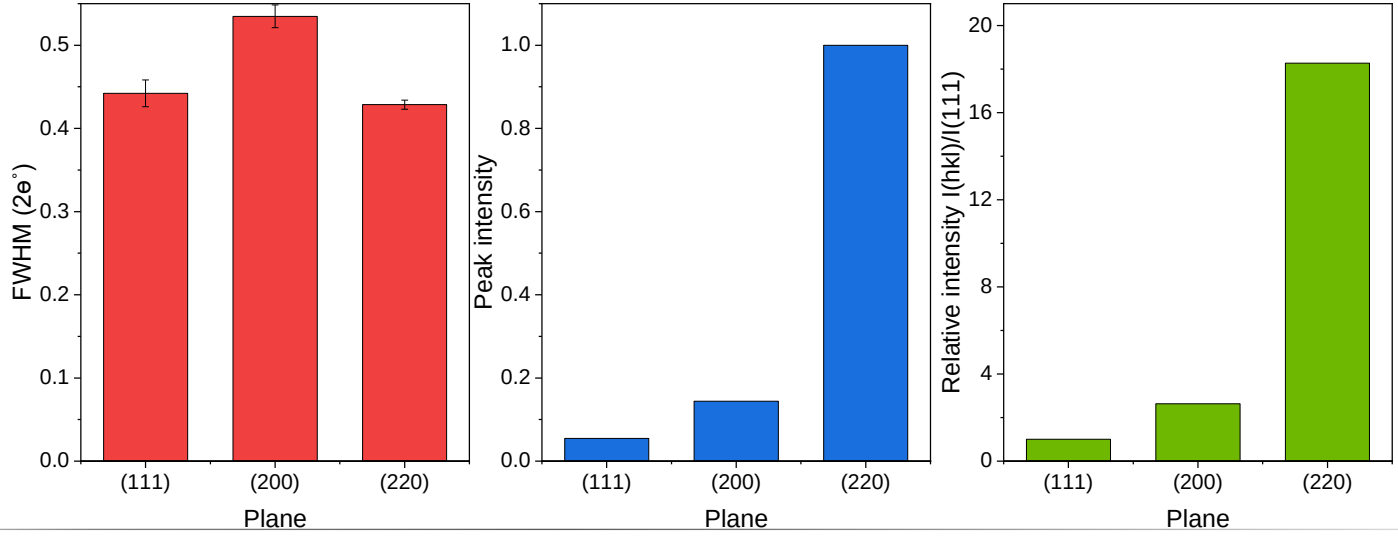
Summarized Data

Sample S5

Position 1



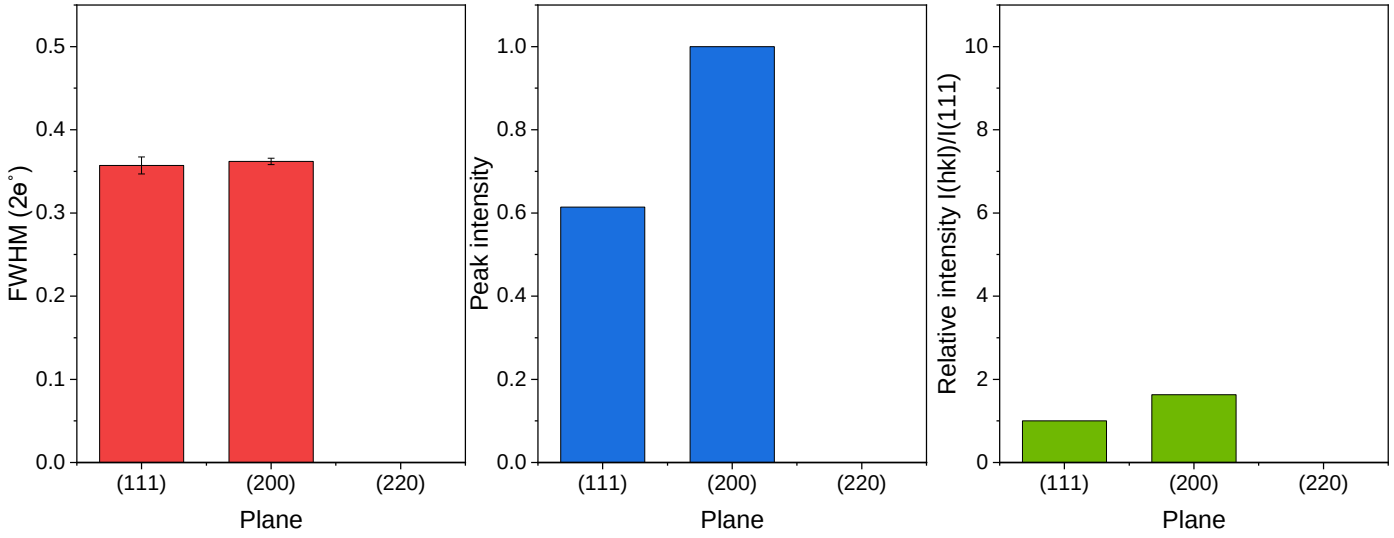
Position 2



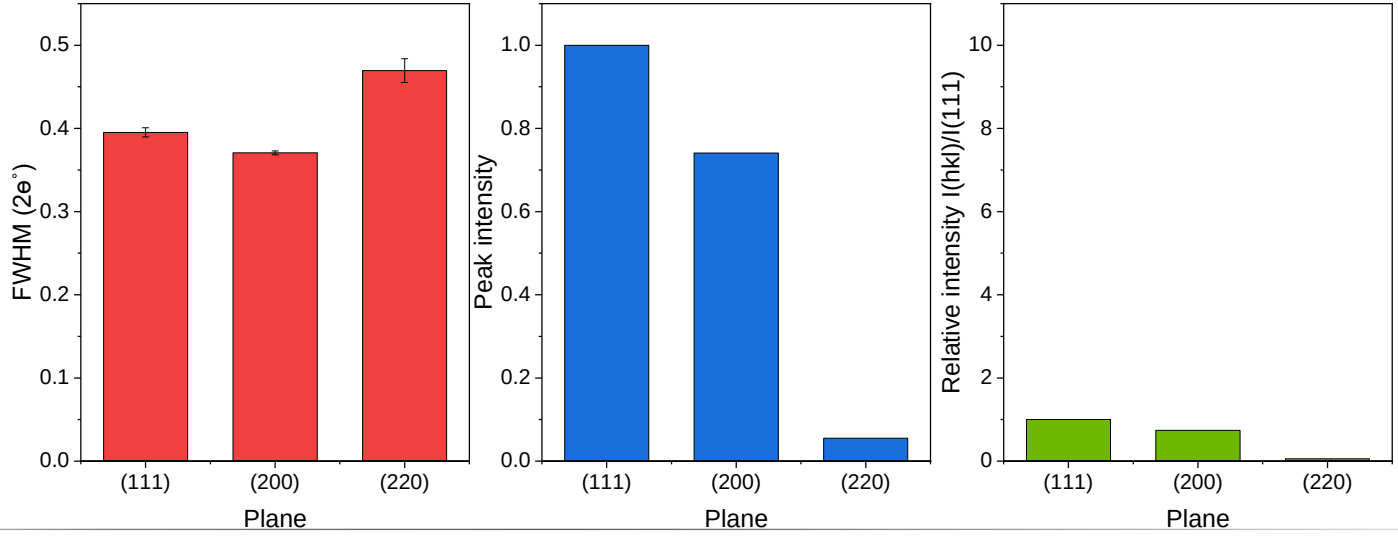
Summarized Data

Sample S6

Position 1



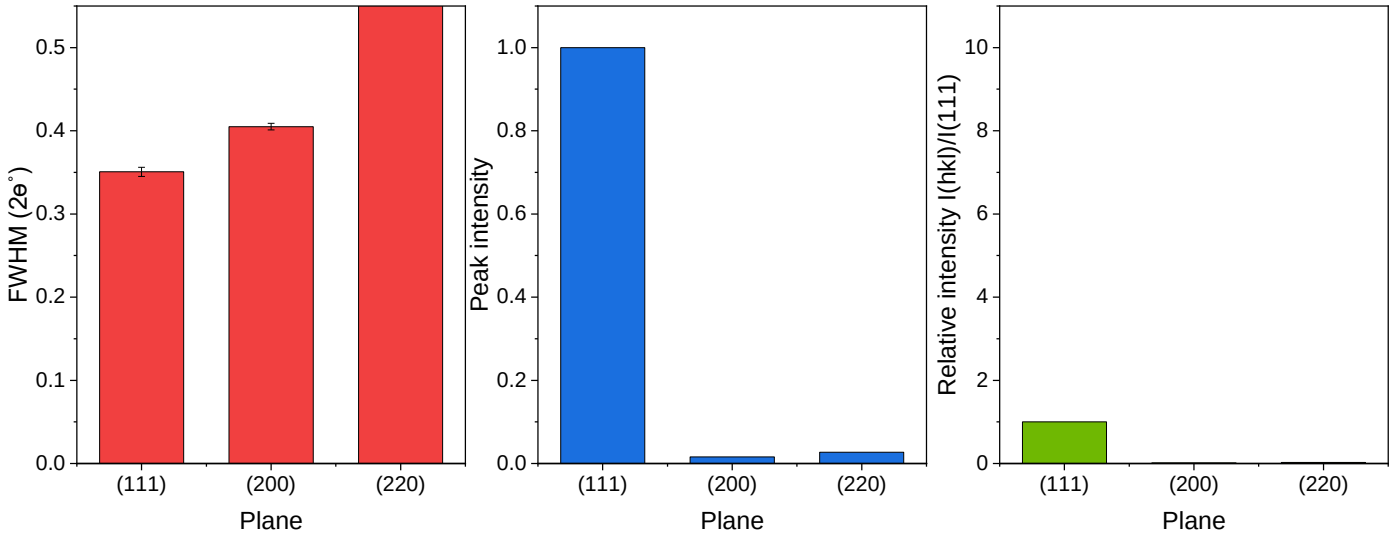
Position 2



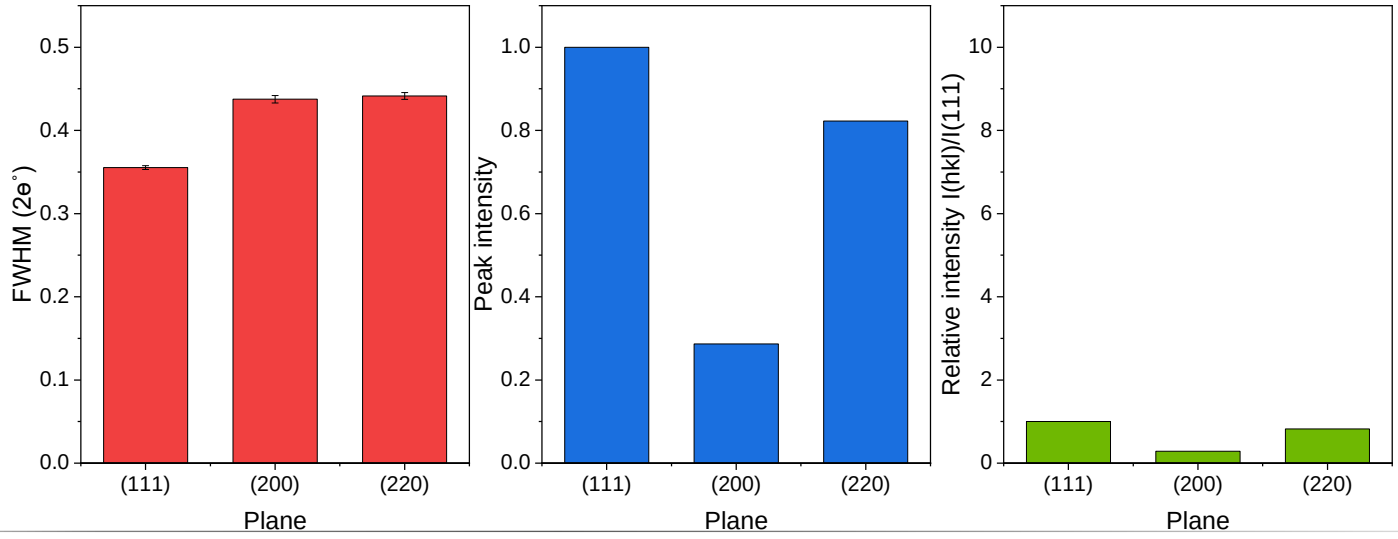
Summarized Data

Sample S7

Position 1



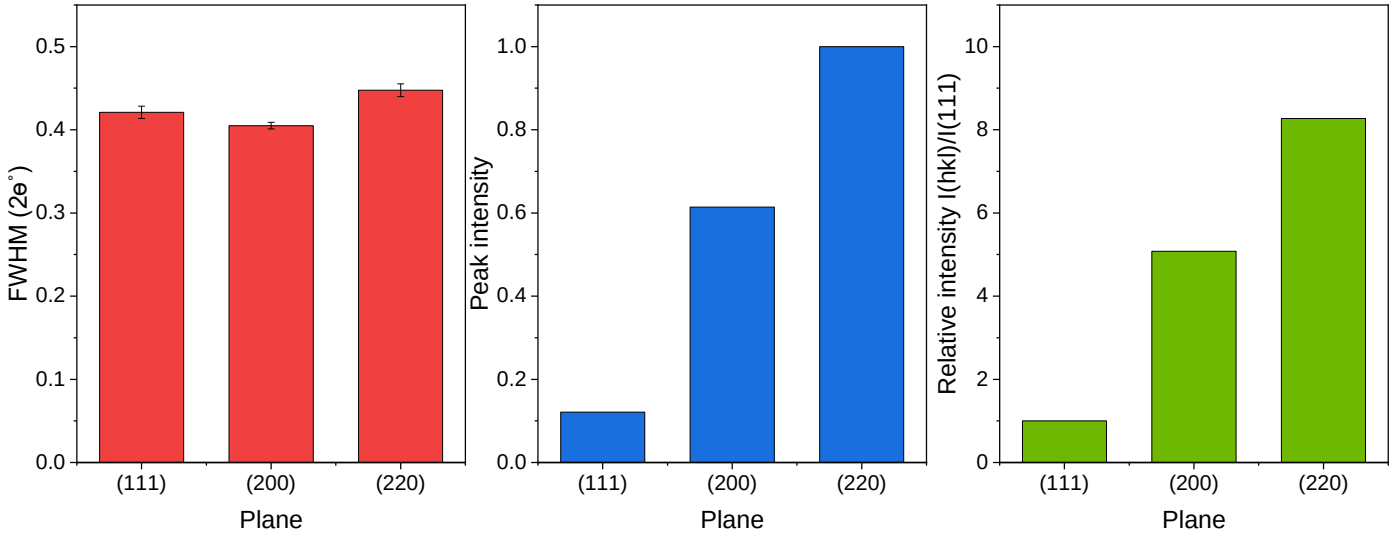
Position 2



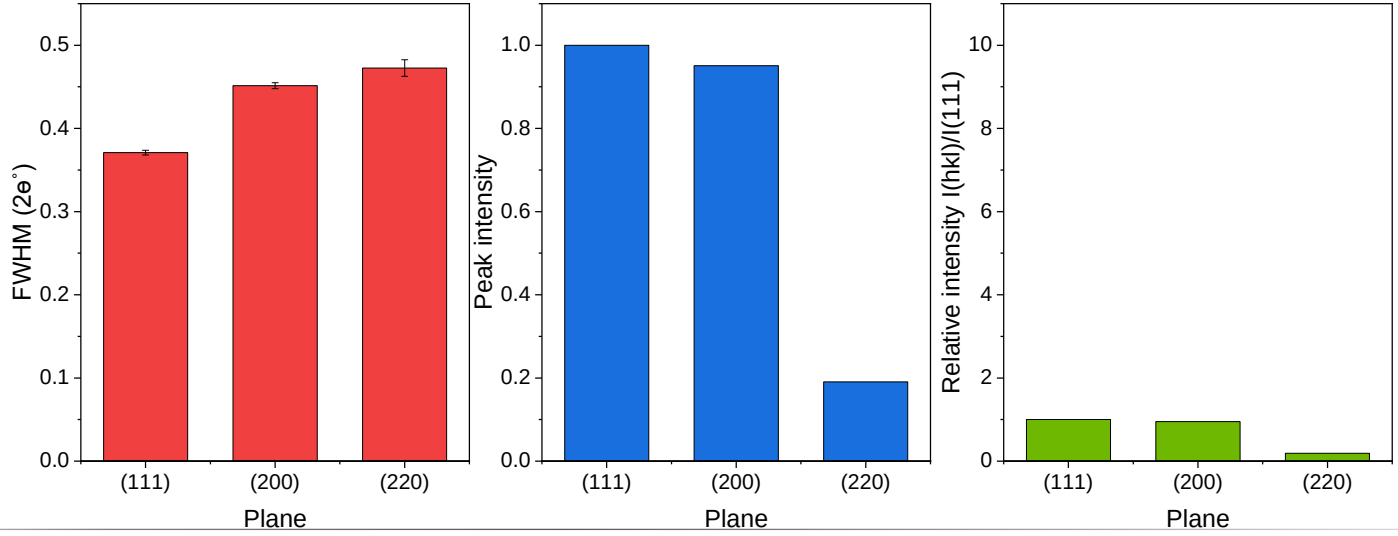
Summarized Data

Sample S8

Position 1



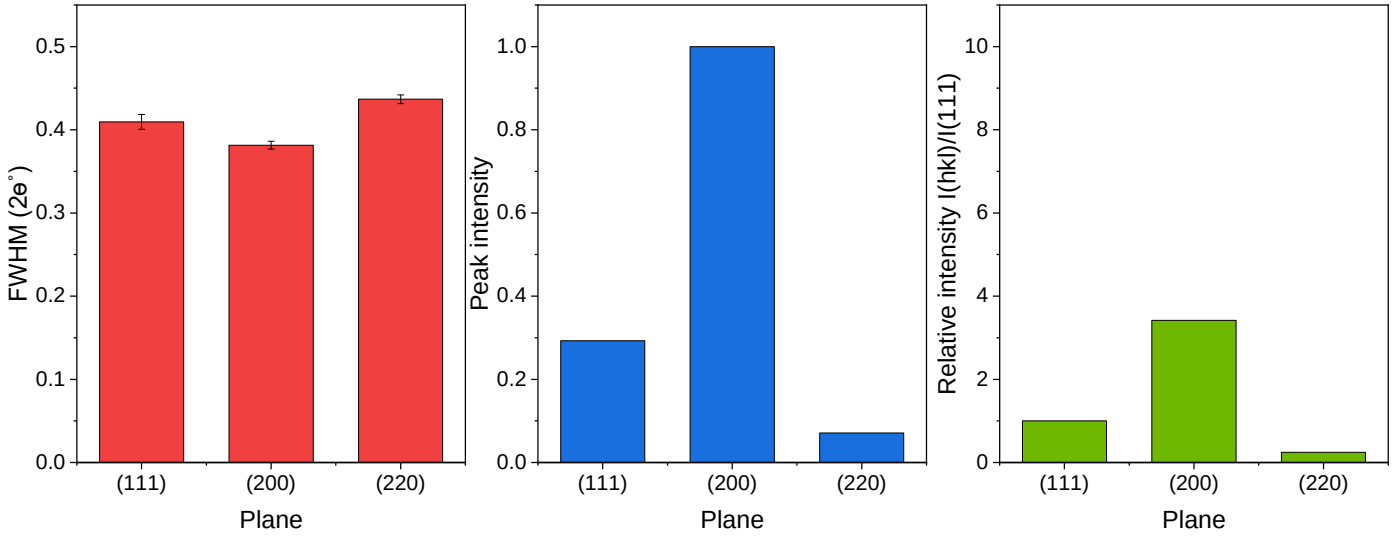
Position 2



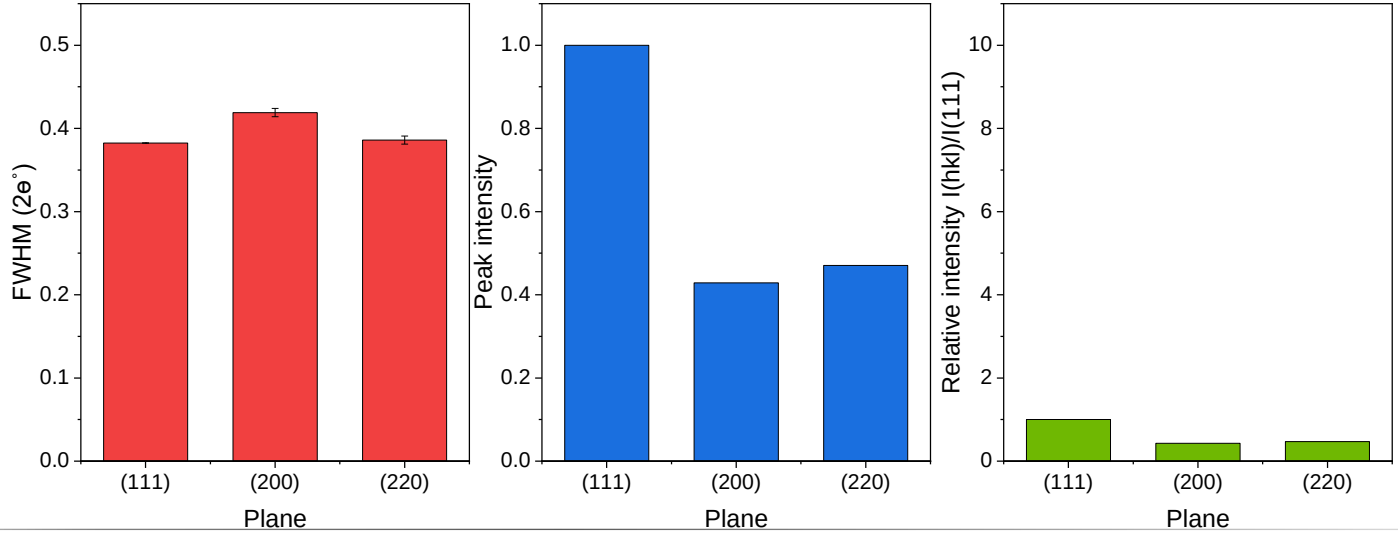
Summarized Data

Sample S9

Position 1



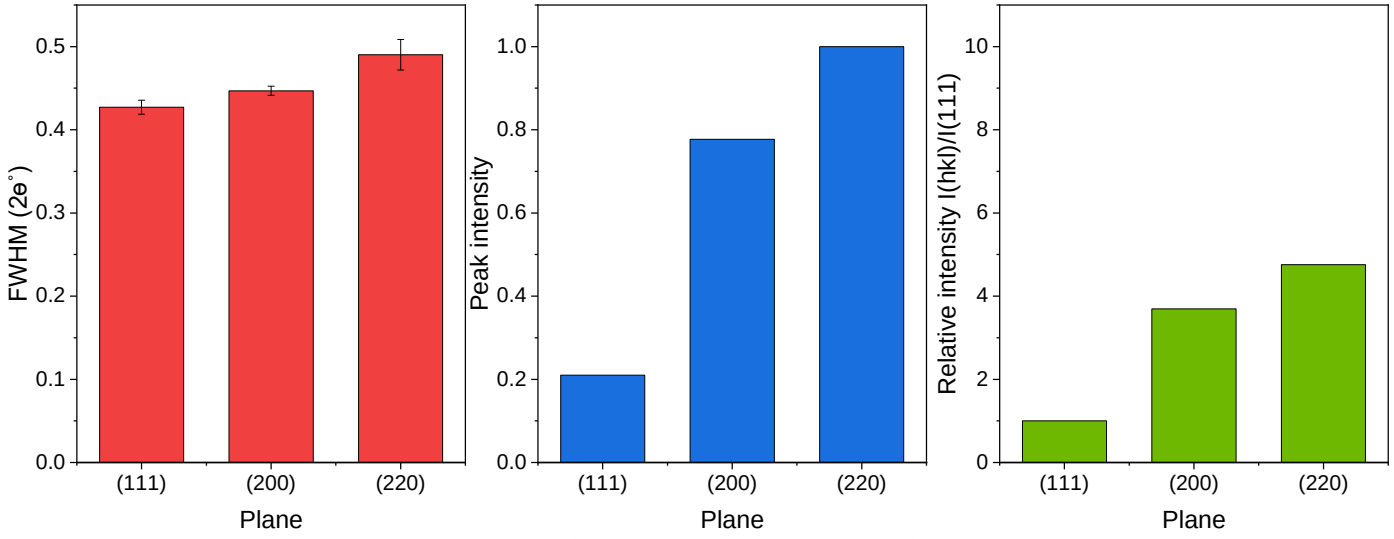
Position 2



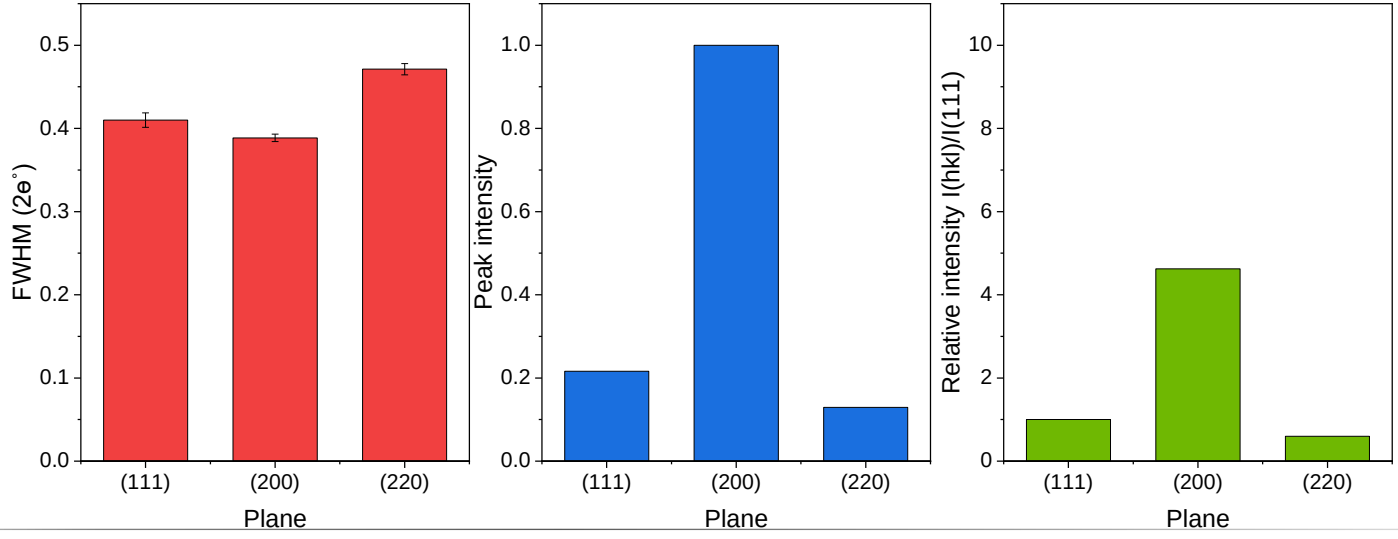
Summarized Data

Sample S10

Position 1



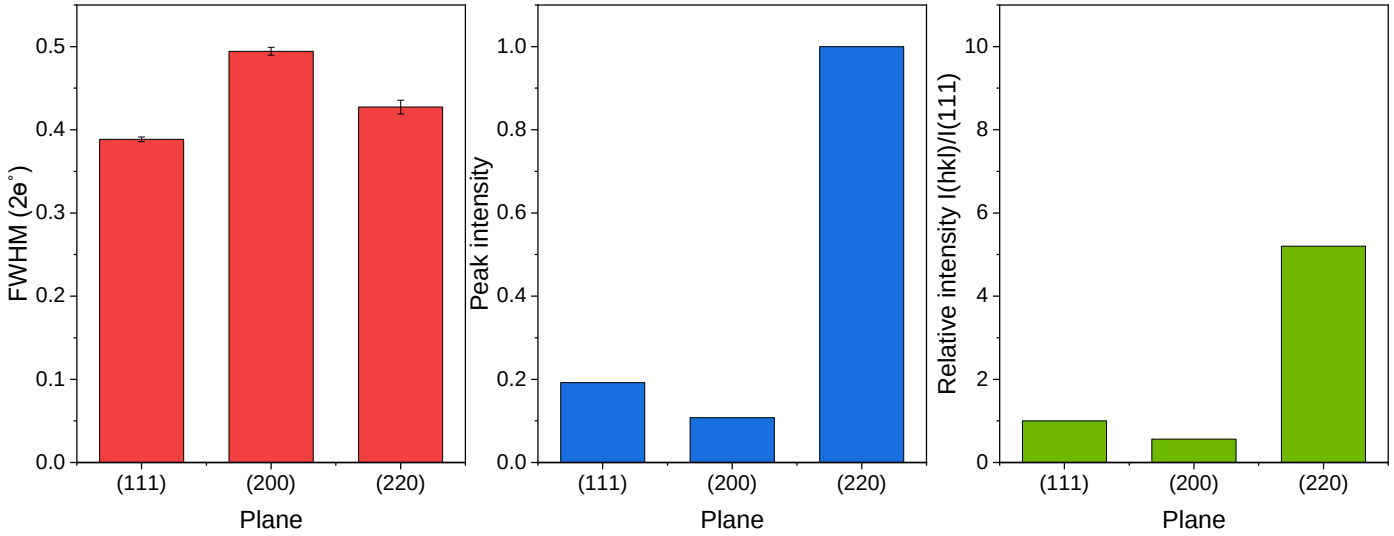
Position 2



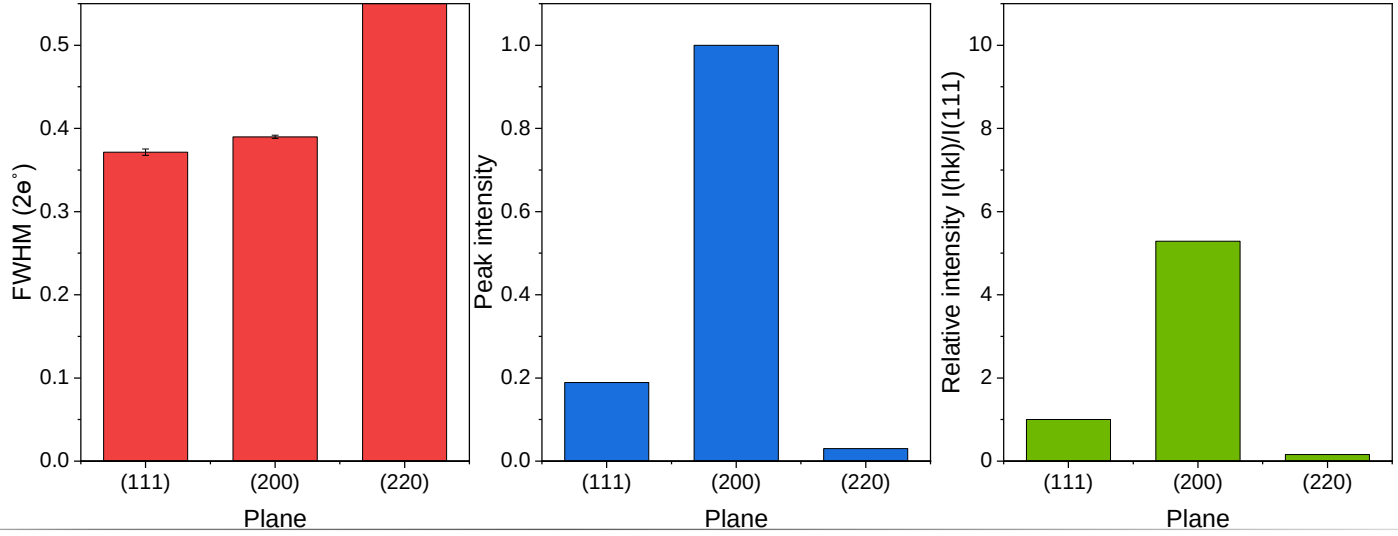
Summarized Data

Sample S11

Position 1



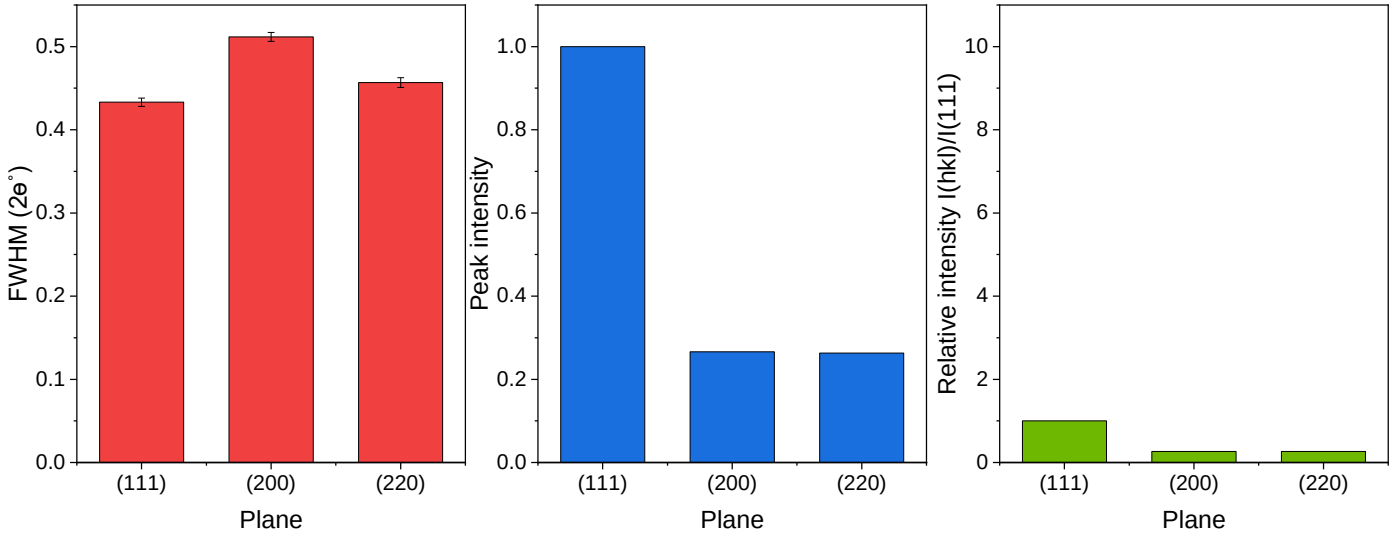
Position 2



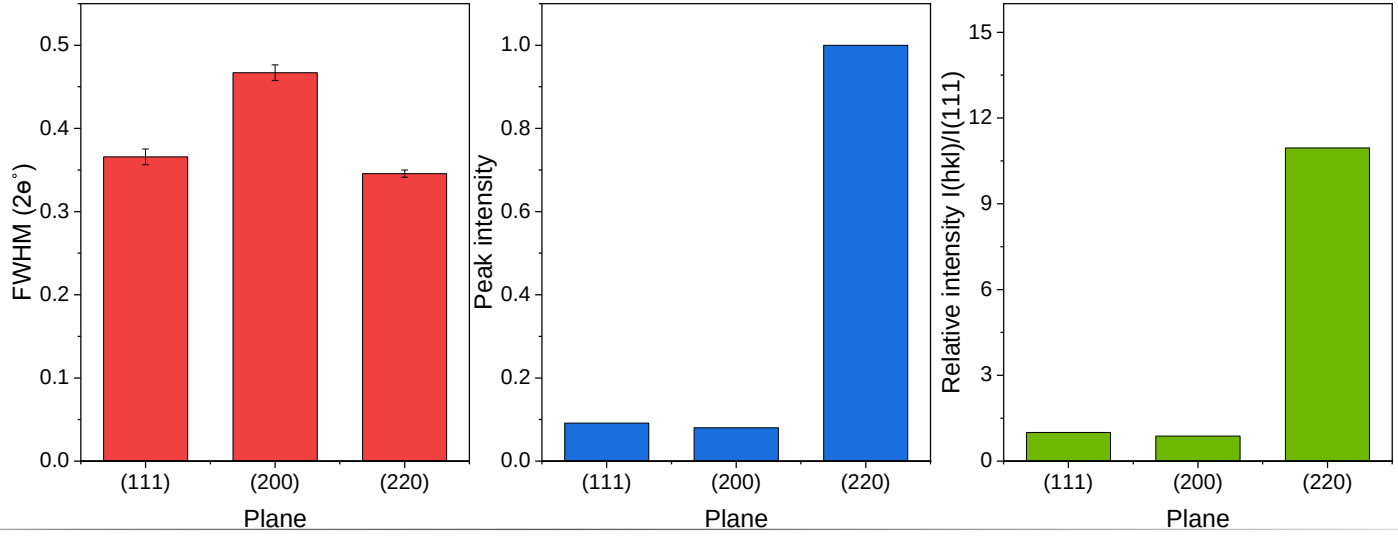
Summarized Data

Sample S12

Position 1



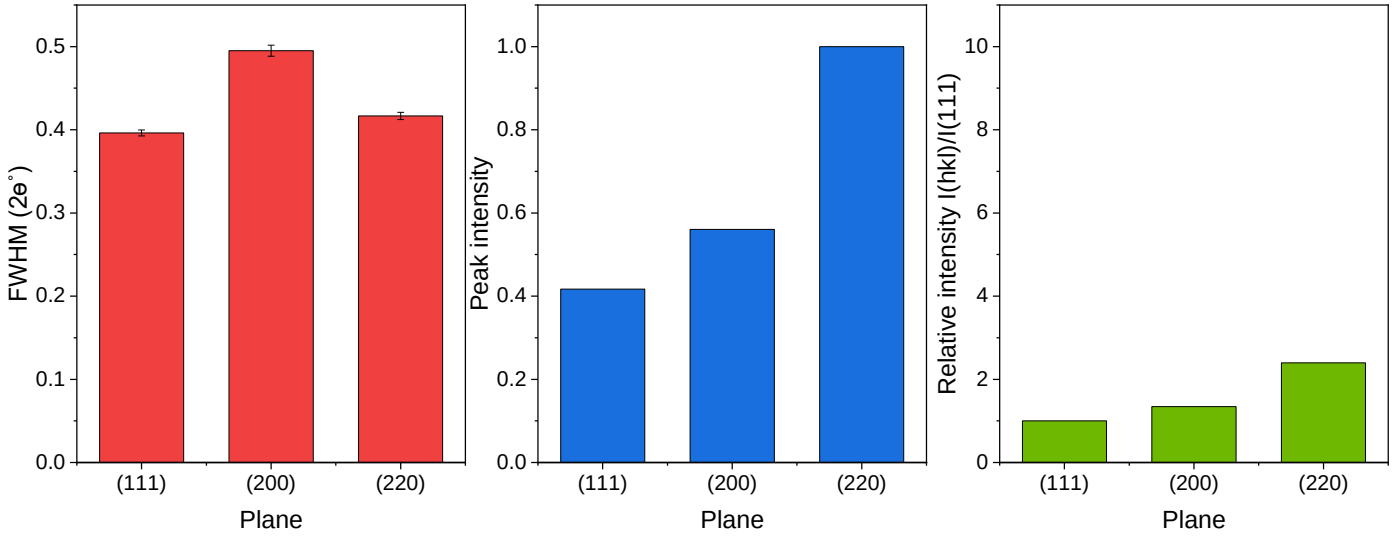
Position 2



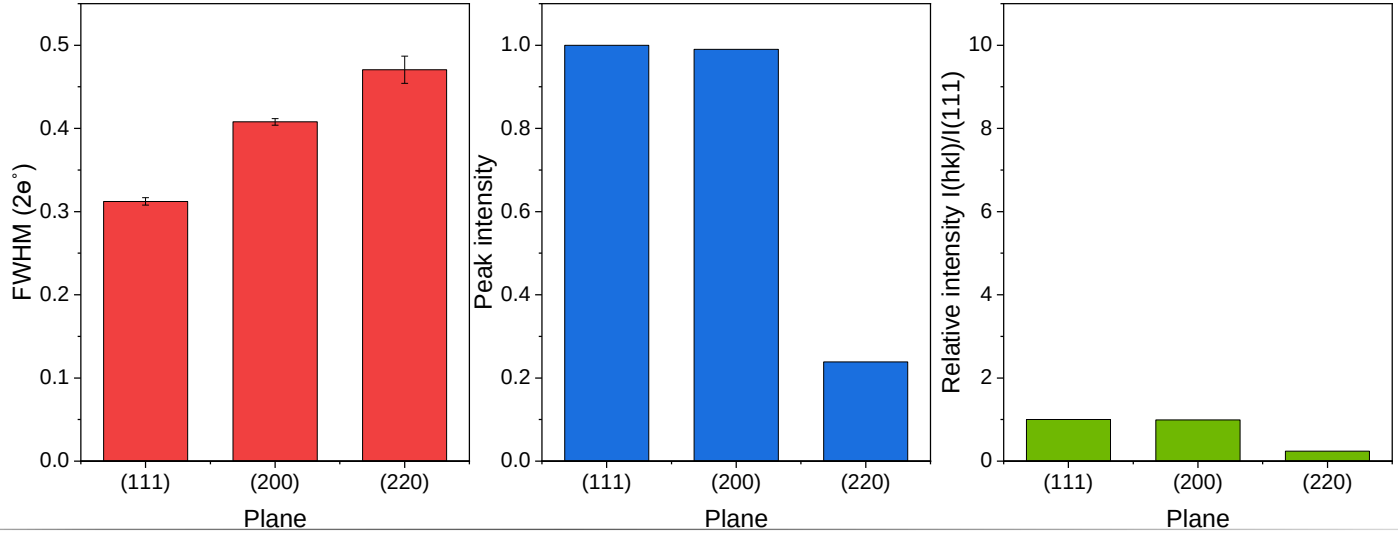
Summarized Data

Sample S13

Position 1



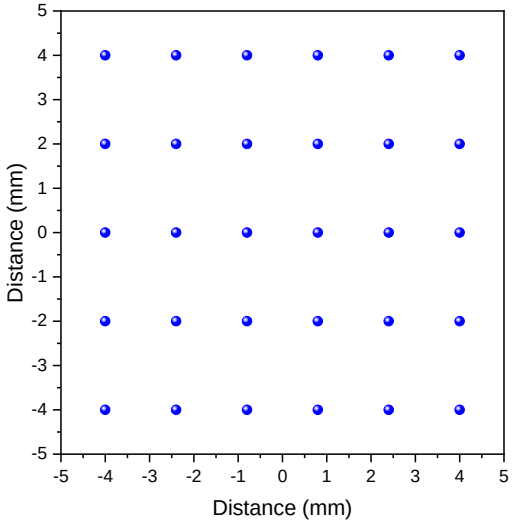
Position 2

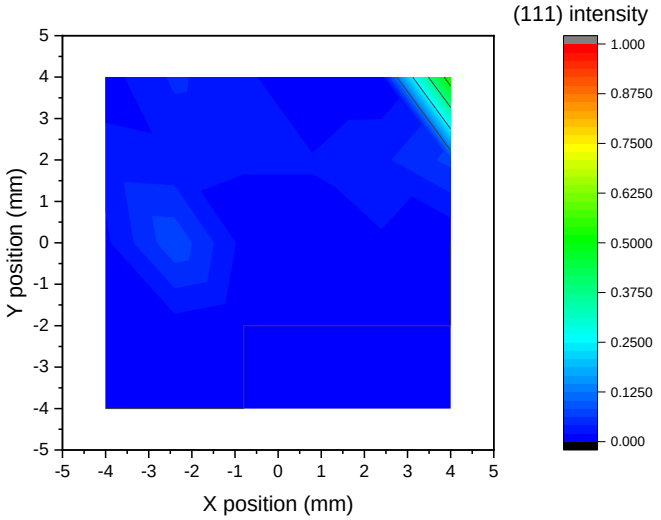


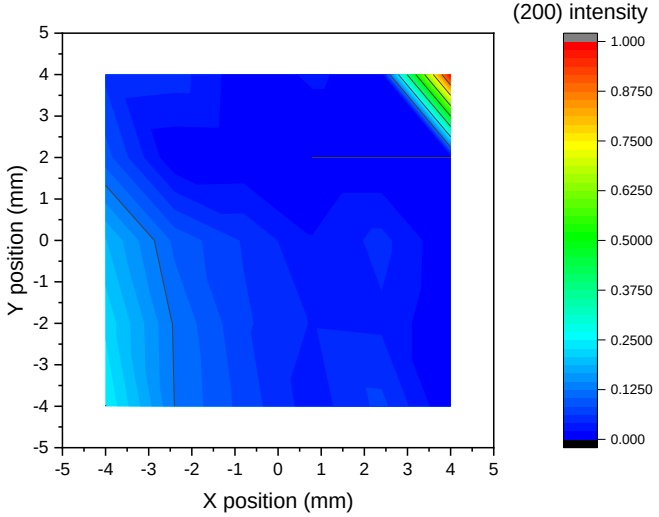
Conclusion

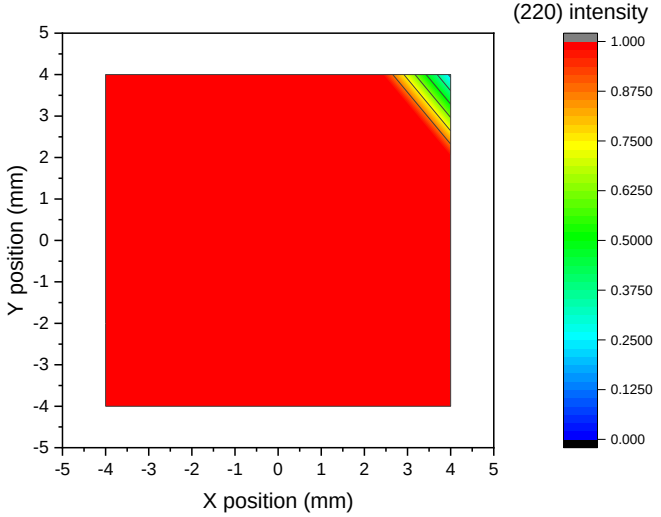
- ✓ The characteristic peak positions of the Cu (111), (002), and (022) planes did not change among the samples. However, their intensities changed significantly between samples and measurements.
- ✓ Some Cu plane peaks showed very low intensity or disappeared in some samples, such as S1 and S6.
- ✓ In general, the characteristic graphene peak around 2θ was not observed in the samples. This may be because the graphene layer was too thin to be detected by XRD. However, a background from the amorphous graphene regions was observed in S1, S2, S4, S5, S6, S10, and S13.
- ✓ In some samples, small additional peaks were observed near the characteristic Cu (002) peak, especially in S2, S4, S7, and S10. These peaks may be related to the CuO (112) plane.

Mapping XRD analysis_S13

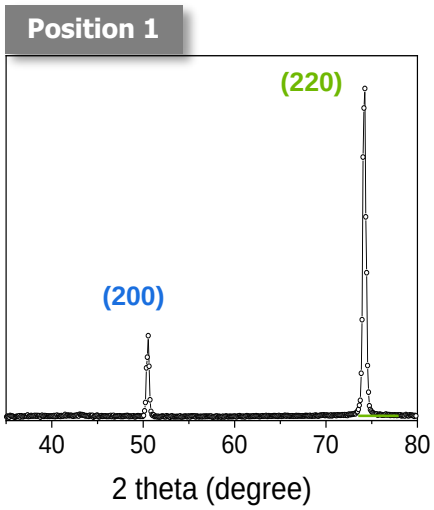


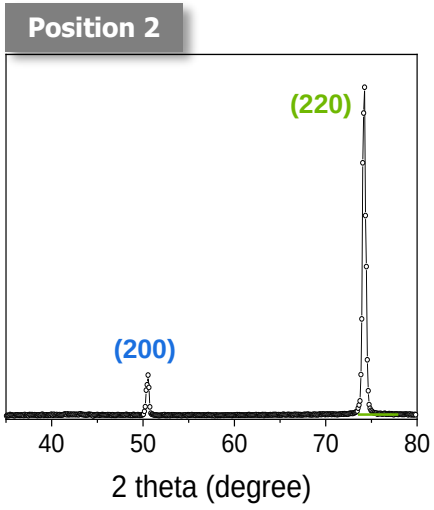


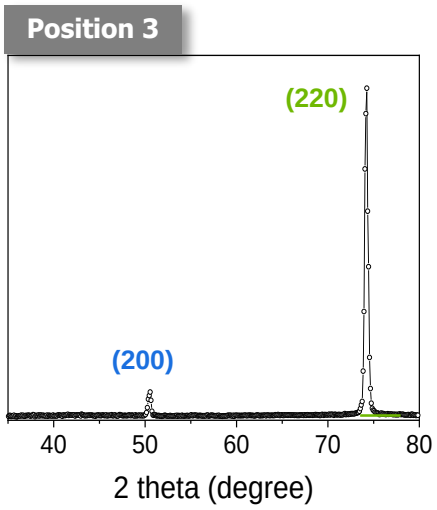


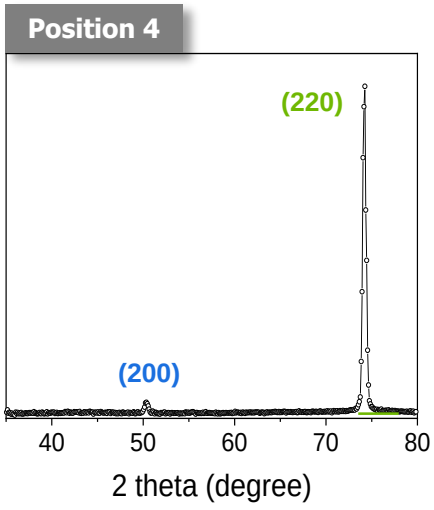


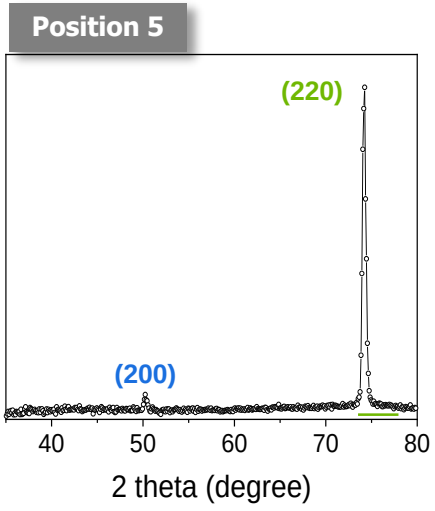
Mapping XRD analysis_S13

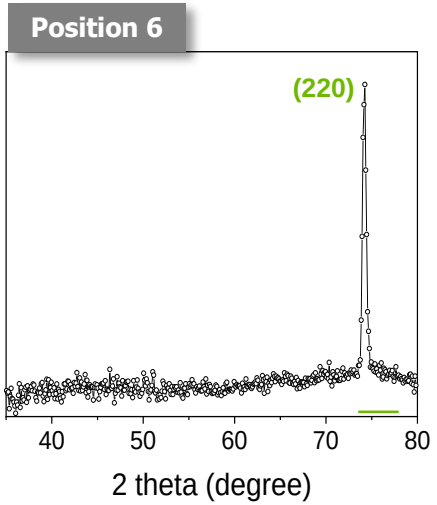




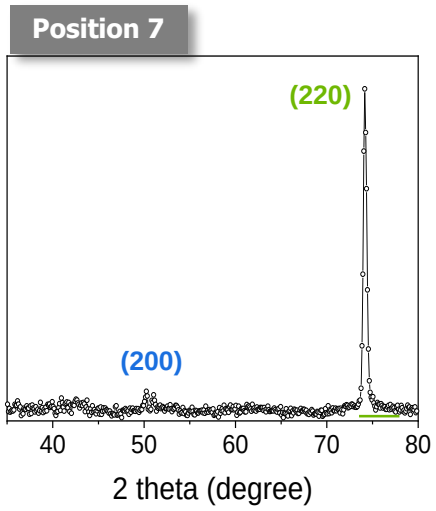


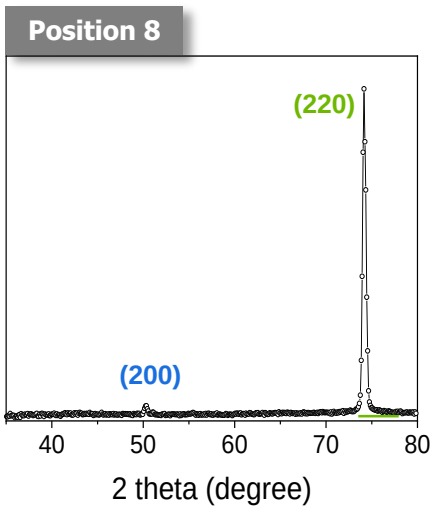


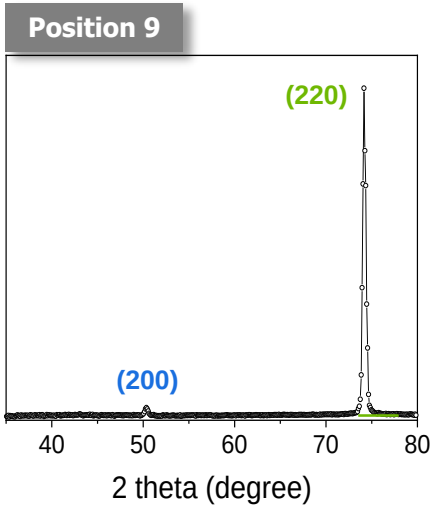


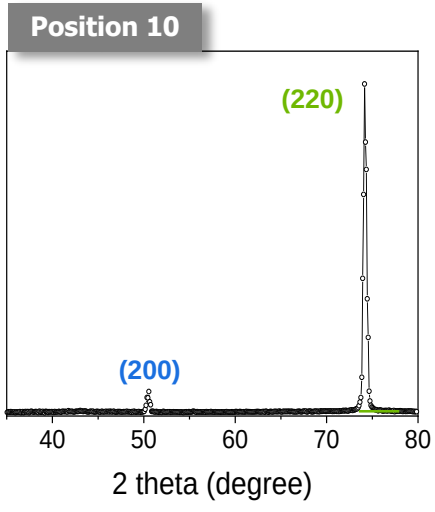


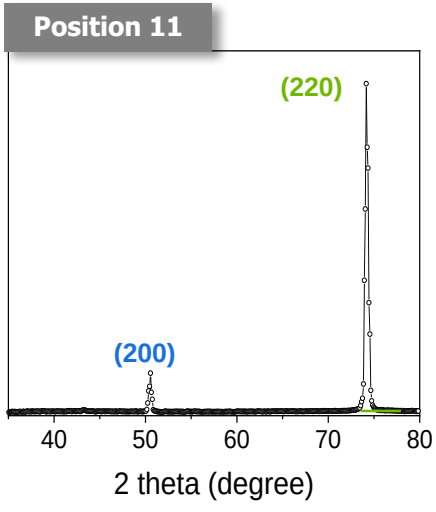
Mapping XRD analysis_S13

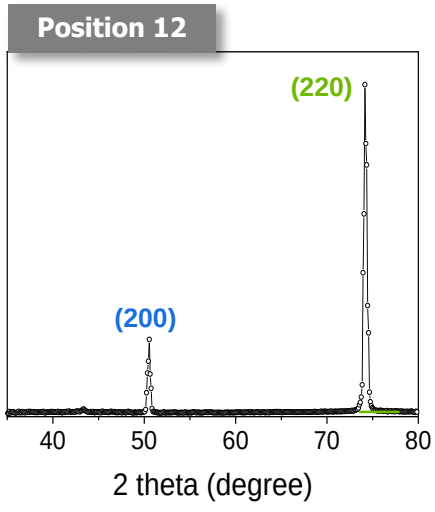




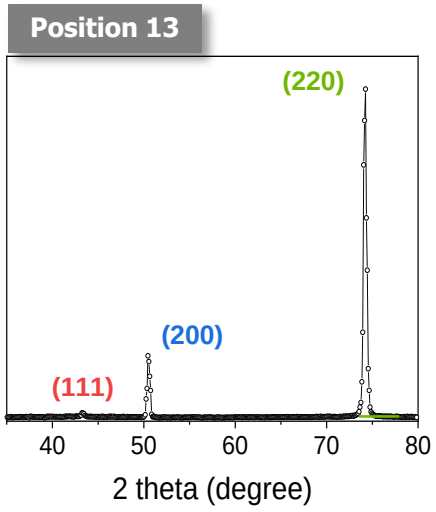


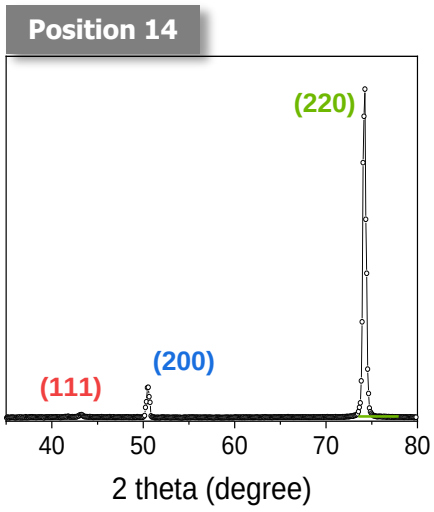


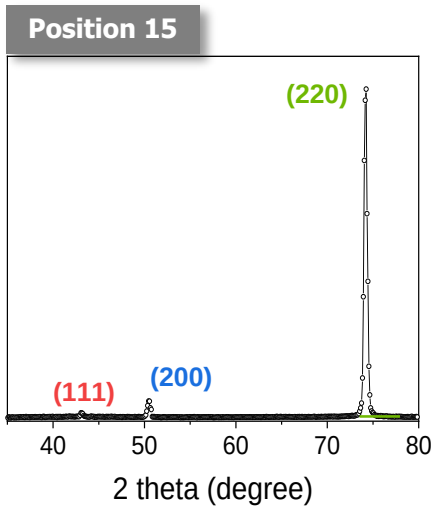


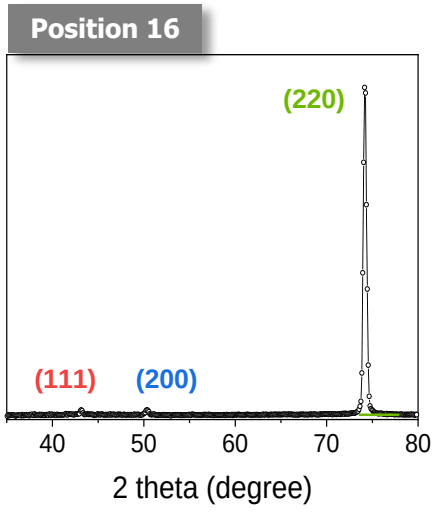


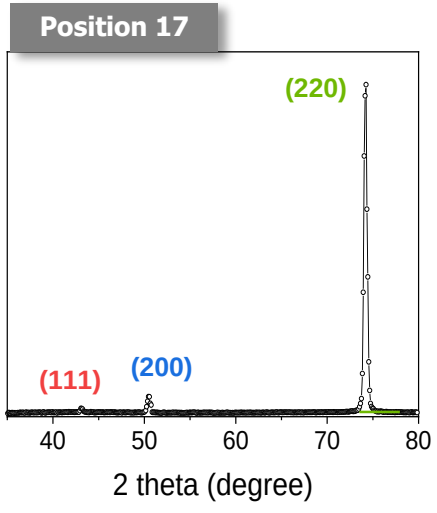
Mapping XRD analysis_S13

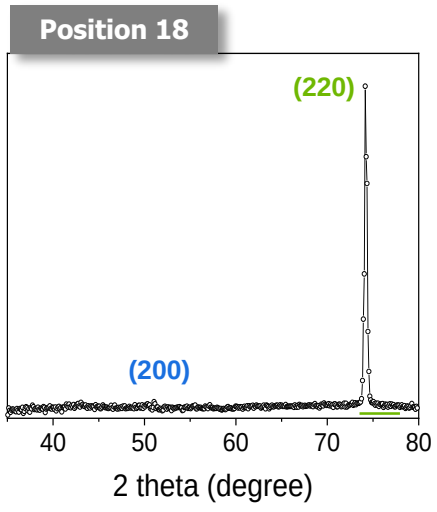




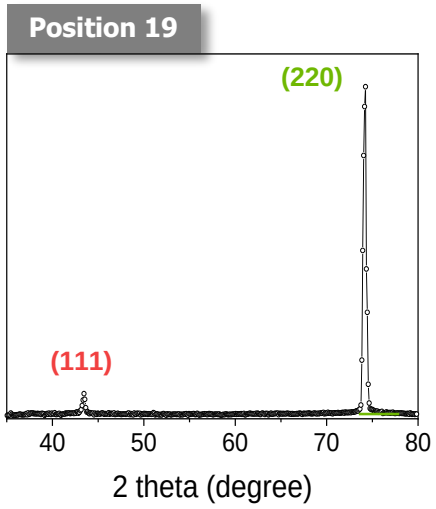


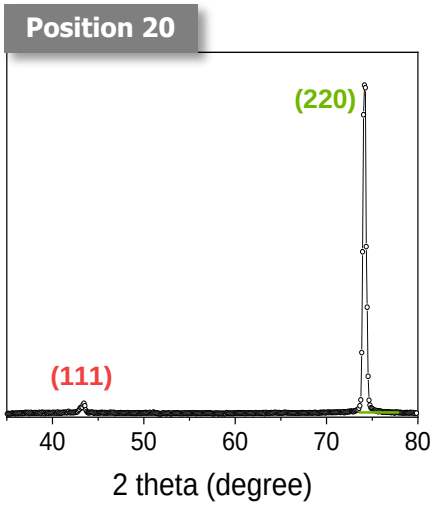


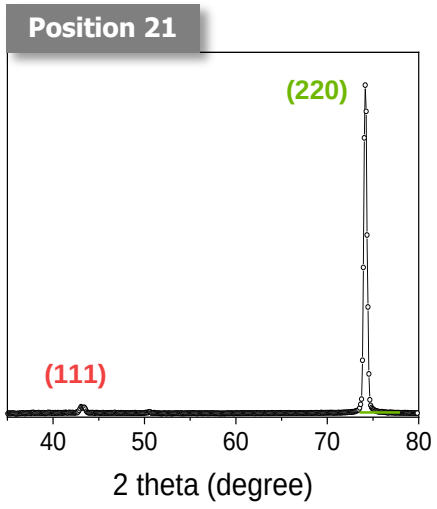


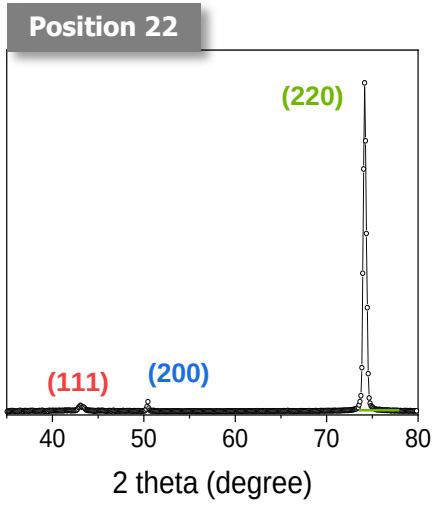


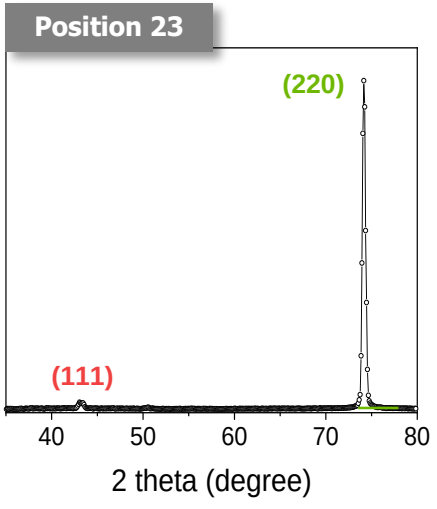
Mapping XRD analysis_S13

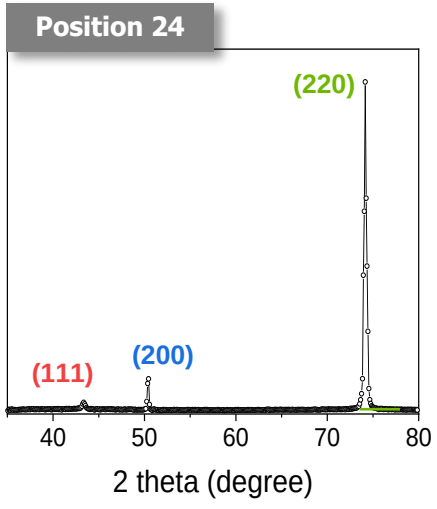




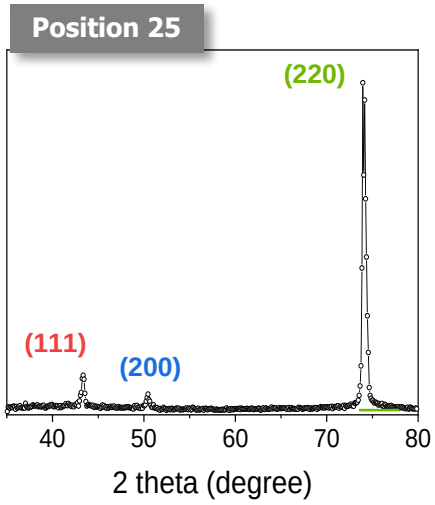


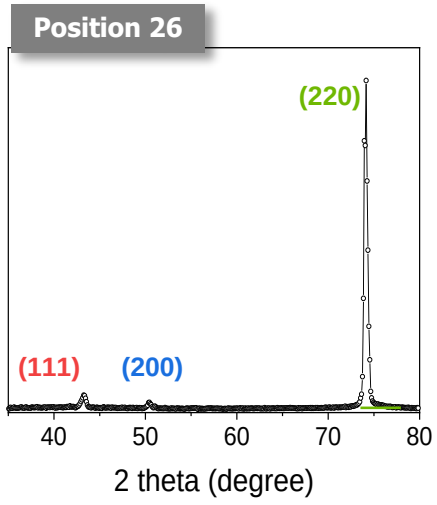


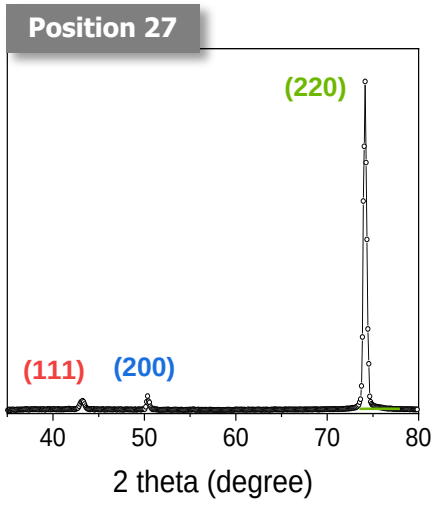


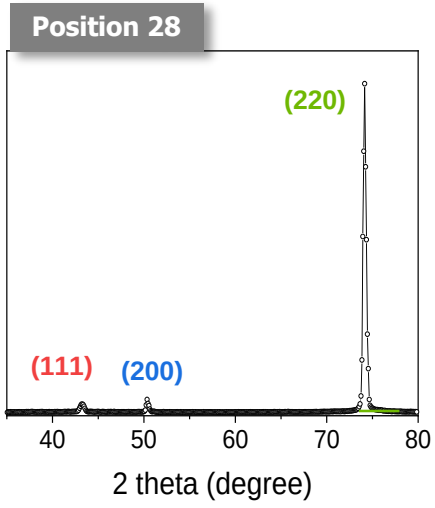


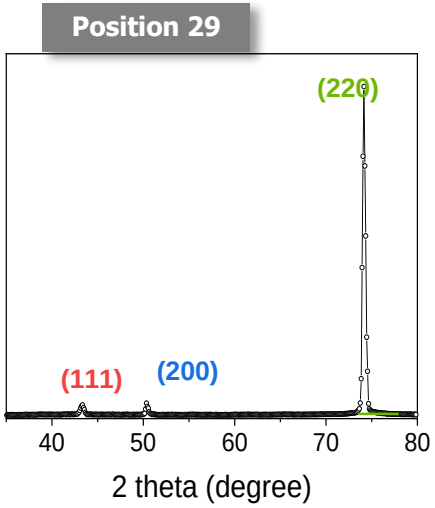
Mapping XRD analysis_S13

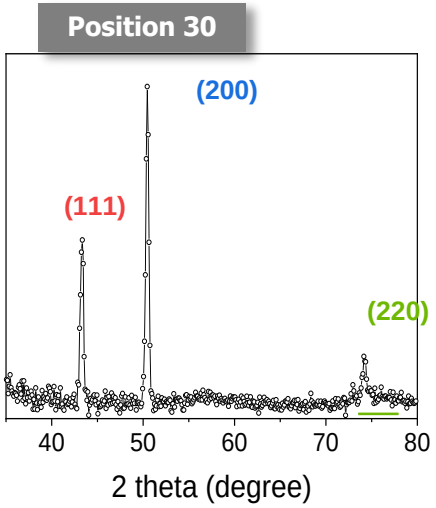




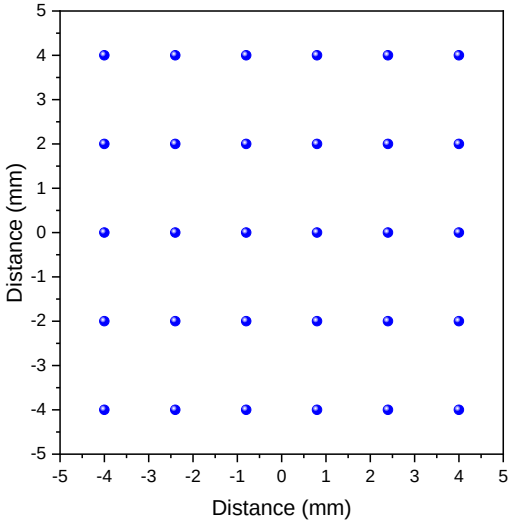


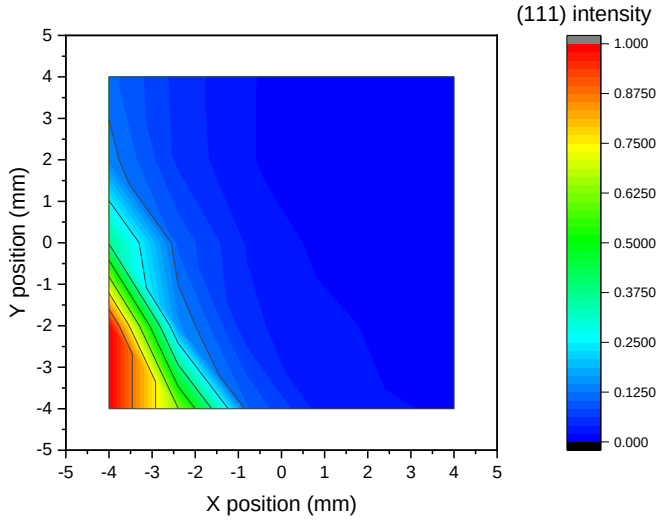


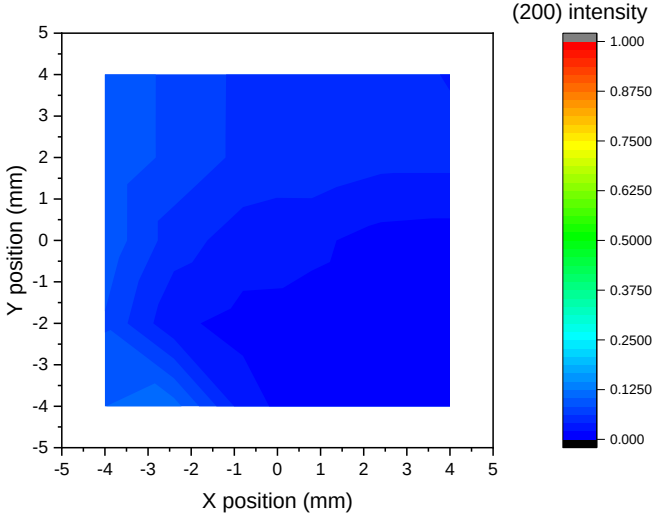


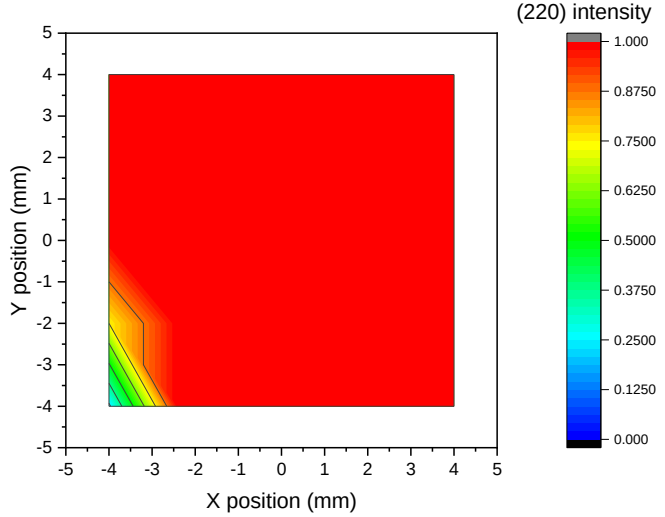


Mapping XRD analysis_S12



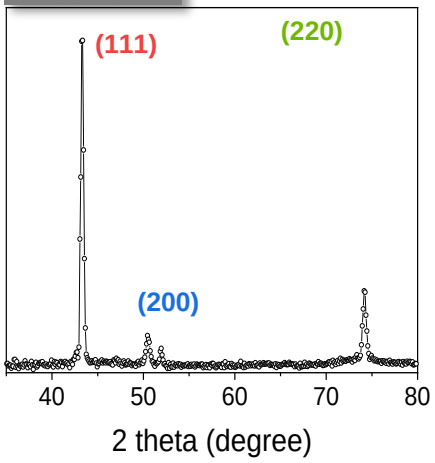




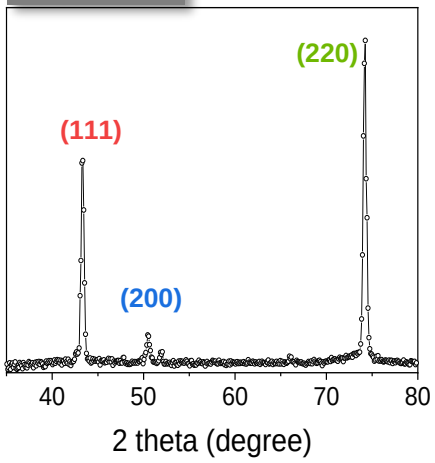


Mapping XRD analysis_S12

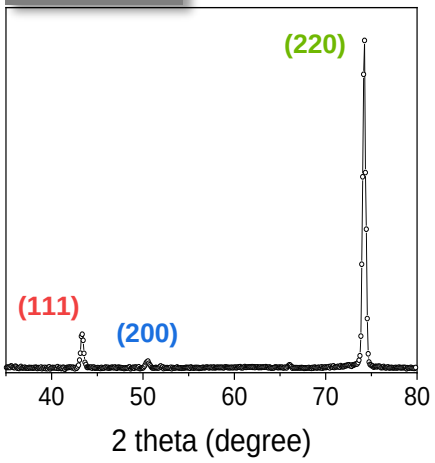
Position 1



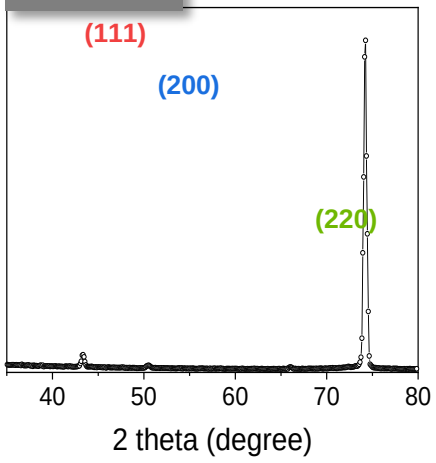
Position 2



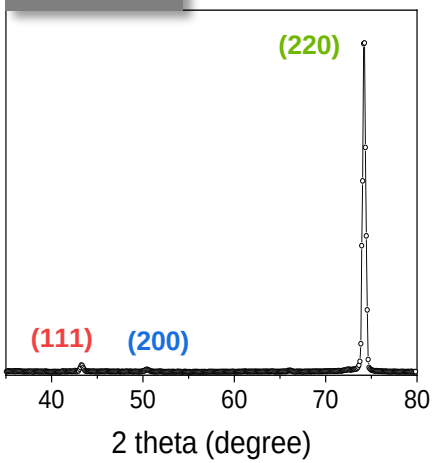
Position 3



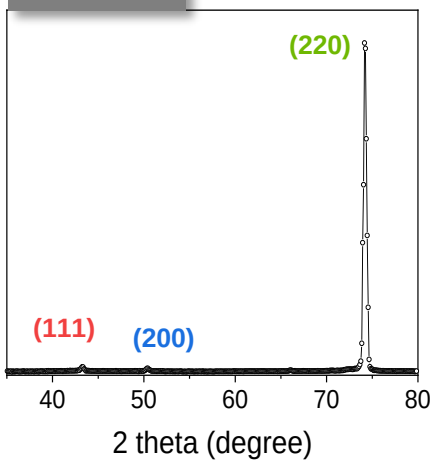
Position 4



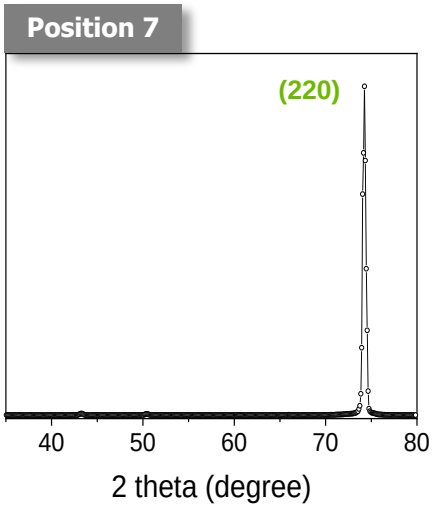
Position 5

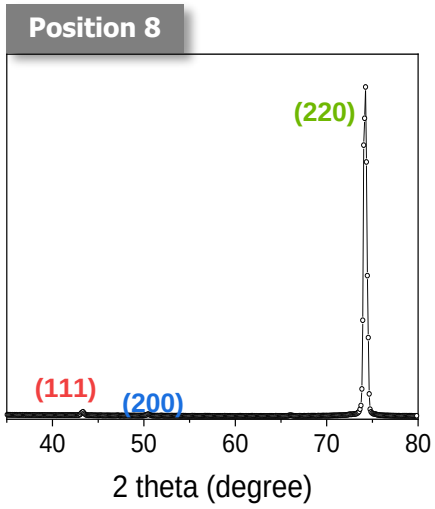


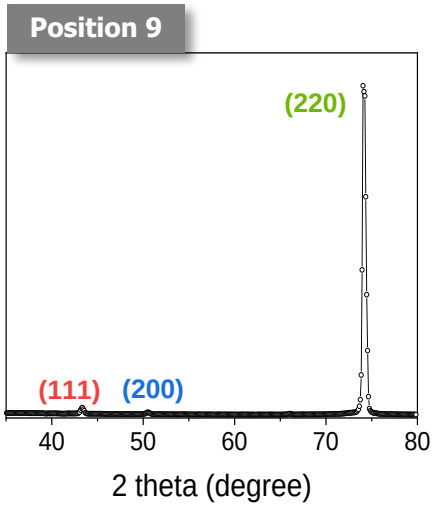
Position 6

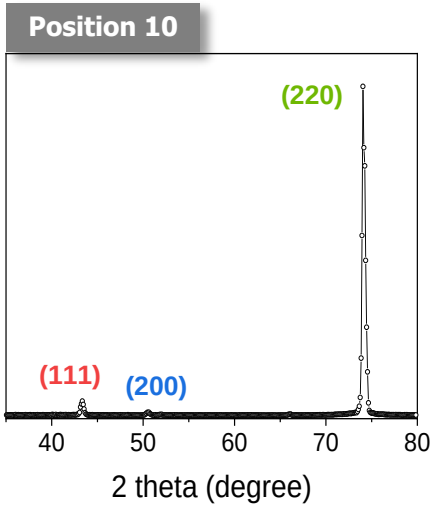


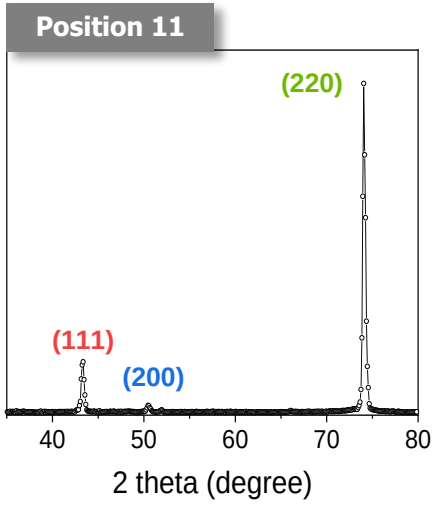
Mapping XRD analysis_S12

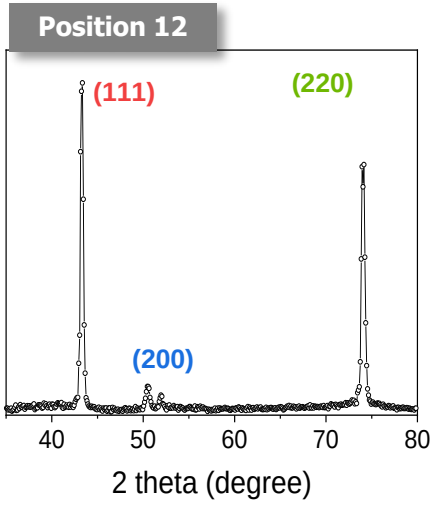




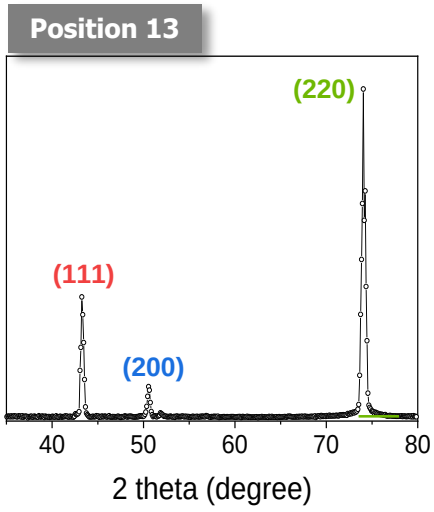


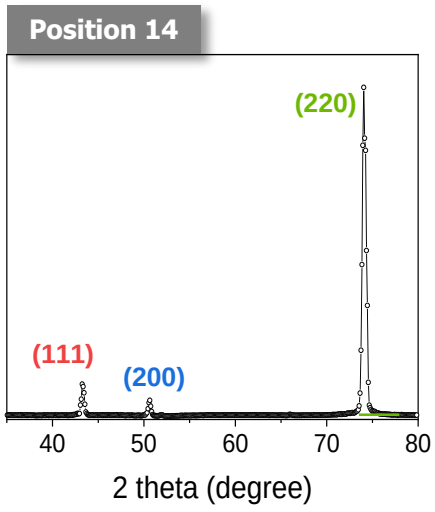


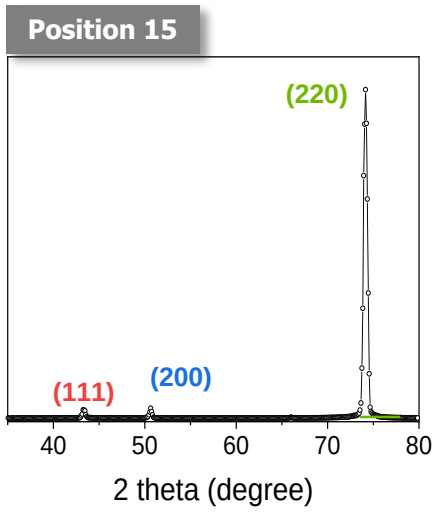


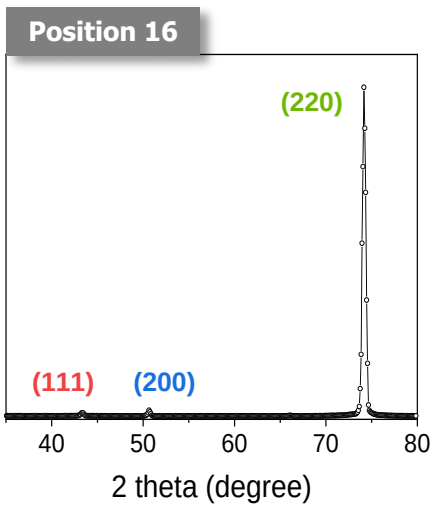


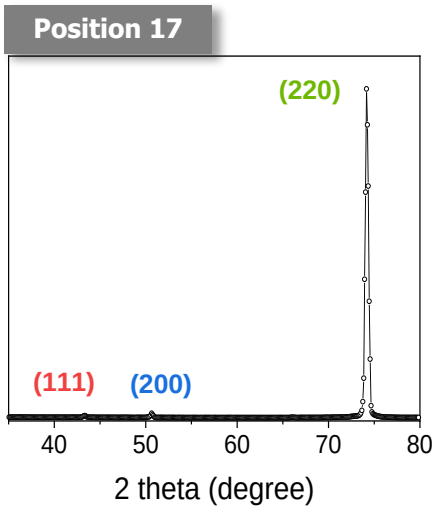
Mapping XRD analysis_S12

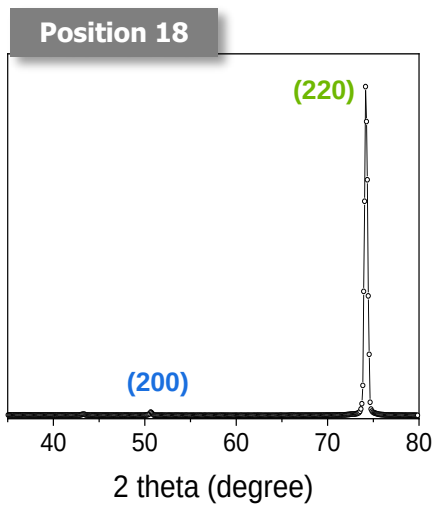




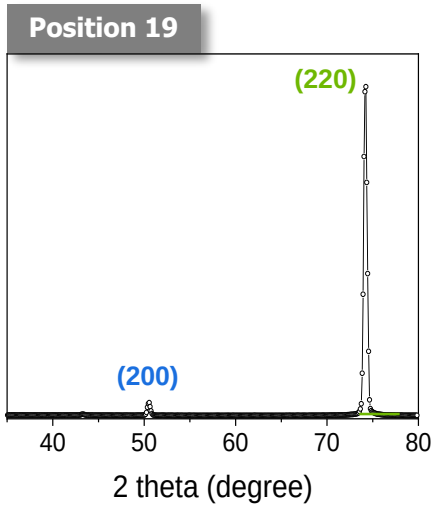


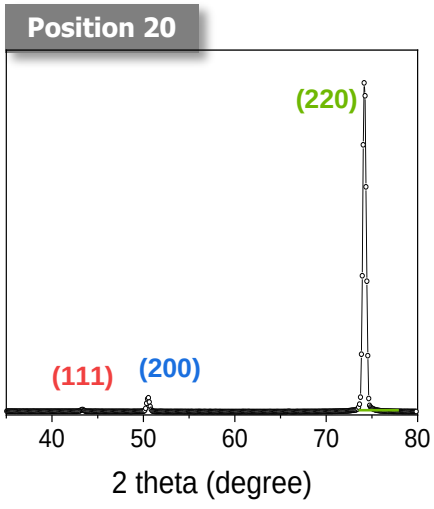


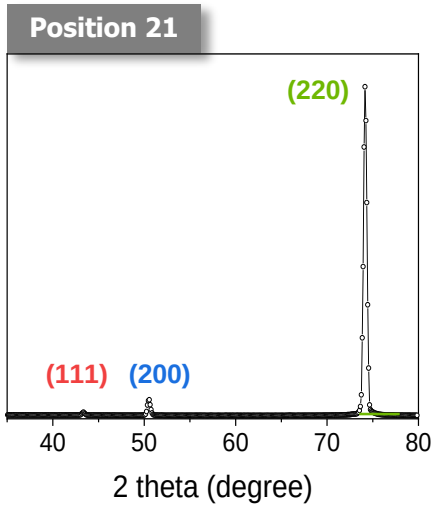


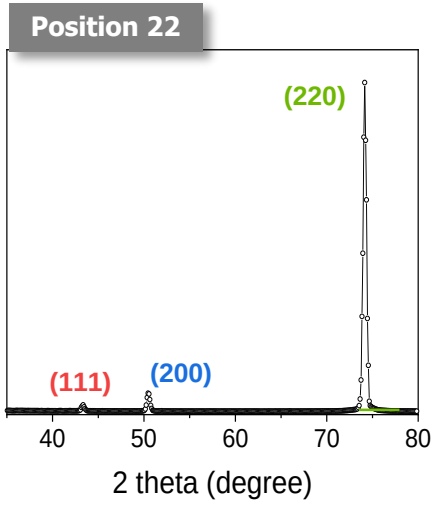


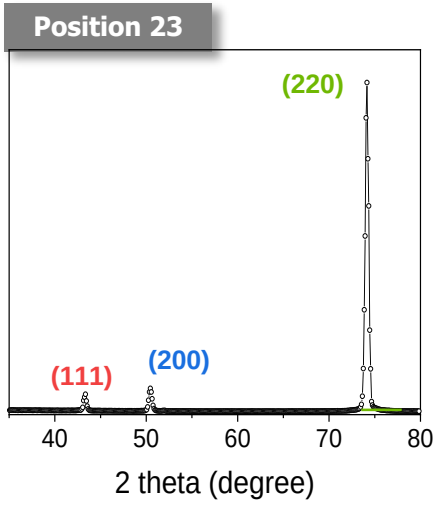
Mapping XRD analysis_S12

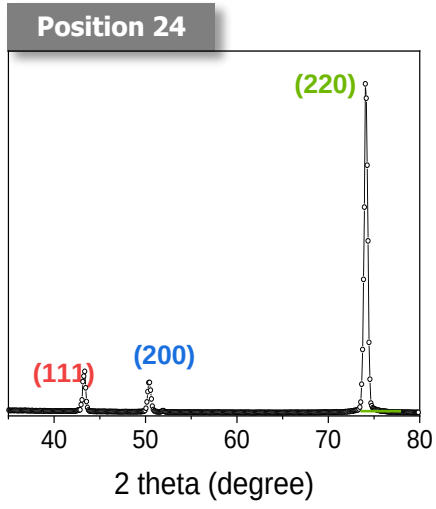












Mapping XRD analysis_S12

